PHYS624 Notes

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Useful Expressions

Euler-Lagrange:

$$\partial_\nu\left(\frac{\partial\mathcal{L}}{\partial\left(\partial_\nu\eta^\rho\right)}\right)=\frac{\partial\mathcal{L}}{\partial\eta^\rho}$$

Energy-momentum tensor:

$$T^\alpha{}_\nu=\frac{\partial\mathcal{L}}{\partial\left(\partial_\alpha\eta^\rho\right)}\partial_\nu\eta^\rho-\mathcal{L}\delta^\alpha{}_\nu$$

Noether current:

$$j^\nu=\frac{\partial\mathcal{L}}{\partial\left(\partial_\nu\eta^\rho\right)}\bar{\delta}\eta^\rho+\mathcal{L}\delta x^\mu$$

Gamma matrices (Chiral basis):

$$\gamma^0=\begin{bmatrix}0 & \mathbb{I}_2\\ \mathbb{I}_2 & 0\end{bmatrix}\qquad\gamma^i=\begin{bmatrix}0 & \sigma^i\\ -\sigma^i & 0\end{bmatrix}\qquad\gamma^5=\begin{bmatrix}-\mathbb{I}_2 & 0\\ 0 & \mathbb{I}_2\end{bmatrix}$$

Transpose relations:

$$\left(\gamma^0\right)^T=\gamma^0\quad\left(\gamma^1\right)^T=-\gamma^1\quad\left(\gamma^2\right)^T=\gamma^2\quad\left(\gamma^3\right)^T=-\gamma^3$$

Complex conjugation relations:

$$\left(\gamma^0\right)^*=\gamma^0\quad\left(\gamma^1\right)^*=\gamma^1\quad\left(\gamma^2\right)^*=-\gamma^2\quad\left(\gamma^3\right)^*=\gamma^3$$

Conjugate transpose relations:

$$\left(\gamma^0\right)^\dagger=\gamma^0\quad\left(\gamma^i\right)^\dagger=-\gamma^i$$

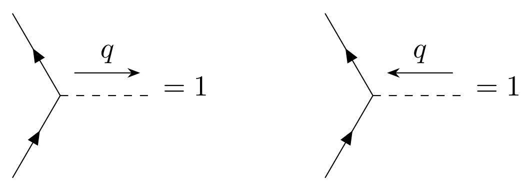
Feynman Rules

ϕ^4 Theory

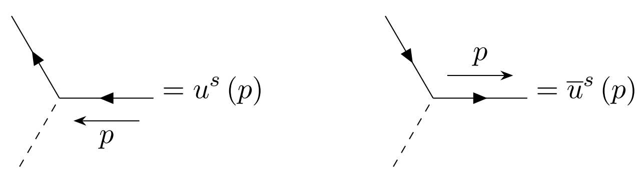
1. Propagators with momentum p contribute $\frac{i}{p^2-m^2+i\varepsilon}$.
2. Each vertex contributes $-i\lambda$.
3. External lines contribute a factor of 1.
4. Impose momentum conservation at each vertex.
5. Integrate over undetermined loop momentum.
6. Divide by the symmetry factor.

Yukawa Theory

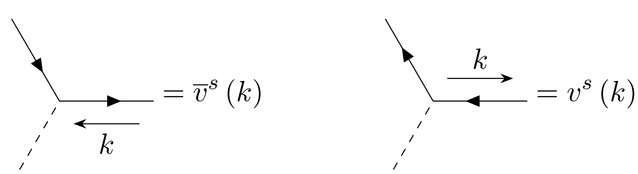
1. The scalar propagator, with momentum q, gives $\frac{i}{q^2-m_\phi^2+i\varepsilon}$
2. The fermion propagator, with momentum p, gives $\frac{i(\not{p})-m_f}{p^2-m_f^2+i\varepsilon}$
3. Vertices give a factor of $-ig$
4. External legs are dependent on the propagator type:
   a) Scalar external legs:



   b) Fermion external legs:

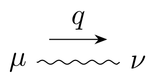


   c) Antifermion external legs:



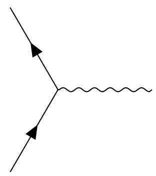
QED

1. Photon propagators:



$$= -\frac{ig_{\mu\nu}}{q^2 + i\varepsilon}$$

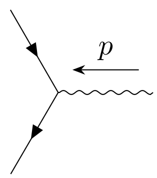
2. Vertices:



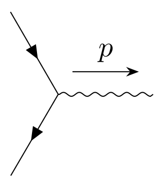
$$= -ie\gamma^{\mu}$$

3. External fermion legs are the same as in the Yukawa theory

4. External photon legs:



$$\to A_\mu \left|p\right\rangle = \varepsilon_\mu \left(p\right)$$



$$\to \left\langle p\right| A_\mu = \varepsilon^*_\mu$$

Motivation

Why do we need Quantum Field Theory? Consider a problem we already know, the Hydrogen atom. Consider an electron in the $2s$ state. If we wait long enough, it will decay to the $1s$ state. How long does it take for the electron to decay to decay from $2s \to 1s$? In QM, we spent time computing the energy splitting between the two states, which sets the frequency of the Lyman line. We never posed the question of how long it takes to transition between the two states. The reason for this is that in non-relativistic QM, the transition time is infinite, these are stationary states, they never decay. Since both mass and energy are conserved separately, the decay process

$$2s \to 1s + \text{photons}$$

can never occur, since particle number is conserved in non-relativistic QM. Usually in non-relativistic QM, we begin with a particle and we maintain that particle, the norm of the particle's wavefunction is conserved in time. In this case, we have photons that come into being, and we need some formalism that allows us to describe processes like this. Let us write down the Schrodinger equation for the Hydrogen atom:

$$\frac{\hbar^2}{2m}\boldsymbol{\nabla}^2\psi\left(\boldsymbol{r},t\right) - \frac{e^2}{r}\psi\left(\boldsymbol{r},t\right) = i\hbar\frac{\partial\psi}{\partial t}$$

We have a wavefunction that describes the electron, and the $\frac{e^2}{r}$ term comes from the electromagnetic field, and we are treating this field completely classically, we are using the classical Coulomb potential. The photons are quantum objects, and they are excitations in the EM field, which means we need to treat them quantum mechanically. Just like we quantized the motion of the electron into $\psi\left(\boldsymbol{r},t\right)$, we need to quantize the EM field in order to obtain a quantum mechanical formalism for the decay process.

QFT has many subtleties, but there is a central idea that we want to highlight. In QM, we discuss wave-particle duality: if we quantize the motion of particles, we observe wave behavior. What we will see in this course is that if we quantize the motion of waves, we get particles, the duality holds bidirectionally. Most of this course will be exploring the quantization of waves and how they generate particles.

What does this other direction of the duality mean? For every particle we think of in nature, we can start with a field description, and each particle will be an excitation of the field, i.e. an electron is an excitation of the electron field.

There are three ingredients that go into QFT:

1. Non-relativistic quantum mechanics
2. Special Relativity
3. Classical field theory

With these three things, we can produce relativistic quantum field theory. In fact, we can pick any of two of these, and we have a consistent subject. For example, if we put non-relativistic QM and special relativity together, we get relativistic QM. If we put classical field theory and special relativity together, we get relativistic classical field theory (such as E&M). Finally, if we put non-relativistic QM together with classical field theory, we will get non-relativistic QFT.

In this course, we will choose to discuss classical field theory and special relativity, to obtain relativistic classical field theory. We will then consider relativistic QM, then non-relativistic QFT, and then finally we will put them all together to look at relativistic QFT.

Part I.
Classical Field Theory

This discussion is taken from the last chapter of Goldstein.

1. Discrete Systems

We can think of classical field theory as the mechanics of continuous media. The way we approach a continuous system is to take it as the limit of a discrete system with many degrees of freedom, we take the continuum limit to recover the continuous system.

Consider an infinitely long elastic rod that undergoes longitudinal vibrations, that is, compression waves. We will approximate this as an infinite chain of point masses spaced a distance a apart, connected by massless springs with spring constant k:

Suppose we have a vibration along this chain, we displace the masses from their equilibrium positions. We label the displacements by η, and we index each mass, that is, the displacement of the ith mass is η_i. At equilibrium, $\eta_i = 0$ for all i.

We can write down the kinetic energy in the chain:

$$T = \frac{1}{2} m \sum_i \dot{\eta}_i^2$$

And the potential energy:

$$V = \frac{1}{2} k \sum_i \left(\eta_{i+1} - \eta_i\right)^2$$

And then write down the Lagrangian:

$$\begin{aligned} L &= T - V \\ &= \frac{1}{2} \sum_i \left[m\dot{\eta}_i^2 - k\left(\eta_{i+1} - \eta_i\right)^2 \right] \end{aligned}$$

We can rewrite this to introduce the chain spacing:

$$L = \sum a L_i$$

Where

$$L_i = \frac{1}{2}\frac{m}{a}\dot{\eta}_i^2 - \frac{1}{2}ka\left(\frac{\eta_{i+1} - \eta_i}{a}\right)^2$$

We now want to relate the quantity ka to the Young's modulus of the material, Y. To do this, we first note that for an elastic rod, from Hooke's Law, the force is equal to the Young's modulus times ξ, the extension per unit length:

$$F = Y\xi$$

Let us apply this to our system. Consider a constant force being applied to one end of the rod. In this case, we have a uniform tension being applied to the springs, which is given by Hooke's Law for a spring:

$$F = k\left(\eta_{i+1} - \eta_i\right)$$

We can rewrite this:

$$F = ka\left(\frac{\eta_{i+1} - \eta_i}{a}\right)$$

Now recall that a is the chain spacing, and therefore ka is the Young's modulus, and the remaining term is exactly the displacement per unit length.

Now let us take the continuum limit of our discrete system. To do this, we move from the discrete index i to a continuous index x. When we make this replacement, we have that η_i becomes $\eta\left(x\right)$, and η_{i+1} becomes $\eta\left(x + a\right)$. Previously, we labelled each mass by its counted number. Instead, we now label each mass by its location at equilibrium. $\eta\left(x\right)$ is the displacement of the mass that, when the system is in equilibrium, would be sitting at location x. Note that x is not a dynamical variable, it is just a constant that labels the equilibrium locations, $\eta\left(x\right)$ is the dynamical variable. This is essentially downgrading x to the level of t, instead of a dynamical variable, it is a parameter that the actual dynamical variables depend on, which is foreshadowing the introduction of relativity, but everything here is completely classical.

Now if we look at how our expressions change when we make this continuum limit:

$$\frac{\eta_{i+1} - \eta_i}{a} \to \frac{\eta\left(x + a\right) - \eta\left(x\right)}{a}$$

Now we note that in the continuum limit, this becomes $d\eta/dx$:

$$\frac{\eta\left(x + a\right) - \eta\left(x\right)}{a} = \frac{d\eta}{dx}$$

Now if we look at our summation in the continuum limit:

$$a\sum_i \to \int dx$$

And at our m/a, which now becomes the mass per unit length, μ:

$$\frac{m}{a} \to \mu$$

Putting all of these together, we find that the full continuum Lagrangian is given by

$$L = \frac{1}{2}\int \mathrm{d}x \left[\mu\dot{\eta}^2 - Y\left(\frac{\partial\eta}{\partial x}\right)^2\right]$$

Now using the Lagrangian, we can obtain the equation of motion. Let us first look at the discrete equations of motion for the ith mass:

$$m\ddot{\eta}_i - k\left(\eta_{i+1} - \eta_i\right) - k\left(\eta_i - \eta_{i-1}\right) = 0$$

Suppose we now take the continuum limit of this equation. In this case, we have that:

$$\eta_{i+1} - \eta_i \to a\left(\frac{\partial\eta}{\partial x}\right)\bigg|_x$$
$$\eta_i - \eta_{i-1} \to a\left(\frac{\partial\eta}{\partial x}\right)\bigg|_{x-a}$$

This gives us the continuum expression:

$$a\left[\mu\frac{\partial^2\eta}{\partial t^2} - ka\frac{\partial^2\eta}{\partial x^2}\right] = 0$$

Now recall that $Y = ka$, so we have the continuum equation of motion:

$$\mu\frac{\partial^2\eta}{\partial t^2} - Y\frac{\partial^2\eta}{\partial x^2} = 0$$

Which is the wave equation, and our wave velocity will be $v = \sqrt{Y/\mu}$. We obtained this by taking the continuum limit of the discrete equation of motion, but let us now recover this directly from the continuum Lagrangian that we derived earlier, rather than first discussing the discrete case.

2. Continuous Lagrangian Formalism

Usually, when we do particle mechanics, we write down an action, which is the time integral of a Lagrangian. In this case our Lagrangian is itself an integral over a variable. We denote the integrand as the Lagrangian density, $\mathcal{L}$:

$$L = \frac{1}{2}\int \mathrm{d}x \left[\mu\dot{\eta}^2 - Y\left(\frac{\partial\eta}{\partial x}\right)^2\right]$$
$$\mathcal{L} = \frac{1}{2}\left[\mu\dot{\eta}^2 - Y\left(\frac{\partial\eta}{\partial x}\right)^2\right]$$

Using this denotation, the action is the time integral and the spatial integral of $\mathcal{L}$.

We want to obtain the equation of motion directly from $\mathcal{L}$. In general, the Lagrangian density is a function of η and it its partials[1], along with the parameters that we have, x and t:

$$\mathcal{L} = \mathcal{L}\left(\eta, \frac{\partial\eta}{\partial x}, \frac{\partial\eta}{\partial t}, x, t\right)$$

[1] Note that we are not technically restricted to just the first order partials, but for higher order partials, we end up with differential equations that are harder to solve and produce spurious solutions.

Starting from this, we define the action[2]:

$$S = \int_{t_1}^{t_2} \int_{x_1}^{x_2} \mathrm{d}x \, \mathrm{d}t \, \mathcal{L}$$

We want to extremize the action with respect to variations of the dynamical variable, η. Note that we fix the endpoints in t and x, at both ends of the trajectory. We thus fix the variation of η at the endpoints to be zero.

Suppose the variation is of the form:

$$\eta\left(x,t\right) = \eta_0\left(x,t\right) + \delta\eta\left(x,t\right)$$

In this form, our previous fixing of the variation is written as:

$$\begin{aligned}\delta\eta\left(x_1,t\right) = \delta\eta\left(x_2,t\right) = 0\\ \delta\eta\left(x,t_1\right) = \delta\eta\left(x,t_2\right) = 0\end{aligned}$$

We can now write out the variation in the action:

$$\delta S = \iint \mathrm{d}x \, \mathrm{d}t \left[\frac{\partial \mathcal{L}}{\partial \eta}\delta\eta + \frac{\partial \mathcal{L}}{\partial\left(\frac{\partial \eta}{\partial x}\right)}\delta\left(\frac{\partial \eta}{\partial x}\right) + \frac{\partial \mathcal{L}}{\partial\left(\frac{\partial \eta}{\partial t}\right)}\delta\left(\frac{\partial \eta}{\partial t}\right)\right]$$

Now thinking back to Lagrangian dynamics, we note that

$$\begin{aligned}\delta\left(\frac{\partial \eta}{\partial x}\right) &= \frac{\partial\left(\eta_0 + \delta\eta\right)}{\partial x} - \frac{\partial \eta_0}{\partial x}\\ &= \frac{\partial}{\partial x}\delta\eta\end{aligned}$$

And similarly for $\frac{\partial \eta}{\partial t}$. This allows us to rewrite the change in our action:

$$\delta S = \iint \mathrm{d}x \, \mathrm{d}t \left[\frac{\partial \mathcal{L}}{\partial \eta}\delta\eta + \frac{\partial \mathcal{L}}{\partial\left(\frac{\partial \eta}{\partial x}\right)}\underbrace{\delta\left(\frac{\partial \eta}{\partial x}\right)}_{\frac{\partial}{\partial x}\delta\eta} + \frac{\partial \mathcal{L}}{\partial\left(\frac{\partial \eta}{\partial t}\right)}\underbrace{\delta\left(\frac{\partial \eta}{\partial t}\right)}_{\frac{\partial}{\partial t}\delta\eta}\right]$$

Now integrating by parts, and noting that the boundary terms vanish because of the fixed boundary conditions:

$$\delta S = \iint \mathrm{d}x \, \mathrm{d}t \left[\frac{\partial \mathcal{L}}{\partial \eta} - \frac{\partial}{\partial x}\left(\frac{\partial \mathcal{L}}{\partial\left(\frac{\partial \eta}{\partial x}\right)}\right) - \frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial\left(\frac{\partial \eta}{\partial t}\right)}\right)\right]\delta\eta$$

If we set $\delta S = 0$, then we see that the only way for this to be true is if everything in the square brackets is zero, which is the same as the usual Lagrangian argument. This leaves us with the Euler-Lagrange equation for the continuous case:

$$\frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial\left(\frac{\partial \eta}{\partial t}\right)}\right) + \frac{\partial}{\partial x}\left(\frac{\partial \mathcal{L}}{\partial\left(\frac{\partial \eta}{\partial x}\right)}\right) - \frac{\partial \mathcal{L}}{\partial \eta} = 0$$

[2] Chacko uses I to denote the action.

This is the equation that must be satisfied on the classical trajectory, the one that extremizes the action. Note that in the discrete case, we had a set of coupled ODEs, but in the continuous case we have a single PDE.

Let us apply this to our elastic rod, and see if this recovers the previously obtained equation of motion. We have our Lagrangian density:

$$\mathcal{L} = \frac{1}{2}\mu\left(\frac{\partial\eta}{\partial t}\right)^2 - \frac{1}{2}Y\left(\frac{\partial\eta}{\partial x}\right)^2$$

Now computing our partials:

$$\frac{\partial\mathcal{L}}{\partial\left(\frac{\partial\eta}{\partial x}\right)} = -Y\frac{\partial\eta}{\partial x}$$
$$\frac{\partial\mathcal{L}}{\partial\left(\frac{\partial\eta}{\partial t}\right)} = \mu\frac{\partial\eta}{\partial t}$$

Similarly, we can look at $\frac{\partial\mathcal{L}}{\partial\eta}$:

$$\frac{\partial\mathcal{L}}{\partial\eta} = 0$$

Which, when inserted into our equation, gives us:

$$\mu\frac{\partial^2\eta}{\partial t^2} - Y\frac{\partial^2\eta}{\partial x^2} = 0$$

Which is exactly what we obtained from taking the discrete system to the continuum limit.

In this case, we have only used a single dynamical field, η. How do we generalize this to multiple fields?

Suppose we are now working with more spatial dimensions. In this case, we move from t, x to x^μ, where μ is an index, $\mu = 0, 1, 2, 3$, where $x^0 = t$, $x^1 = x$, $x^2 = y$, and $x^3 = z$. Suppose we have a general number of fields, so η becomes η^ρ, where ρ indexes over some arbitrary number of indices, it may be a Lorentz index, or it could be any number of scalar fields. We keep this arbitrary so that we can derive all cases at once.

We can write down the general Lagrangian density:

$$\mathcal{L} = \mathcal{L}\left(\eta^\rho, \partial_\nu\eta^\rho, x^\nu\right)$$

We want to extremize the action, which is now an integral over all spacetime:

$$S = \int \mathrm{d}^4x\,\mathcal{L}$$

Note that in this formalism, space and time are on equal footing, so it will be easy to generalize to relativity.

Now looking at variations in η^ρ:

$$\eta^\rho = {\eta_0}^\rho + \delta\eta^\rho$$

We again fix the endpoints, $\delta\eta^\rho = 0$ at the endpoints in spacetime.

We can look at variations in the action, where we use Einstein notation, summation over repeated indices is implied:

$$\delta S = \int \mathrm{d}^4 x \, \left[\frac{\partial \mathcal{L}}{\partial \eta^\rho}\delta\eta^\rho + \frac{\partial \mathcal{L}}{\partial\left(\partial_\nu \eta^\rho\right)}\delta\left(\partial_\nu \eta^\rho\right)\right]$$

Again noting that $\delta\left(\partial_n u\eta^\rho\right) = \partial_\nu\left(\delta\eta^\rho\right)$, and integrating by parts, we have that

$$\delta S = \int \mathrm{d}^4 x \, \left[\frac{\partial \mathcal{L}}{\partial \eta^\rho}\delta\eta^\rho - \partial_\nu\left(\frac{\partial \mathcal{L}}{\partial\left(\partial_\nu \eta^\rho\right)}\right)\right]$$

Setting this equal to zero, and using the same argument as the single field case, we have the general Euler-Lagrange equation in the continuous formalism:

$$\partial_\nu\left(\frac{\partial \mathcal{L}}{\partial\left(\partial_\nu \eta^\rho\right)}\right) - \frac{\partial \mathcal{L}}{\partial \eta^\rho} = 0$$

With this, we can take a very general Lagrangian density, and then obtain the equation of motion.

Recall the classical dynamics of a single point particle. In this case, we have the Euler-Lagrange equation:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0$$

We see that this is very similar to our continuous result, we just make space take the same footing as time, and we recover the same form.

2.1. Energy-Momentum Tensor

Recall from point particle mechanics, that we have energy conservation if the Lagrangian does not explicitly depend on time. In our continuous formalism, we will show that if the Lagrangian density does not depend on x^0, we have energy conservation, and if the density does not depend on x^i then p^i is conserved.

Let us first recall the classical proof of this, which we will then generalize to the field formalism.

If we have no explicit dependence of L on t, then we have that

$$L = L\left(q_i, \dot{q}_i\right)$$

If this is the case, then

$$\begin{aligned}\frac{dL}{dt} &= \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\frac{d}{dt}\left(\dot{q}_i\right) + \frac{\partial L}{\partial q_i}\frac{dq_i}{dt} \\ &= \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\dot{q}_i\right) - \dot{q}_i\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) + \frac{\partial L}{\partial q_i}\frac{dq_i}{dt}\end{aligned}$$

Where we have rewritten the first term. Now applying the equation of motion, we know that

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = \frac{\partial L}{\partial q_i}$$

Inserting this, we see that the second and third terms cancel:

$$\frac{dL}{dt} = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\dot{q}_i\right)$$

Which can be rewritten:

$$\frac{d}{dt}\left(\dot{q}_i\frac{\partial L}{\partial \dot{q}_i} - L\right) = 0$$

Now nothing that this is just the time derivative of the Hamiltonian:

$$\frac{d}{dt}H = 0$$

We see that the total energy (the Hamiltonian) is a constant of the motion.

Now let us generalize this to the field formalism. We have a Lagrangian density, which is a function of our fields η^ρ, their partials, $\partial_\nu \eta^\rho$, but explicitly not a function of x^μ:

$$\mathcal{L} = \mathcal{L}\left(\eta^\rho, \partial_\nu \eta^\rho\right)$$

We can look at $\frac{\partial L}{\partial x^\nu}$:

$$\frac{\partial \mathcal{L}}{\partial x^\nu} = \frac{\partial \mathcal{L}}{\partial \eta^\rho}\partial_\nu \eta^\rho + \frac{\partial \mathcal{L}}{\partial\left(\partial_\alpha \eta^\rho\right)}\partial_\nu\left(\partial_\alpha \eta^\rho\right)$$

Now rewriting the second term, just as we did in the classical derivation:

$$\frac{\partial \mathcal{L}}{\partial\left(\partial_\alpha \eta^\rho\right)}\partial_\nu\partial_\alpha \eta^\rho = \partial_\alpha\left[\frac{\partial \mathcal{L}}{\partial\left(\partial_\alpha \eta^\rho\right)}\partial_\nu \eta^\rho\right] - \partial_\alpha\left[\frac{\partial \mathcal{L}}{\partial\left(\partial_\alpha \eta^\rho\right)}\right]\partial_\nu \eta^\rho$$

By the equation of motion, we see that the second term here cancels with the first term in the equation above.

Thus we are left with

$$\frac{\partial \mathcal{L}}{\partial x^\nu} = \partial_\alpha\left[\frac{\partial \mathcal{L}}{\partial\left(\partial_\alpha \eta^\rho\right)}\partial_\nu \eta^\rho\right]$$

Which we can rewrite as:

$$\partial_\alpha\left[\frac{\partial \mathcal{L}}{\partial\left(\partial_\alpha \eta^\rho\right)}\partial_\nu \eta^\rho - \mathcal{L}\delta^\alpha{}_\nu\right] = 0$$

We define the quantity in brackets as $T^\alpha{}_\nu$, which is known as the energy-momentum tensor (or the stress-energy tensor):

$$T^\alpha{}_\nu = \frac{\partial \mathcal{L}}{\partial\left(\partial_\alpha \eta^\rho\right)}\partial_\nu \eta^\rho - \mathcal{L}\delta^\alpha{}_\nu$$

Our equation tells us that this tensor is a constant of the motion:

$$\partial_\alpha T^\alpha{}_\nu = 0$$

We can compare this to the continuity equation from electromagnetism:

$$\frac{\partial \rho}{\partial t} = -\boldsymbol{\nabla} \cdot \boldsymbol{J}$$

Which can be rewritten in tensor notation:

$$\partial_\mu j^\mu = 0$$

Where $j^0 = \rho$, and $j^i = \boldsymbol{j}$. This equation looks very similar, except for the fact that our recently derived equation has two indices, rather than 1. The continuity equation tells us that the net current inflow is equal to the charge enclosed in the region. We can show this by integrating the equation over some region:

$$\begin{aligned}\int d^3x \, \frac{\partial \rho}{\partial t} &= \int d^3x \, (-\boldsymbol{\nabla} \cdot \boldsymbol{J}) \\ \frac{\partial}{\partial t} \int d^3x \, \rho &= -\int d^3x \, \boldsymbol{\nabla} \cdot \boldsymbol{J} \\ \frac{\partial}{\partial t} Q_{\text{enclosed}} &= -\int d\boldsymbol{A} \cdot \boldsymbol{J}\end{aligned}$$

We see that this equates the change in enclosed charge to the flux of the current flowing out of the region. In the case where the volume is all of space, current vanishes at the boundary, and we have global conservation of charge:

$$\frac{\partial}{\partial t} Q_{\text{enclosed}} = 0$$

We will show that the energy-momentum tensor relation that we derived behaves exactly the same, but with 4 independent relations, due to the extra index. In the case where $\nu = 0$, we get energy conservation, and when $\nu = 1, 2, 3$, we get momentum conservation in the respective direction. If we look at some volume, we get similar continuity relations for energy and momentum. Let us show this. We can write out the relation and then integrate over a volume:

$$\begin{aligned}\partial_t T^0{}_\nu + \partial_j T^j{}_\nu &= 0 \\ \partial_t \int d^3x \, T^0{}_\nu + \int d^3x \, \partial_j T^j{}_\nu &= 0\end{aligned}$$

If we integrate over all of space, then the second term becomes a surface integral, and vanishes:

$$\int d^3x \, \partial_j T^j{}_\nu \to \int ds \, n_j T^j{}_\nu = 0$$

Then we have just the first term:

$$\partial_t \underbrace{\int d^3x \, T^0{}_\nu}_{R_\nu} = 0$$

We denote the 4 quantities R_ν the "conserved charges". We denote $T^0{}_\nu$ as the "charge densities", and $T^j{}_\nu$ are the "charge current densities". These are all named via direct analogy to electromagnetism. We have shown that these quantities are conserved, but we have not brought in any ideas of energy or momentum.

Let us consider again the elastic rod. In this case, we can compute ${T^0}_0$:

$$ {T^0}_0 = \frac{1}{2}\mu\left(\frac{\partial\eta}{\partial t}\right)^2 + \frac{1}{2}Y\left(\frac{\partial\eta}{\partial x}\right)^2 $$

We see that these are the kinetic energy density and potential energy density, respectively. Thus ${T^0}_0$ is the total energy density of the system. Thus we see that the conservation of R_0 over all of space indicates that energy is conserved over all of space.

We could also consider ${T^j}_0$, which are energy current densities[3]. Instead, let us come to ${T^0}_x$ in the case of the elastic rod:

$$ {T^0}_x = \mu\dot{\eta}\frac{\partial\eta}{\partial x} $$

Let us consider the momentum density of the elastic rod. There is a contribution $\mu\dot{\eta}$ (mass times velocity), which is there even for rigid body motion. We want the contribution to the momentum from wave motion inside the rod. We first notice that when wave motion happens, there is a net change in the mass of the element of the rod between x and $x+dx$. If $\eta(x) > \eta(x+dx)$, then the mass between the two points has decreased. This change is given by:

$$ \mu\left[\eta(x) - \eta(x+dx)\right] \to -\mu\frac{\partial\eta}{\partial x}\,dx $$

Where we have taken the continuum limit. The net change in momentum between x and $x+dx$ associated with this motion is:

$$ \left(-\mu\frac{\partial\eta}{\partial x}\,dx\right)\dot{\eta} = \left(-\mu\dot{\eta}\frac{\partial\eta}{\partial x}\right)\,dx $$

Which matches exactly what we got for ${T^0}_x$ (with a minus sign). This is known as the wave (or field) momentum density, and the integral of this over all space is conserved. Thus we have connected our conserved R_ν quantities to the conservation of energy and wave momentum in the elastic rod case.

2.2. Hamiltonian Formulation

Up until now, we have been working with the Lagrangian formalism, but at various points we will consider the Hamiltonian formulation. Let us once again consider the discrete chain of masses and springs. Conjugate to each η_i, we have a canonical momentum, p_i:

$$ \begin{aligned} p_i &= \frac{\partial L}{\partial\dot{\eta}_i} \\ &= a\frac{\partial L_i}{\partial\dot{\eta}_i} \end{aligned} $$

When we take the continuum limit of this, we see that p_i vanishes, since $L_i \sim \frac{1}{2}\mu\dot{\eta}_i^2$, and $a \to 0$. The derivative is finite, but a goes to zero, so $p_i = 0$. However, we can define the momentum density π, which is nonzero in the continuum limit:

$$ \pi = \lim_{a\to 0}\frac{p_i}{a} $$

[3] Goldstein shows these in the case of the elastic rod, and discusses the physical intuition for them, as well as the momentum current densities.

$$= \frac{\partial \mathcal{L}}{\partial \dot{\eta}}$$

Note that in general, π is a function of x. Now that we have the conjugate momentum, we can write down the Hamiltonian:

$$\begin{aligned} H &= \sum_i p_i \dot{\eta}_i - L \\ &= a \sum_i \left(\frac{\partial L_i}{\partial \dot{\eta}_i} \dot{\eta}_i - L_i \right) \end{aligned}$$

Writing this in the continuum limit:

$$H = \int dx \, \left(\pi \left(x \right) \dot{\eta} \left(x \right) - \mathcal{L} \right)$$

Which defines the Hamiltonian density, $\mathcal{H}$:

$$\mathcal{H} = \pi \left(x \right) \dot{\eta} \left(x \right) - \mathcal{L}$$

Essentially, when we define a valid continuum momentum, the Hamiltonian density is exactly what we would expect. We can generalize this to more than 1 field, where we now have multiple momenta:

$$\pi_\rho = \frac{\partial \mathcal{L}}{\partial \left(\dot{\eta}^\rho \right)}$$

For each field, we have a conjugate momentum. We can then write the multiple-field Hamiltonian density:

$$\mathcal{H} = \sum_\rho \pi_\rho \dot{\eta}^\rho - \mathcal{L}$$

Note that this is exactly what we found for ${T^0}_0$:

$$\mathcal{H} = {T^0}_0$$

This should not be very surprising, since this is exactly the same result we find in classical mechanics, as long as our Lagrangian is not explicitly time-dependent, the Hamiltonian is a constant of the motion.

2.3. Noether's Theorem

Let us now discuss Noether's theorem, which is a connection between the symmetry properties of the Lagrangian, and conserved quantities, known as currents. We have seen this theorem from classical mechanics, let us now discuss the form that this theorem takes in classical field theory.

Symmetry in Physics

For some historical background, at the turn of the 20th century, we had several revolutionary ideas. The first of these was special relativity, which not only introduced a new set of physical laws, but also changed the way that physicists do physics. When Maxwell's equations were written down, these laws came out of a lot of experimentation, a long series of experiments.

The way that Einstein derived $E = mc^2$ was to realize that Maxwell's equations have a symmetry property, they are invariant under Lorentz transformations. He then imposed the requirement of Lorentz symmetry onto the other laws of physics and explored what happened. He was the first physicist to put symmetry first. After Einstein did this, Dirac required that the laws of quantum mechanics be invariant under Lorentz transformations, and derived the existence of the positron. Pauli predicted the neutrino via the symmetry of beta decays, Gell-Mann generalized isospin symmetry to discover the Ω^-. This method was used extensively since Einstein in the 20th century, the application of symmetries to the laws of physics.

Theorem 2.1. *Noether's Theorem. For every continuous symmetry transformation of the Lagrangian, there exists a conserved current.*

Note that this is only in one direction, symmetries imply conserved quantities, but conserved quantities do not imply symmetries. Another thing to note is that the transformations that produce conserved currents are necessarily continuous, there must be some infinitesimal generator of the symmetry transformation.

Let us now prove Noether's Theorem. Let us first consider a transformation[4]:

$$x^\mu \to x^{\mu'} = x^\mu + \delta x^\mu \left(x^\nu\right)$$

Where the second term is a function x^μ of x^ν. This can be any general transformation. This transformations affects our field:

$$\eta^\rho \left(x^\mu\right) \to \eta^{\rho'} \left(x^{\mu'}\right) = \eta^\rho \left(x^\mu\right) + \delta\eta^\rho \left(x^\mu\right)$$

This changes our Lagrangian density, but we can claim that the functional form is the same if the transformation is a symmetry of the Lagrangian:

$$\mathcal{L}' \left[\eta'^\rho \left(x'^\mu\right), \partial'_\nu \eta'^\rho \left(x'^\mu\right), x'^\mu\right] = \mathcal{L} \left[\eta'^\rho \left(x'^\mu\right), \partial'_\nu \eta'^\rho \left(x'^\mu\right), x'^\mu\right] \tag{1}$$

This is known as *form invariance*, the form of the Lagrangian does not change. If the transformation is not a symmetry, it would be $\mathcal{L}'$ on the right side, rather than the original $\mathcal{L}$. The other claim that we could make is that the action remains the same:

$$S' = \int_{\Omega'} dx^{\mu'}\, \mathcal{L}' \left[\eta^\rho \left(x^{\mu'}\right), \partial'_\nu \eta^{\rho'} \left(x^{\mu'}\right), x^{\mu'}\right] = \int_\Omega dx^\mu\, \mathcal{L} \left[\eta^\rho \left(x^\mu\right), \partial_\nu \eta^\rho \left(x^\mu\right), x^\mu\right] \tag{2}$$

This is known as *scale invariance*. Note that the Jacobian determinant in the left integral is not present, this is what makes the transformation a symmetry. If we had explicitly written out the Jacobian determinant then this statement would be true for all transformations. According to Goldstein, a transformation must require both of these invariances for the transformation to be a symmetry. However, according to Chacko, there is a weaker claim that still holds. Consider the case where form invariance is not met, but scale invariance is. The argument for this case existing is that a scaling transformation might change d^4x, but the functional form of the Lagrangian might be changed in the exact opposite way, in order to keep the action integral invariant. Thus we would not have the exact same functional form of the Lagrangian, breaking form invariance, but the action would be invariant, giving us scale invariance. However, let us just go with what Goldstein says, because the most common case is that both conditions are met.

[4]Note that this is a *passive* transformation, but can be done in the language of active transformations as well.

Let us do an example of a transformation and Lagrangian. Consider the Lagrangian:

$$\mathcal{L} = \partial_\mu \phi^\dagger \partial^\mu \phi - m^2 \phi^\dagger \phi + m^2 \left(\phi^2 + \phi^{\dagger 2}\right)$$

This is a scalar field, the dagger represents complex conjugation. Now consider a field transformtaion:

$$\phi \to \phi' = e^{i\alpha}\phi$$

In this case, we can look at $\phi'^\dagger$:

$$\phi'^\dagger = e^{-i\alpha}\phi^\dagger$$

With this, we can rewrite our Lagrangian density in terms of ϕ' and $\phi'^\dagger$:

$$\begin{aligned}\mathcal{L}' &= \partial_\mu \left[e^{i\alpha}\phi'^\dagger\right] \partial^\mu \left[e^{-i\alpha}\phi'\right] - m^2 \left[e^{i\alpha}\phi'^\dagger\right]\left[e^{-i\alpha}\phi'\right] + m^2 \left(e^{-2i\alpha}\phi'^2 + e^{2i\alpha}\phi'^{\dagger 2}\right) \\ &= \partial_\mu \phi'^\dagger \partial^\mu \phi' - m^2 \phi'^\dagger \phi' + m^2 \left(e^{-2i\alpha}\phi'^2 + e^{2i\alpha}\phi'^{\dagger 2}\right)\end{aligned}$$

We see that this transformation is not a symmetry transformation, since we have an extra $e^{\pm 2i\alpha}$ in the third term. If we removed the last term from the Lagrangian, then this would be a symmetry transformation of the modified Lagrangian, since the first two terms remain form invariant.

Let us now continue proving Noether's theorem. Suppose we satisfy both Defn. 1 and Defn. 2, we have both form and scale invariance. In this case, the statement of scale invariance can be rewritten as:

$$\int_{\Omega'} dx^\mu \, \mathcal{L}\left[\eta^{\rho'}\left(x^\mu\right), \partial_\nu \eta^{\rho'}\left(x^\mu\right), x^\mu\right] - \int_\Omega dx^\mu \, \mathcal{L}\left[\eta^\rho\left(x^\mu\right), \partial_\nu \eta^\rho\left(x^\mu\right), x^\mu\right] = 0$$

Where we have changed the Lagrangian in the left integral to $\mathcal{L}$, via form invariance. Note that we have also changed the dummy variable of integration of the left integral from x' to x. These integrals differ in their region of integration, as well as the fields that are in the Lagrangian. In other words, we have an of $\Delta\mathcal{L}$ over the region Ω, and we have an integral of $\mathcal{L}$ over the region $\Omega' - \Omega$. Thus we can rewrite the equation as:

$$\underbrace{\int_\Omega dx^\mu \, \left(\mathcal{L}\left[\eta^{\rho'}, \partial_\nu \eta^{\rho'}, x^\mu\right] - \mathcal{L}\left[\eta^\rho, \partial_\nu \eta^\rho, x^\mu\right]\right)}_{\Delta\mathcal{L} \text{ in } \Omega} + \underbrace{\int_{\Omega'-\Omega} dx^\mu \, \mathcal{L}\left[\eta^{\rho'}, \partial_\nu \eta^{\rho'}, x^\mu\right]}_{\mathcal{L} \text{ in disjoint region}} = 0 \tag{3}$$

The first integral seems workable, but what do we do with the second integral? We will not treat this integral honestly, instead we will deal with it in 1 dimension and then claim that it generalizes to the 4D integral. Consider the analogous 1D case of our equation:

$$\int_{a+\delta a}^{b+\delta b} dx \, \left[f\left(x\right) + \delta f\left(x\right)\right] - \int_a^b dx \, f\left(x\right) = 0$$

$$\int_a^b dx \, \delta f\left(x\right) + \int_b^{b+\delta b} dx \, \left[f\left(x\right) + \delta f\left(x\right)\right] - \int_a^{a+\delta a} dx \, \left[f\left(x\right) + \delta f\left(x\right)\right] = 0$$

Now dropping the $\delta f\left(x\right)$ in the integrals that are over infinitesimal regions, and claiming that $f\left(x\right)$ is approximately constant over such small regions:

$$\begin{aligned}\int_a^b dx\,\delta f\left(x\right)+\int_b^{b+\delta b}dx\,\left[f\left(x\right)+\delta f\left(x\right)\right]-\int_a^{a+\delta a}dx\,\left[f\left(x\right)+\delta f\left(x\right)\right]=0\\ \int_a^b dx\,\delta f\left(x\right)+f\left(b\right)\delta b-f\left(a\right)\delta a=0\end{aligned}$$

Now rewriting this to "undo" an integration:

$$\int_a^b dx\,\left(\delta f\left(x\right)+\frac{d}{dx}\left[f\left(x\right)\delta x\right]\right)=0$$

Where δx is any smooth function of x that satisfies the boundary conditions of the integral. Now what is the higher dimensional analog of our newly derived 1D equation? We claim that it is:

$$\int_{\Omega'-\Omega}dx^\mu\,\mathcal{L}\left[\eta^{\rho'},\partial_\nu\eta^{\rho'},x^\mu\right]=\int_S dS_\mu\,\mathcal{L}\left[\eta^\rho,\partial_\nu\eta^\rho,x^\mu\right]\delta x^\mu$$

Where the term on the right is the result of the integral over the infinitesimal disjoint region. Similarly to the 1D case, we can write this as a total derivative:

$$\int_S dS^\mu\,\mathcal{L}\left[\eta^\rho,\partial_\nu\eta^\rho,x\right]\delta x_\mu=\int_\Omega dx^\nu\,\frac{\partial}{\partial x^\nu}\left[\mathcal{L}\left[\eta^\rho,\partial_\nu\eta^\rho,x^\mu\right]\delta x^\mu\right]$$

This is essentially Gauss's Law in 3+1 dimensions. We are saying that the Lagrangian does not have enough time to change over the small disjoint region, and we can integrate it over the surface of the disjoint region. Now let us return to Equation 3, where we have now dealt with the second integral. Let us now consider the first term. We can rewrite this as:

$$\int_\Omega dx^\mu\,\left(\mathcal{L}\left[\eta^{\rho'},\partial_\nu\eta^{\rho'},x^\mu\right]-\mathcal{L}\left[\eta^\rho,\partial_\nu\eta^\rho,x^\mu\right]\right)=\int_\Omega dx^\mu\left[\frac{\partial\mathcal{L}}{\partial\eta^\rho}\bar{\delta}\eta^\rho+\frac{\partial\mathcal{L}}{\partial\left(\partial_\nu\eta^\rho\right)}\bar{\delta}\left(\partial_\nu\eta^\rho\right)\right]$$

Where $\bar{\delta}\eta$ is the change in η at the point with coordinates x^μ (as opposed to $\delta\eta$, which is the change in η at the same physical point). To demonstrate the difference between our two deltas, consider a field $\phi\left(x\right)$, which is a scalar under rotations. We take some physical point x, and we rotate our coordinate system, and the new coordinate of the same physical point is now x'. The field at the physical point must always be the same, regardless of the coordinate system:

$$\phi\left(x\right)=\phi'\left(x'\right)$$

Knowing this, we can find ϕ':

$$\begin{aligned}\phi\left(x\right)&=\phi'\left(x'\right)\\ \phi\left(x\right)&=\phi'\left(R^{-1}x\right)\end{aligned}$$

Recall that

$$\eta^{\rho'}\left(x^{\mu'}\right)=\eta^\rho\left(x^\mu\right)+\delta\eta^\rho\left(x^\mu\right)$$

For a scalar under rotations, the value at the same physical point does not change, so $\delta\eta^\rho\left(x^\mu\right)=0$. However, there is a point in our new coordinate system that has the same label as our physical point in the original coordinates. This is a *different* physical point, but they have the same coordinates. This is how we define $\bar{\delta}\eta^\rho$:

$$\bar{\delta}\eta^\rho = \eta^{\rho'}(x) - \eta^\rho(x)$$

Note that for our scalar field, this is *not* zero, unlike $\delta\eta^\rho$. We can write out what $\bar{\delta}\eta^\rho$ is:

$$\begin{aligned}\bar{\delta}\eta^\rho &= \eta^{\rho'}(x) - \eta^\rho(x)\\ &= \eta^{\rho'}\left(x' - \delta x\right) - \eta^\rho(x)\\ &= \eta^{\rho'}\left(x'\right) - \delta x^\alpha \frac{\partial \eta^{\rho'}}{\partial x'^\alpha} - \eta^\rho(x)\\ &= -\delta x^\alpha \frac{\partial \eta^\rho}{\partial x^\alpha}\end{aligned}$$

Note that we drop the primes in the derivative because we already have a δx, the difference between the primed and unprimed coordinates is subleading relative to δx. We can now compute $\bar{\delta}(\partial_\nu \eta^\rho)$ via analogy to $\bar{\delta}\eta^\rho$:

$$\begin{aligned}\bar{\delta}(\partial_\nu\eta^\rho) &= \partial_\nu \eta^{\rho'}(x) - \partial_\nu\eta^\rho(x)\\ &= \partial_\nu\left[\bar{\delta}\eta^\rho(x)\right]\end{aligned}$$

With these two quantities computed, we now look at the integral we had, and we can replace the $\frac{\partial\mathcal{L}}{\partial\eta^\rho}$ using the equations of motion:

$$\int_\Omega dx^\mu \left[\frac{\partial\mathcal{L}}{\partial\eta^\rho}\bar{\delta}\eta^\rho + \frac{\partial\mathcal{L}}{\partial(\partial_\nu\eta^\rho)}\bar{\delta}(\partial_\nu\eta^\rho)\right] = \int_\Omega dx^\mu \left[\frac{\partial}{\partial x^\nu}\left(\frac{\partial\mathcal{L}}{\partial(\partial_\nu\eta^\rho)}\bar{\delta}\eta^\rho\right)\right]$$

Now recombining this term with the disjoint integral term:

$$\int_\Omega dx^\mu \frac{\partial}{\partial x^\nu}\left[\frac{\partial\mathcal{L}}{\partial(\partial_\nu\eta^\rho)}\bar{\delta}\eta^\rho + \mathcal{L}\delta x^\mu\right] = 0$$

Now we recall that we never specified what Ω was, it can be any 4-volume. Thus the integrand must vanish for all volumes, and therefore the quantity in the total derivative must be constant. This is denoted j:

$$j^\nu = \frac{\partial\mathcal{L}}{\partial(\partial_\nu\eta^\rho)}\bar{\delta}\eta^\rho + \mathcal{L}\delta x^\mu \tag{4}$$

This is the Noether current, and is conserved, $\partial_\mu j^\mu = 0$.

Noether Current Examples

The first example we will look at is translations. Consider a Lagrangian density that is invariant under translations. In this case, the value of the field at the physical point does not change, regardless of whether it is a scalar field or a vector field. Thus we have that

$$\eta^{\rho'}\left(x'\right) = \eta^\rho(x)$$

Which gives us that $\delta\eta^\rho(x) = 0$. We can explicitly define our translation:

$$x'^\nu = x^\nu + a^\nu$$

and so $\delta x^\nu = a^\nu$. We now want to find $\bar{\delta}\eta^\rho$:

$$\begin{aligned}\bar{\delta}\eta^\rho &= \eta^{\rho'}(x) - \eta^\rho(x) \\ &= -a^\nu \partial_\nu \eta^\rho\end{aligned}$$

From this, we can write out the Noether current:

$$\begin{aligned}j^\nu &= \frac{\partial \mathcal{L}}{\partial(\partial_\nu \eta^\rho)}\left[-a^\alpha \partial_\alpha \eta^\rho\right] + \mathcal{L} a^\nu \\ &= -a^\alpha \underbrace{\left[\frac{\partial \mathcal{L}}{\partial(\partial_\nu \eta^\rho)}\partial_\alpha \eta^\rho - \mathcal{L}\delta^\nu{}_\alpha\right]}_{T^\nu{}_\alpha}\end{aligned}$$

We see that we recover the energy-momentum tensor as our conserved quantity, spacetime translational invariance leads to the conservation of the energy-momentum tensor.
Let us do another example. Consider the Lagrangian density:

$$\mathcal{L} = \partial_\mu \phi^\dagger \partial^\mu \phi - m^2 \phi^\dagger \phi$$

Now consider the transformation:

$$\begin{aligned}\phi &\to e^{i\alpha}\phi = \phi' \\ \phi^\dagger &\to e^{-i\alpha}\phi^\dagger = \phi'^\dagger\end{aligned}$$

We can first show that this is a symmetry, and then we can find the Noether current. To show that it is a symmetry, we must show form invariance:

$$\begin{aligned}\mathcal{L} &= \partial_\mu\left(e^{i\alpha}\phi'^\dagger\right)\partial^\mu\left(e^{-\alpha}\phi'\right) - m^2\left(e^{i\alpha\phi'^\dagger}e^{-i\alpha}\phi'\right) \\ &= \partial_\mu \phi'^\dagger \partial^\mu \phi' - m^2 \phi'^\dagger \phi'\end{aligned}$$

We see that if we insert the transformed coordinates, we obtain the same functional form, giving us form invariance. Scale invariance in this case is trivially obtained, since the transformation does not depend on the coordinates. Thus this transformation is indeed a symmetry.
Now let us compute the Noether current. We can compute $\bar{\delta}\phi$, which, since we haven't changed the coordinates, is the same as $\delta\phi$:

$$\begin{aligned}\bar{\delta}\phi &= \phi'(x) - \phi(x) \\ &= e^{i\alpha}\phi(x) - \phi(x) \\ &= i\alpha\phi(x)\end{aligned}$$

Where we have Taylor expanded the exponential, since we consider an infinitesimal transformation. We can do the same for $\phi^\dagger$, and we find that

$$\bar{\delta}\phi^\dagger = -i\alpha\phi^\dagger$$

We can now insert these into the definition of the Noether current:

$$\begin{aligned}j^\nu &= \frac{\partial \mathcal{L}}{\partial(\partial_\nu \phi)}\bar{\delta}\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\nu \phi^\dagger)}\bar{\delta}\phi^\dagger \\ &= \left(\partial^\nu \phi^\dagger\right)(i\alpha\phi) + \partial_\nu \phi\left(-i\alpha\phi^\dagger\right) \\ &= -i\alpha\left[\phi^\dagger \partial^\nu \phi - \phi \partial^\nu \phi^\dagger\right]\end{aligned}$$

3. Relativistic Wave Equations

"I found a very suspicious copy of Sakurai in China twelve years ago... Had to hide it in my suitcase so the customs guys wouldn't find it."

Chacko

This discussion comes from two books, Sakurai's Advanced Quantum Mechanics, and Relativistic Quantum Mechanics, by Bjorken and Drell.

Suppose we want to write down a wave equation that obeys special relativity, that is, how do we generalize the Schrödinger equation? We have an energy relation:

$$E = \frac{p^2}{2m} + V(x)$$

Which then turns into a wave equation, E and t become operators:

$$i\hbar \frac{\partial}{\partial t}\phi = -\frac{\hbar^2}{2m}\boldsymbol{\nabla}^2\psi + V\psi$$

This equation is only invariant under Galilean symmetries, not Lorentz transformations. The naive generalization is to use the relativistic energy momentum relation:

$$E = \sqrt{c^2p^2 + m^2c^4}$$

Which turns our wave equation into:

$$i\hbar\frac{\partial\psi}{\partial t} = \sqrt{-\hbar^2c^2\boldsymbol{\nabla}^2 + m^2c^4}\psi$$

How do we deal with this square root? The first idea is to square both sides of our energy momentum relation:

$$E^2 = c^2p^2 + m^2c^4$$

We can then write out the wave equation:

$$-\hbar^2\frac{\partial^2\psi}{\partial t^2} = -\hbar^2c^2\boldsymbol{\nabla}^2\psi + m^2c^4\psi \tag{5}$$

This is known as the relativistic Schrödinger equation[5], but is more commonly known as the Klein-Gordon equation. Writing this equation in relativistic notation:

$$\partial_\alpha\partial^\alpha\psi + \frac{m^2c^2}{\hbar^2}\psi = 0$$

Let us look for plane wave solutions of this equation:

$$\psi(\boldsymbol{x}, t) = e^{i(\boldsymbol{k}\cdot\boldsymbol{x} - \omega t)}$$

[5]Schrödinger actually wrote this equation down before the non-relativistic equation, but he abandoned it due to some of the issues we will see shortly.

$$= \exp\left[\frac{\boldsymbol{p}\cdot\boldsymbol{x} - Et}{\hbar}\right]$$

Where $\boldsymbol{p} = \hbar\boldsymbol{k}$ and $E = \hbar\omega$. If we insert this ansatz into our differential equation:

$$-\frac{\omega^2}{c^2} + k^2 + \left(\frac{mc}{\hbar}\right)^2 = 0$$

From this, we have that

$$E = \pm\sqrt{c^2p^2 + m^2c^4}$$

Immediately, we have a problem. We see that we have negative energy solutions, which implies that we can have arbitrarily negative energies, there is an infinite number of these negative energy states! We ignore this and proceed (as physicists usually do).

In non-relativistic quantum mechanics, we have conservation of probability, does the Klein-Gordon wave equation also have this property? Looking back at the regular quantum mechanical wave equation:

$$-i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\boldsymbol{\nabla}^2\psi + V\psi$$

Taking the complex conjugate of both sides:

$$-i\hbar\frac{\partial\psi^*}{\partial t} = -\frac{\hbar^2}{2m}\boldsymbol{\nabla}^2\psi * + V\psi *$$

If we take the wave equation and multiply both sides by ψ^*, and subtract the product of this conjugated wave equation and ψ (taking the Wronskian), we find that:

$$i\hbar\frac{\partial}{\partial t}\left(\psi\psi^*\right) = -\frac{\hbar^2}{2m}\boldsymbol{\nabla}\cdot\left(\psi^*\boldsymbol{\nabla}\psi - \psi\boldsymbol{\nabla}\psi^*\right)$$

This is a continuity equation (compare this to $\frac{\partial\rho}{\partial t} = -\boldsymbol{\nabla}\cdot\boldsymbol{j}$ from electromagnetism). We can write this in a form that makes it more clear:

$$\frac{\partial\rho}{\partial t} + \boldsymbol{\nabla}\cdot\boldsymbol{S} = 0$$

Where $\rho = \psi^*\psi$ is the conserved probability density. Note that ρ is positive definite, which is necessary if we want to interpret ρ as a probability[6]. From this continuity equation, we have that the probability over all space is a constant in time.

Now let us consider the Klein-Gordon theory. In this case, we can again take the Wronskian, after which we find:

$$\frac{1}{c^2}\frac{\partial}{\partial t}\left(\psi^*\frac{\partial\psi}{\partial t} - \psi\frac{\partial\psi^*}{\partial t}\right) = \boldsymbol{\nabla}\cdot\left(\psi^*\boldsymbol{\nabla}\psi - \psi\boldsymbol{\nabla}\psi^*\right)$$

Where we have dropped factors of $\hbar$ and c. We see that we again find a continuity equation, with the same $\boldsymbol{S}$, but the ρ is not the same:

$$\frac{\partial\tilde{\rho}}{\partial t} + \boldsymbol{\nabla}\cdot\boldsymbol{S} = 0$$

[6]Famously, this was first done by Max Born, for which he received a Nobel Prize.

Here $\tilde{\rho}$ is no longer $\psi^*\psi$:

$$\tilde{\rho} = \frac{i\hbar}{2mc^2}\left(\psi^*\frac{\partial\psi}{\partial t} - \psi\frac{\partial\psi^*}{\partial t}\right)$$

This is real, but is no longer positive definite, we cannot identify this with a probability density. People ignored this and proceeded. What finally killed the Klein-Gordon equation was the calculation of the fine structure of Hydrogen using this theory, which disagreed with experiment, and the theory was *then* abandoned as a wave equation.

What do we do now? The Klein-Gordon equation does not work as a relativistic wave equation. From this, we move to the Dirac equation.

3.1. The Dirac Equation

Dirac looked at the Schrodinger equation and the Klein-Gordon equation, and noted that in the Schrodinger equation, we have only 1 order of time derivatives, while the Klein-Gordon equation is second order in time. So he attempted to write down an equation that was first order in time. To do this, we have to deal with the square root that we ran into earlier:

$$E = \sqrt{c^2p^2 + m^2c^4}$$

To deal with this, Dirac rewrote this as:

$$E = c\boldsymbol{\alpha}\cdot\boldsymbol{p} + mc^2\beta$$

Where α_i and β are matrices. Note that this makes the energy a matrix, and the wavefunction a column vector. To do this, if we square this, we better get the right expression:

$$\begin{aligned}
E^2 &= \left(c\boldsymbol{\alpha}\cdot\boldsymbol{p} + mc^2\beta\right)^2 \\
&= \left(c\left(\alpha_x p_x + \alpha_y p_y + \alpha_z p_z\right) + mc^2\beta\right)^2 \\
&= c^2\left(\alpha_x^2 p_x^2 + \alpha_y^2 p_y^2 + \alpha_z^2 p_z^2 + \beta^2 m^2 c^2\right) \\
&+ c^2\left[\left(\alpha_x\alpha_y + \alpha_y\alpha_x\right)p_x p_y + \left(\alpha_y\alpha_z + \alpha_z\alpha_y\right)p_y p_z + \left(\alpha_z\alpha_x + \alpha_x\alpha_z\right)p_z p_x\right] \\
&+ \left[\left(\alpha_x\beta + \beta\alpha_x\right)p_x mc^2 + \left(\alpha_y\beta + \beta\alpha_y\right)p_y mc^2 + \left(\alpha_z\beta + \beta\alpha_z\right)p_z mc^2\right]
\end{aligned}$$

Now compare this mess to the correct expression:

$$E^2 = c^2p_x^2 + c^2p_y^2 + c^2p_z^2 + m^2c^4$$

To have these match, we have many conditions that must be met. The first is that

$$\alpha_x^2 = \alpha_y^2 = \alpha_z^2 = \mathbb{I}$$

We also have an anticommutation relation:

$$\alpha_i\alpha_j + \alpha_j\alpha_i = 0$$

for $i \neq j$. We also have that

$$\beta\alpha_i + \alpha_i\beta = 0$$

And $\beta^2 = \mathbb{I}$.

We must find 4 matrices $\{\alpha_x, \alpha_y, \alpha_z, \beta\}$ that satisfy these properties. Let us first postpone a discussion of what these matrices are. Suppose we have a set of matrices that satisfy these properties, what does our theory look like? We can write our Hamiltonian:

$$\hat{H} = c\boldsymbol{\alpha} \cdot \hat{\boldsymbol{p}} + mc^2\beta$$

If we now enforce that the Hamiltonian is Hermitian, $\hat{H} = \hat{H}^\dagger$, we have another condition on the matrices:

$$\begin{aligned}\alpha_i &= \alpha_i^\dagger \\ \beta &= \beta^\dagger\end{aligned}$$

We see we need 4 Hermitian, idempotent, and anticommuting matrices. We can then write out our wave equation:

$$i\hbar\frac{\partial\psi}{\partial t} = -i\hbar c\boldsymbol{\alpha} \cdot \boldsymbol{\nabla}\psi + mc^2\beta\psi \tag{6}$$

We see that this is indeed first order in time, so we have a decent chance of getting a conserved positive definite probability density. Let us determine if this is indeed the case. We can take the Hermitian conjugate of our wave equation:

$$-i\hbar\frac{\partial\psi^\dagger}{\partial t} = i\hbar c\boldsymbol{\nabla}\psi^\dagger \cdot \boldsymbol{\alpha} + mc^2\psi^\dagger\beta$$

Multiplying this by ψ, and then multiplying our wave equation by $\psi^\dagger$, and then subtracting the two, we find that

$$\begin{aligned}i\hbar\frac{\partial}{\partial t}\left(\psi^\dagger\psi\right) &= -i\hbar c\left(\psi^\dagger\boldsymbol{\alpha} \cdot \boldsymbol{\nabla}\psi + \boldsymbol{\nabla}\psi^\dagger \cdot \boldsymbol{\alpha}\psi\right) \\ &= -i\hbar c\boldsymbol{\nabla} \cdot \left(\psi^\dagger\boldsymbol{\alpha}\psi\right)\end{aligned}$$

Thus we have our probability current density $\boldsymbol{S} = c\psi^\dagger\boldsymbol{\alpha}\psi$, and our probability density $\rho = \psi^\dagger\psi$.

Up until now, we have assumed that there exists a set of 4 matrices that satisfy these properties. Let us now see if we can in fact find such a set of matrices. If we wanted a set of 3 matrices, we could use the Pauli matrices, since they satisfy these conditions. However, we need 4! In fact, if we were to work in 2+1D, we would be able to use the Pauli matrices, since we would only need a set of 3 matrices. This implies that we might be able to generalize from the Pauli matrices to something that works in 3+1D.

This leads us to the Dirac matrices. We want to find whether or not we can find four 2×2 matrices that are Hermitian, mutually anticommuting, and idempotent. Since the Pauli matrices and the identity span all 2×2 Hermitian matrices, we can actually prove that there is no such fourth matrix. What about 3×3 matrices? We can actually prove that the Dirac matrices must be *even* dimensional, so 3×3 will not work.

Proof. The α_i matrices must be Hermitian. Thus we can diagonalize one of them, via some unitary transformation. Note that $\alpha_i^2 = \mathbb{I}$ must be true before and after the diagonalization. Also note that since α_i is idempotent, it has eigenvalues of ± 1. This argument applies to all of the 4 matrices, they all have eigenvalues ± 1. Now $\alpha_i\beta + \beta\alpha_i = 0$ (one of our conditions), which implies that $\beta^{-1}\alpha_i\beta = -\alpha_i$. Now taking the trace, and applying the cyclic property of the trace, we find that

$$\mathrm{Tr}\left[\alpha_i\right] = 0$$

We have shown that the eigenvalues are ± 1, and the trace is zero, so there must be an equal number of $+1$ eigenvalues and -1 eigenvalues, and therefore the matrix must be even dimensional. $\square$

From this, we know that we need the Dirac matrices to be 4×4 at the very least. There is no unique set of matrices, but here is one set:

$$\alpha_i = \begin{bmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{bmatrix} \qquad \beta = \begin{bmatrix} 0 & \mathbb{I}_2 \\ \mathbb{I}_2 & 0 \end{bmatrix}$$

Where $\mathbb{I}_2$ is the 2×2 identity matrix, and σ_i is a 2×2 Pauli matrix. Since this set is not unique, different books use different choices. Some choices work better for the relativistic limit, and others in the non-relativistic limit. Peskin for example, chooses a set that is better for relativistic computations. We can transition between different sets of these matrices by a similarity transformation:

$$\alpha_i' = S\alpha_i S^{-1} \qquad \beta' = S\beta S^{-1}$$

Where S is a unitary matrix, because the αs and β are Hermitian.

We have now found an explicit set of matrices that satisfy our conditions, so let us look for plane wave solutions of the Dirac equation. Rewriting our equation:

$$\left(-i\hbar c\boldsymbol{\alpha} \cdot \boldsymbol{\nabla} + mc^2\beta\right)\psi = i\hbar\frac{\partial\psi}{\partial t}$$

This is not the usual way we write the Dirac equation. This equation (as we will see) is invariant under Lorentz transformations, but this is not very clearly seen. To make this more clear, we multiply both sides by β, and divide by c:

$$\left(-i\hbar\beta\boldsymbol{\alpha} \cdot \boldsymbol{\nabla} + mc\right)\psi = i\hbar\frac{\beta}{c}\frac{\partial\psi}{\partial t}$$

This equation has the form

$$i\hbar\gamma^\mu\partial_\mu\psi - mc\psi = 0$$

Where we define[7] $\gamma^0 = \beta$, and $\gamma^i = \beta\alpha^i$. From this, we can write the Dirac equation using natural units, where $\hbar = c = 1$:

$$\left(i\gamma^\mu\partial_\mu - m\right)\psi = 0 \tag{7}$$

In natural units, everything is now described in terms of energy (eV). Note that we will also be using Heaviside-Lorentz units[8], in which Coulomb's Law is written as

$$F_C = \frac{q_1q_2}{4\pi r^2}$$

[7] Note that we define $x^0 = ct$

[8] Gaussian units are easier in the case where we start with Coulomb's Law itself, but Heaviside-Lorentz is nicer when starting from the Lagrangian, which is what we will be doing. SI units are never useful.

Heaviside-Lorentz is essentially SI units, except we don't have the ε_0.

We can rewrite the α and β commuatation relations in terms of the γ matrices:

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$$

Now let us return to looking for plane wave solutions. We want solutions of the form:

$$\psi(x,t) = w(\boldsymbol{p})\, e^{i(\boldsymbol{p}\cdot\boldsymbol{x}-Et)}$$

For the sake of simplicity, we will choose $\boldsymbol{p} = p\hat{z}$. w is a four component matrix, but it is convenient to write this as a two-component object (since the γ matrices can be separated into 2×2 block matrices):

$$w = \begin{pmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \end{pmatrix} = \begin{pmatrix} w_A \\ w_B \end{pmatrix}$$

If we insert our plane wave solution into the Dirac equation, since the derivative $\partial_\mu\psi$ is zero along the x and y direction, we have

$$\left(\gamma^0 - \gamma^3 p - m\right) w = 0$$

Now inserting the γ matrices in block form:

$$\left[E\begin{pmatrix} 0 & \mathbb{I}_2 \\ \mathbb{I}_2 & 0 \end{pmatrix} - p\begin{pmatrix} 0 & \sigma_3 \\ -\sigma_3 & 0 \end{pmatrix} - m\right]\begin{pmatrix} w_A \\ w_B \end{pmatrix} = 0$$

This gives us two equations:

$$Ew_B - p\sigma_3 w_B - mw_A = 0 \tag{8}$$
$$Ew_A + p\sigma_3 w_A - mw_B = 0 \tag{9}$$

Now we can rewrite Eqn 8:

$$w_A = \frac{(E - p\sigma_3)}{m} w_B$$

Inserting this into Eqn 9:

$$\frac{(E + p\sigma_3)(E - p\sigma_3)}{m} w_B - mw_B = 0$$

Using the fact that $\sigma_3^2 = \mathbb{I}$, we are left with

$$\left(E^2 - p^2 - m^2\right) w_B = 0$$

From which we have

$$E = \pm\sqrt{p^2 + m^2}$$

Let us make a few comments about what we have seen. The first is that w_A and w_B are not independent, if we know one of them, we can find the other. The other thing to note is that unless the final condition on the energy (the energy dispersion relation) is satisfied, we have a trivial solution. Perhaps the most glaring issue is that we have a nontrivial negative energy solution, we were not able to avoid the issue that we first saw in the Klein-Gordon theory.

Note that w_B can take two independent values:

$$w_B = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

and since E can be either positive or negative, we have a total of four independent plane wave solutions. Listing them out:

$$\begin{aligned}
w_{\uparrow +}: &\quad w_B = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad w_A = \begin{pmatrix} \frac{E_+ - p}{m} \\ 0 \end{pmatrix} \\
w_{\downarrow +}: &\quad w_B = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \qquad w_A = \begin{pmatrix} 0 \\ \frac{E_+ + p}{m} \end{pmatrix} \\
w_{\uparrow -}: &\quad w_B = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad w_A = \begin{pmatrix} \frac{E_- - p}{m} \\ 0 \end{pmatrix} \\
w_{\downarrow -}: &\quad w_B = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \qquad w_A = \begin{pmatrix} 0 \\ \frac{E_- + p}{m} \end{pmatrix}
\end{aligned}$$

We label these independent solutions (for reasons that will become clear later) with spin indices, as well as the sign of the energy. These solutions are orthogonal:

$$w_{\uparrow +} \cdot w_{\uparrow -} = w_{\uparrow +} \cdot w_{\downarrow +} = \cdots = 0$$

These are generally not written in this way, people have come up with notation that allows for easier computation. We can alter the normalizations of these solutions:

$$\begin{aligned}
u_1(p): &\quad w_B = \begin{pmatrix} \sqrt{E_+ + p} \\ 0 \end{pmatrix} \qquad w_A = \begin{pmatrix} \sqrt{E_+ - p} \\ 0 \end{pmatrix} \\
u_2(p): &\quad w_B = \begin{pmatrix} 0 \\ \sqrt{E_+ - p} \end{pmatrix} \qquad w_A = \begin{pmatrix} 0 \\ \sqrt{E_+ + p} \end{pmatrix}
\end{aligned}$$

These are the two positive energy solutions ($w_{\uparrow +}$ and $w_{\downarrow +}$). For the negative energy solutions, we use a different convention, we look for solutions of the Dirac equation of the form

$$\psi(x) = \tilde{w}(p) e^{i(Et - p\hat{z})}$$

with $E > 0$. Essentially, the negative energy solutions propagating in a certain direction can be written in the same form as *positive* energy solutions propagating in the *opposite* direction.

We can again break $\tilde{w}$ into a two-component object:

$$\tilde{w} = \begin{pmatrix} \tilde{w}_A \\ \tilde{w}_B \end{pmatrix}$$

We can then rewrite our negative energy solutions:

$$w_{\uparrow -}: \quad \tilde{w}_B = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad \tilde{w}_A = \begin{pmatrix} -\frac{E_+ + p}{m} \\ 0 \end{pmatrix}$$

$$w_{\downarrow -}: \quad \tilde{w}_B = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \qquad \tilde{w}_A = \begin{pmatrix} 0 \\ -\frac{E_+ - p}{m} \end{pmatrix}$$

We can again change the normalization, and write down the negative energy solutions:

$$v_1(p): \quad \tilde{w}_B = \begin{pmatrix} \sqrt{E_+ + p} \\ 0 \end{pmatrix} \qquad \tilde{w}_A = \begin{pmatrix} -\sqrt{E_+ - p} \\ 0 \end{pmatrix}$$
$$v_2(p): \quad \tilde{w}_B = \begin{pmatrix} 0 \\ \sqrt{E_+ - p} \end{pmatrix} \qquad \tilde{w}_A = \begin{pmatrix} 0 \\ -\sqrt{E_+ + p} \end{pmatrix}$$

With this new notation, we have that $u_i(p)$ represent positive energy solutions moving along $\boldsymbol{p}$, and $v_i(p)$ are negative energy solutions moving opposite $\boldsymbol{p}$.

There are certain standard properties of these solutions that are useful when doing calculations. We can begin with another definition:

$$\overline{w} = w^\dagger \gamma^0$$

For any four component object w. From this, we have that

$$\overline{u}^r(p)\, u^s(p) = 2m\delta^{rs}$$
$$u^{r\dagger}(p)\, u^s(p) = 2E_p\delta^{rs}$$

And for the vs:

$$\overline{v}^r(p)\, v^s(p) = -2m\delta^{rs}$$
$$v^{p\dagger}(p)\, v^s(p) = 2E_p\delta^{rs}$$

We also have that

$$\overline{v}^r(p)\, u^s(p) = 0$$
$$\overline{u}^r(p)\, v^s(p) = 0$$

However,

$$v^{r\dagger}(p)\, u^s(p) \neq 0$$
$$u^{r\dagger}(p)\, v^s(p) \neq 0$$

The above property arises because the us and vs are propagating in opposite directions, so in fact:

$$v^{r\dagger}(p)\, u^s(-p) = 0$$
$$u^{r\dagger}(p)\, v^s(-p) = 0$$

We can now also write down a summation property:

$$\sum_{s=1,2} u^s(p)\, \overline{u}^s(p) = \not{p} - m$$

Where $\not{p} = p_\mu \gamma^\mu$[9]. Note that this is not an inner product, this is an outer product (column vector times a row vector), and thus we are left with a matrix, rather than a number.

Let us again write down $u^1(p)$:

$$u_1(p) : \begin{pmatrix} \sqrt{E-p} \\ 0 \\ \sqrt{E+p} \\ 0 \end{pmatrix}$$

Where $E = +\sqrt{p^2+m^2}$. What is the right way to think about this? Let us draw an analogy to non-relativistic quantum mechanics. We can decompose wavefunctions into a spatial component and a spin part, they nicely factorize. Looking at plane waves solutions of a free particle, we have solutions of the form:

$$e^{i(\boldsymbol{p}\cdot\boldsymbol{x}-Et)} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Where this is a solution for a spin up free particle. We can make a linear combination of these plane waves to make a general wavefunction:

$$\underbrace{\int d^3p\, f(p)\, e^{i(\boldsymbol{p}\cdot\boldsymbol{x}-Et)}}_{\psi(\boldsymbol{x},t)} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Suppose we wanted to do the same thing for the plane wave solutions of the Dirac equation:

$$\int d^3p\, f(p)\, u_1(p)\, e^{i(\boldsymbol{p}\cdot\boldsymbol{x}-Et)}$$

However, the components of $u_1(p)$ are not the same, they have different dependences on p, and thus we would be left with something of the form:

$$\begin{pmatrix} \phi_1(x,t) \\ 0 \\ \phi_2(x,t) \\ 0 \end{pmatrix}$$

The spin/spatial factorization is no longer present in the relativistic theory. In fact, if we take the non-relativistic limit, where $p \to 0$, we see that we recover the factorization.

Let us now discuss the negative energy states. What stops an electron in a finite positive energy state from falling into a negative energy state? According to Dirac, the answer is the Pauli Exclusion Principle. All the negative energy states are normally fully occupied by electrons, which is known as the *Dirac sea*. If this is the case, an electron in a positive energy state, by the Pauli Exclusion Principle, cannot occupy a negative energy state, and thus the negative energy states have no observable consequences. This is a strange concept, but if we think about condensed matter, we often think about the Fermi surface. If the Fermi surface is full, and we insert another electron, the electron cannot interact with the electrons at the Fermi surface, since it is fully occupied. Only if we can introduce enough energy to knock an electron out of the Fermi surface can we see interactions.

[9] This is known as the Feynman slash notation.

Dirac of course did not know about all of this. Instead, his depiction was an infinite band of negative energy states and an infinite band of positive energy states, separated by a band gap of $2mc^2$. The idea was the negative energy band was completely filled, and observable states lived in the positive energy band. The only way to drive interactions between an external electron and the negative band would be to transfer enough energy to knock a negative energy electron into a positive energy state. Only high momentum transfer processes that can cross the “band gap” can interact with the Dirac sea.

Suppose we do knock a negative energy electron out of the sea. In this case, the sea has a net positive charge (since we have a “hole” with a missing electron), and so the sea would appear as a positively charged, positive energy particle. Dirac hated this, but shortly after he developed this idea, the positron was observed, which is exactly the hole state, a positive charge electron.

To summarize, the implications of Dirac's theory are the aforementioned existence of positrons, and the possibility of pair creation, high energy processes can create electron-positron pairs.

3.2. Spin in the Dirac Theory

Let us now return to the plane wave solutions and the stated but unproven connection to spin. We expect that the angular momentum of an isolated system is conserved. Let us consider the angular momentum operator $\boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{p}$ for a single particle, and see if it is conserved. In the Heisenberg picture, we can look at the evolution of the orbital angular momentum in a particular direction:

$$\begin{aligned} i\hbar \frac{dL_z}{dt} &= [L_z, H] \\ &= \left[(xp_y - yp_x), \left(c\boldsymbol{\alpha} \cdot \boldsymbol{p} + mc^2\beta\right)\right] \\ &= [x, p_x]\, c\alpha_x p_y - [y, p_y]\, c\alpha_y p_x \\ &= i\hbar c\left(\alpha_x p_y - \alpha_y p_x\right) \end{aligned}$$

This is not zero! The orbital angular momentum evolves over time, which, if we assume that total angular momentum must be conserved, implies that there is another source of angular momentum. This is the spin angular momentum. The combination of the spin and orbital angular momenta will be conserved, but not the two individually.

Consider the operator $\frac{1}{2}\hbar\Sigma_z$, where Σ_z is defined as

$$\Sigma_z = -i\alpha_x\alpha_y$$

We will show that

$$S_z = \frac{1}{2}\hbar\Sigma_z$$

is the spin angular momentum operator in the z direction, and that the sume of the orbital and spin angular momentum will be conserved:

$$\frac{d}{dt}\left(L_z + S_z\right) = 0$$

And via symmetry, we can define the same type of operator for the x and y directions.

Let us now prove this. We can look at the Heisenberg evolution of our defined operator:

$$i\hbar \frac{d}{dt}\left(\frac{1}{2}\hbar\Sigma_z\right) = \frac{1}{2}\hbar\left[\Sigma_z, H\right]$$

$$
\begin{aligned}
&= -\frac{1}{2} i\hbar \left[\alpha_x \alpha_y, c\boldsymbol{\alpha} \cdot \boldsymbol{p} + mc^2 \beta\right] \\
&= -\frac{1}{2} i\hbar \left[\left(c\alpha_x \alpha_y\right)\left[\alpha_y, \alpha_x\right] + \left[\alpha_x, \alpha_y\right] c\alpha_y p_y\right] \\
&= i\hbar c \left(\alpha_y p_x - \alpha_x p_y\right)
\end{aligned}
$$

Where we have used the fact that $\alpha_x \alpha_y \alpha_z = \alpha_z \alpha_x \alpha_y$, and $\alpha_x \alpha_y \beta = \beta \alpha_x \alpha_y$. Note that our result exactly cancels out the result for the time evolution of L_z. Thus, this choice of S_z, the time derivative of the total angular momentum is zero, and thus this is indeed our spin component. This is historically how the existence of spin-1/2 particles was determined, while we take it for granted that electrons have spin.

More generally, we can define the spin operator:

$$
\boldsymbol{S} = \frac{1}{2}\hbar \boldsymbol{\Sigma}
$$

Where Σ has a cylic property:

$$
\begin{aligned}
\Sigma_x &= -i\alpha_y \alpha_z \\
\Sigma_y &= -i\alpha_z \alpha_x \\
\Sigma_z &= -i\alpha_x \alpha_y
\end{aligned}
$$

We can look at the square of the spin operator, noting that $\Sigma_i^2 = 1$:

$$
\begin{aligned}
\boldsymbol{S}^2 &= \frac{1}{4}\hbar^2 \left(\Sigma_x^2 + \Sigma_y^2 + \Sigma_z^2\right) \\
&= \frac{3}{4}\hbar^2
\end{aligned}
$$

We can explicitly write out Σ_z:

$$
\begin{aligned}
\Sigma_z &= i\alpha_x \alpha_y \\
&= \begin{pmatrix} \sigma_3 & 0 \\ 0 & \sigma_3 \end{pmatrix}
\end{aligned}
$$

If we apply our spin operator to what we had labelled the spin up and down solutions:

$$
\begin{aligned}
S_z w_{\pm\uparrow} &= \frac{1}{2}\hbar w_{\pm\uparrow} \\
S_z w_{\pm\downarrow} &= -\frac{1}{2}\hbar w_{\pm\downarrow}
\end{aligned}
$$

Which validates our choice of labelling, these are indeed spin up and down.

3.3. Lorentz Covariance of the Dirac Equation

Let us now discuss the Lorentz covariance of the Dirac equation. Suppose we write down Newton's Second Law:

$$
\boldsymbol{F} = m\boldsymbol{a}
$$

What does it mean for us to say that $\boldsymbol{a}$ and $\boldsymbol{F}$ are vectors? This is a statement about how they transform under rotations. When we say something is a vector, we say that it transforms like the coordinates do under rotations. We would say that Newton's Second Law is "rotationally covariant". What does this mean? Suppose we have $\boldsymbol{F}$ and $\boldsymbol{a}$ in a particular coordinate system. If we do a coordinate transformation, the equation will still remain true, while the components of $\boldsymbol{F}$ and $\boldsymbol{a}$ are not invariant, both will transform covariantly, the form of the equation will remain invariant.

Now let us do the same for Lorentz transformations of the Dirac equation. We have the Dirac equation:

$$(i\gamma^\mu \partial_\mu - m)\,\psi(x) = 0$$

If someone is in a Lorentz transformed frame, $\psi(x)$ will not be the same as in your frame. However, the new $\psi(x)$ in the boosted frame will also satisfy the Dirac equation, the Dirac equation behaves covariantly under rotations and boosts.

Let us now prove this Lorentz covariance. Suppose we have a Lorentz transformation:

$$x^\mu \to x'^\mu = \Lambda^\mu{}_\nu x^\nu$$

We want to show that this leads to $\psi'(x')$ which satisfies:

$$(i\gamma^\mu \partial_\mu - m)\,\psi'\left(x'\right) = 0$$

We can look for a transformation S, which depends on Λ, that transforms between our old and new field:

$$\begin{aligned}\psi'\left(x'\right) &= S(\Lambda)\,\psi(x) \\ &= S(\Lambda)\,\psi\left(\Lambda^{-1}x'\right)\end{aligned}$$

Inverting our original equation, we have that

$$\psi(x) = S^{-1}(\Lambda)\,\psi'\left(x'\right)$$

Similarly, had we done the opposite Lorentz transform Λ^{-1}, we would have that:

$$\psi(x) = S\left(\Lambda^{-1}\right)\psi'\left(x'\right)$$

These two give us the fact that

$$S^{-1}(\Lambda) = S\left(\Lambda^{-1}\right)$$

Then, the Dirac equation can be written as:

$$(i\gamma^\mu \partial_\mu - m)\overbrace{S^{-1}(\Lambda)\,\psi'\left(x'\right)}^{\psi(x)} = 0$$

Now using the chain rule, we can rewrite our ∂_μ:

$$\begin{aligned}\partial_\mu &= \frac{\partial x'^\nu}{\partial x^\mu}\frac{\partial}{\partial x'^\nu} \\ &= \Lambda^\nu{}_\mu \partial'_\nu\end{aligned}$$

Thus the Dirac equation becomes:

$$\left(i\gamma^\mu \Lambda^\nu{}_\mu \partial'_\nu - m\right)S^{-1}(\Lambda)\,\psi'\left(x'\right) = 0$$

Left multiplying by $S(\Lambda)$, we are left with:

$$\left(iS(\Lambda)\gamma^\mu S^{-1}(\Lambda)\Lambda^\nu{}_\mu\partial'_\nu - m\right)\psi'(x') = 0$$

From this, we obtain Lorentz covariance if the following condition on $S(\Lambda)$ is met:

$$S(\Lambda)\gamma^\mu S^{-1}(\Lambda)\Lambda^\nu{}_\mu = \gamma^\nu$$

How can we find such a $S(\Lambda)$? In this case, it is simpler to look at the infinitesimal case of a transformation. Consider an infinitesimal Lorentz transformation:

$$\Lambda^\nu{}_\mu = \delta^\nu{}_\mu + \Delta w^\nu{}_\mu$$

Which has some associated $S(\Lambda)$, which has some unknown matrix $\sigma_{\mu\nu}$:

$$S = \mathbb{I} - \frac{i}{4}\sigma_{\mu\nu}\Delta w^{\mu\nu}$$

Note that the factor of $\frac{i}{4}$ is placed there with malice aforethought, we will see that there is a convention for $\sigma_{\mu\nu}$, which factors out the $\frac{i}{4}$.

Now let us take our two equations and substitute them into the expression for the requirement on S, and drop all terms of order higher than Δw:

$$\begin{aligned}\left(1 - \frac{i}{4}\sigma_{\mu\nu}\Delta w^{\mu\nu}\right)\gamma^\alpha\left(1 + \frac{i}{4}\sigma_{\lambda\sigma}\Delta w^{\lambda\sigma}\right)\left(\delta^\beta_\alpha + \Delta w^\beta{}_\alpha\right) &= \gamma^\beta \\ \left(-\frac{i}{4}\sigma_{\mu\nu}\Delta w^{\mu\nu}\right)\gamma^\beta + \gamma^\beta\left(\frac{i}{4}\sigma_{\mu\nu}\Delta w^{\mu\nu}\right) &= -\gamma_\alpha\Delta w^{\beta\alpha} \\ \left(-\frac{i}{4}\sigma_{\mu\nu}\Delta w^{\mu\nu}\right)\gamma^\beta + \gamma^\beta\left(\frac{i}{4}\sigma_{\mu\nu}\Delta w^{\mu\nu}\right) &= \frac{1}{2}\left[-\gamma_\mu\Delta w^{\nu\mu}\delta^\beta_\nu - \gamma_\nu\Delta w^{\mu\nu}\delta^\beta_\mu\right]\end{aligned}$$

Now using the fact that $\Delta w^{\nu\mu} = -\Delta w^{\mu\nu}$, and dividing out by $\Delta w^{\mu\nu}$ on both sides, and rearranging some terms, we are left with:

$$\left[\gamma^\beta, \sigma_{\mu\nu}\right] = -2i\left[\gamma_\mu\delta^\beta_\nu - \gamma_\nu\delta^\beta_\mu\right]$$

Thus we need a $\sigma_{\mu\nu}$ that satisfies this condition, which is independent of the Lorentz transformation that we use.

We claim that the solution is given by:

$$\sigma_{\mu\nu} = \frac{i}{2}\left[\gamma_\mu, \gamma_\nu\right]$$

Let us now verify that this is true. To do this, we need to compute this commutator:

$$\left[\gamma^\beta, \gamma_\mu\gamma_\nu - \gamma_\nu\gamma_\mu\right]$$

Now using a useful property of commutators:

$$[C, AB] = \{A, C\}B - A\{B, C\}$$

Using this, we can compute both parts of the commutator:

$$\begin{aligned}\left[\gamma^\beta, \gamma_\nu\gamma_\mu\right] &= 2\delta^\beta_\mu\gamma_\nu - 2\gamma_\mu\delta^\beta_\nu \\ \left[\gamma^\beta, \gamma_\nu\gamma_\mu\right] &= 2\delta^\beta_\nu\gamma_\mu + 2\gamma_\nu\delta^\beta_\mu\end{aligned}$$

From which we have:

$$\left[\gamma^\beta, \gamma_\mu\gamma_\nu - \gamma_\nu\gamma_\mu\right] = 4\gamma_\nu\delta^\beta_\mu - 4\gamma_\mu\delta^\beta_\nu$$

If we now multiply this by $\frac{i}{2}$, we recover the expected result:

$$\left[\gamma^\beta, \sigma_{\mu\nu}\right] = -2i\left[\gamma_\mu\delta^\beta_\nu - \gamma_\nu\delta^\beta_\mu\right]$$

Thus we have shown that this choice of $\sigma_{\mu\nu}$ works for infinitesimal Lorentz transformations.

Now let us extend this to finite transformations. To do this, we want to build up a finite transformation out of infinitesimal ones. Suppose Δw represents the strength of the transformation, and $G^\nu{}_\mu$ tells us the direction of the transformation:

$$\Delta w^\nu{}_\mu = \Delta w G^\nu{}_\mu$$

For example, for a Lorentz transformation along z:

$$G^\nu{}_\mu = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}$$

Now let us consider G^2:

$$G^2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

And let us also note that $G^3 = G$[10]. We can obtain a finite transformation via the limit of a series of the infinitesimal transformations:

$$\begin{aligned}x'^\nu &= \lim_{N\to\infty}\left[(\mathbb{I} + \Delta w G)\right]^N x^\mu \\ &= \left[e^{N\Delta w G}\right]^\nu{}_\mu x^\mu \\ &= \left[e^{wG}\right]^\nu{}_\mu x^\mu\end{aligned}$$

Where $N\Delta w = w$. Now, rewriting the exponential in terms of hyperbolic sines and cosines:

$$e^{wG} = \cosh(wG) + \sinh(wG)$$

Due to the properties of G, we can show that

$$\cosh(wG) = 1 - G^2 + G^2\cosh(w)$$

[10]Note the similarity to $SU(3)$ rotations!

$$\sinh(wG) = G\sinh(w)$$

To do this, we can series expand the two of them:

$$\cosh(wG) = 1 + \frac{1}{2}w^2G^2 + \frac{1}{4!}w^4G^4 + \dots$$

We can then note that $G^4 = G^2$, since $G^2 = G$, and we are left with:

$$\begin{aligned}\cosh(wG) &= 1 + G^2\left[\frac{1}{2}w^2 + \frac{1}{4!}w^4 + \dots\right] \\ &= 1 + G^2[\cosh w - 1] \\ &= 1 - G^2 + G^2\cosh(w)\end{aligned}$$

We can use exactly the same logic for the sinh case.

Using these two relations, we can rewrite the exponential:

$$e^{wG} = \begin{pmatrix} \cosh(w) & 0 & 0 & -\sinh(w) \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\sinh(w) & 0 & 0 & \cosh(w) \end{pmatrix}$$

This is similar to the rotation matrices, except hyperbolic sines and cosines. This is the structure for a finite rotation in coordinate space. Now looking at the field transformation:

$$\begin{aligned}\psi'(x') &= S\psi(x) \\ &= \lim_{N\to\infty}\left(1 - \frac{i}{4}\Delta w\sigma_{\mu\nu}G^{\mu\nu}\right)^N\psi(x) \\ &= \exp\left[-\frac{i}{4}(N\Delta w)\sigma_{\mu\nu}G^{\mu\nu}\right]\psi(x) \\ &= \exp\left[-\frac{i}{4}\omega\sigma_{\mu\nu}G^{\mu\nu}\right]\psi(x)\end{aligned}$$

This expression is known as $S(\Lambda)$:

$$S(\Lambda) = \exp\left[-\frac{i}{4}\omega\sigma_{\mu\nu}G^{\mu\nu}\right]$$

This dictates how spinor fields transform:

$$\psi'(x') = S(\Lambda)\psi(x)$$

3.4. Bilinears of the Dirac Equation

Before we discuss bilinear covariance, let us list some of the important properties of the γ matrices, in the basis that we are working in.

We have the defining property:

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$$

As well as how we define the dagger:

$$\left(\gamma^0\right)^\dagger = \gamma^0$$
$$\left(\gamma^i\right)^\dagger = -\gamma^i$$

And we can also write this as:

$$(\gamma^\mu)^\dagger = \gamma^0\gamma^\mu\gamma^0$$

Now let us consider $\overline{\psi} = \psi^\dagger\gamma^0$. We saw how ψ transforms:

$$\psi \to \exp\left[-\frac{i}{4}w\sigma_{\mu\nu}G^{\mu\nu}\right]\psi$$

We can see how $\psi^\dagger$ transforms:

$$\psi^\dagger \to \psi^\dagger \exp\left[\frac{i}{4}w\sigma^\dagger_{\mu\nu}G^{\mu\nu}\right]$$

Using the last property of the gamma matrices we wrote down, we can see that

$$\sigma^\dagger_{\mu\nu} = \gamma^0\sigma_{\mu\nu}\gamma^0$$

After which we can find how $\overline{\psi}$ transforms:

$$\overline{\psi} \to \overline{\psi}\exp\left[\frac{i}{4}wG^{\mu\nu}\sigma_{\mu\nu}\right]$$

We see that ψ and $\overline{\psi}$ transform in opposite ways, one picks up a negative exponential, and the other picks up a positive exponential. From this, we see that

$$\overline{\psi}\psi$$

is Lorentz invariant.

Let us now define[11] a matrix γ^5:

$$\begin{aligned}\gamma^5 &= i\gamma^0\gamma^1\gamma^2\gamma^3 \\ &= -\frac{i}{4!}\varepsilon^{\mu\nu\lambda\phi}\gamma_\mu\gamma_\nu\gamma_\lambda\gamma_\phi\end{aligned}$$

We can list some properties of the γ^5 matrix:

$$\left(\gamma^5\right)^\dagger = \gamma^5$$
$$\left(\gamma^5\right)^2 = \mathbb{I}$$
$$\{\gamma^5, \gamma^\mu\} = 0$$

In our choice of basis, we have the nice property that γ^5 is diagonal:

$$\gamma^5 = \begin{pmatrix} -\mathbb{I} & 0 \\ 0 & \mathbb{I} \end{pmatrix}$$

[11] Note that the prefactor for γ^5 is very convention dependent.

Suppose that instead of 3+1D, we were working in 1+1D. In this case, we could just use two of the Pauli matrices as the gamma matrices, in which case γ^5 would be the third Pauli matrix. If instead we were working in 2+1D, we would have 3 gamma matrices, the Paulis, and there would not be a γ^5. This has to do with the structure of the Lorentz group, and is why fermions change drastically as we change the dimensionality of our theory.

Let us now list the bilinear covariants of the Dirac equation.

1. $\overline{\psi}\psi$ is a scalar, and there is 1 of these.
2. $\overline{\psi}\gamma^\mu\psi$ is a vector, and there are 4 of these.
3. $\overline{\psi}\sigma^{\mu\nu}\psi$ is a tensor, and there are 6 of these.
4. $\overline{\psi}\gamma^\mu\gamma^5\psi$ is a pseudovector[12], and there are 4 of these.
5. $\overline{\psi}\gamma^5\psi$ is a pseudoscalar, and there is 1 of these.

Thus we have a total of 16 bilinear covariants, combinations of $\overline{\psi}$ and ψ that transform in specific ways under Lorentz transformations. We can also write down a set of basis matrices for all 4×4 matrices:

$$\{\mathbb{I}, \gamma^\mu, \sigma^{\mu\nu}, \gamma^\mu\gamma^5, \gamma^5\}$$

This is useful, because we can take a look at something very ugly:

$$\overline{\psi}\gamma^\mu\gamma^\nu\gamma^\lambda\gamma^\sigma\gamma^\delta\psi$$

This will reduce to some combination of our irreducible representations (irreps).

[12] pseudo- indicates that it picks up a sign under parity transformations.

Part II.
Quantum Field Theory

4. Schrodinger Field Theory

"Textbooks only ever discuss what worked, not the hundreds of things that didn't work."

Chacko

We shall begin our discussion of quantum field theory via second quantization of the Schrödinger equation.

In a first quantum mechanics course, we take particle motion, quantize, and find wave motion. Suppose instead, we wanted to go the other way, we start with wave motion, quantize, and we will see that we obtain particle motion.

Consider the Lagrangian density:

$$\mathcal{L} = i\hbar\psi^\dagger\dot{\psi} - \frac{\hbar^2}{2m}\boldsymbol{\nabla}\psi^\dagger \cdot \boldsymbol{\nabla}\psi - V(\boldsymbol{r}, t)\,\psi^\dagger\psi \tag{10}$$

Where ψ is a complex scalar field. Let treat this the way we would treat a classical Lagrangian density, and find the equation of motion. If we do this, we find equations for both ψ and $\psi^\dagger$ (note that so far, we are treating ψ and $\psi^\dagger$ as separate fields)

$$\begin{aligned} -i\hbar\dot{\psi}^\dagger &= -\frac{\hbar^2}{2m}\boldsymbol{\nabla}^2\psi^\dagger + V(\boldsymbol{r}, t)\,\psi^\dagger \\ i\hbar\dot{\psi} &= -\frac{\hbar^2}{2m}\boldsymbol{\nabla}^2\psi + V(\boldsymbol{r}, t)\,\psi \end{aligned}$$

This looks exactly like the Schrodinger equation (which is exactly why we chose this Lagrangian density to start with). However, ψ is no longer a wavefunction, it is a field.

The canonical momentum conjugate to ψ can be found to be:

$$\begin{aligned} \pi &= \frac{\partial\mathcal{L}}{\partial\dot{\psi}} \\ &= i\hbar\psi^\dagger \end{aligned}$$

The way we interpret this is that $\psi^\dagger$ is not an independent field, ψ is the field, and is the fundamental object. The reason that this is the case is that our Lagrangian density is first order in time derivatives, and only acting on ψ, there are no $\dot{\psi}^\dagger$ terms in $\mathcal{L}$ [13].

In the Hamiltonian formalism, ψ and π are canonically conjugate variables, and $\psi^\dagger$ is completely determined by π. Essentially, we started with the assumption that ψ and $\psi^\dagger$ were independent fields, and we have shown that actually, $\psi^\dagger$ is completely dependent on ψ.

[13] See the Klein-Gordon theory, in that case, π contains derivatives of ψ.

We can write out the Hamiltonian density:

$$\begin{aligned}\mathcal{H} &= \pi\dot{\psi} - \mathcal{L} \\ &= -\frac{i\hbar}{2m}\boldsymbol{\nabla}\pi\cdot\boldsymbol{\nabla}\psi - \frac{i}{\hbar}V\left(\boldsymbol{r},t\right)\pi\psi\end{aligned}$$

Expressing this in terms of ψ and $\psi^\dagger$:

$$\mathcal{H} = \frac{\hbar^2}{2m}\boldsymbol{\nabla}\psi^\dagger\cdot\boldsymbol{\nabla}\psi + V\left(\boldsymbol{r},t\right)\psi^\dagger\psi$$

So far everything has been classical. How do we now quantize this system? To do this, let us recall the elastic rod. We had a collection of masses and springs, and in the discrete case, we wrote down a Lagrangian:

$$L = \frac{1}{2}\sum_i m_i\dot{\eta}_i^2 - \frac{1}{2}k\sum_i\left(\eta_{i+1}-\eta_i\right)^2$$

How would we quantize this classical system? We would introduce a quantization condition between position and conjugate momentum:

$$\left[\eta_i, m\dot{\eta}_i\right] = i\hbar\delta_{ij}$$

And maintain commutation relations between each set of variables:

$$\left[\eta_i,\eta_j\right] = \left[\dot{\eta}_i,\dot{\eta}_j\right] = 0$$

If we did this, we would find that the quantized excitations in the system are phonons. Suppose we now take the continuum limit of this system. To do this, we first rewrite the discrete case using the lattice spacing a:

$$\left[\eta_i,\frac{m}{a}\dot{\eta}_j\right] = i\hbar\frac{\delta_{ij}}{a}$$

Taking the continuum limit, we have

$$\left[\eta\left(x\right),\pi\left(x'\right)\right] = i\hbar\delta\left(x-x'\right)$$

Where on the right side, the $1/a$ leads to the Kronecker delta blowing up to infinity, and so we recover a Dirac delta. This is the quantization condition for the classical field in the case of the elastic rod.

The generalization to arbitrary numbers of fields and dimensions is:

$$\begin{aligned}\left[\eta^\rho\left(x\right),\pi^\rho\left(x'\right)\right] &= i\hbar\delta^{D-1}\left(x-x'\right) \\ \left[\eta^\alpha\left(x\right),\eta^\beta\left(x'\right)\right] &= 0 \\ \left[\pi^\alpha\left(x\right),\pi^\beta\left(x'\right)\right] &= 0\end{aligned}$$

Where D is the number of dimensions, In 3+1D, we recover a δ^3. The $\eta^\rho\left(x\right)$ and $\pi^\rho\left(x\right)$ are now operators satisfying canonical commutation relations. Let us return to the Schrodinger theory, and apply these commutation relations. In this theory, we have the commutation relations:

$$\left[\hat{\psi}\left(x\right),\hat{\pi}\left(x'\right)\right] = i\hbar\delta^3\left(\boldsymbol{x}-\boldsymbol{x}'\right)$$

$$\left[\hat{\psi}\left(x\right),\hat{\psi}\left(x'\right)\right]=0$$
$$\left[\hat{\psi}^{\dagger}\left(x\right),\hat{\psi}^{\dagger}\left(x'\right)\right]=0$$

Inserting the relation between $\hat{\pi}$ and $\hat{\psi}^{\dagger}$, we can rewrite the first commutation relation:

$$\left[\hat{\psi}\left(x\right),\hat{\psi}^{\dagger}\left(x'\right)\right]=\delta^{3}\left(\boldsymbol{x}-\boldsymbol{x}'\right)$$

These commutation relations are at equal time, $t=t'$. Note that since our operators have time dependence, we are working in the Heisenberg picture. Using the commutation relations, the Heisenberg equation of motion is given by:

$$i\hbar\frac{d\hat{\psi}}{dt}=\left[\hat{\psi},\hat{H}\right]$$

And we recover the same equation of motion that we found earlier, via the Lagrangian and the Euler-Lagrange equation:

$$-\frac{\hbar^{2}}{2m}\boldsymbol{\nabla}^{2}\hat{\psi}+V\left(\boldsymbol{r},t\right)\hat{\psi}=i\hbar\frac{\partial\hat{\psi}}{\partial t}$$

We have spent time discussing what the operators in our theory are, but what about the states? what states do the operators $\hat{\psi}$ and $\hat{\psi}^{\dagger}$ act on? Consider the operator $\hat{N}$:

$$\hat{N}=\int \mathrm{d}^{3}x\,\hat{\psi}^{\dagger}\left(x\right)\hat{\psi}\left(x\right)$$

It can be verified that

$$\begin{aligned}\frac{d\hat{N}}{dt}&=\left[\hat{N},\hat{H}\right]\\&=0\end{aligned}$$

Thus $\hat{N}$ is a constant of the motion. It also means that $\hat{N}$ and $\hat{H}$ can be simultaneously diagonalized in some basis. Let us now go to a basis in which both $\hat{N}$ and $\hat{H}$ are diagonal.

We can expand the field operators in terms of eigenfunctions of the Hamiltonian:

$$\hat{\psi}\left(\boldsymbol{x},t\right)=\sum_{k}\hat{\tilde{a}}_{k}\left(t\right)u_{k}\left(\boldsymbol{x}\right)$$
$$\hat{\psi}^{\dagger}\left(\boldsymbol{x},t\right)=\sum_{k}\hat{\tilde{a}}_{k}\left(t\right)u_{k}^{*}\left(\boldsymbol{x}\right)$$

Where $u_{k}\left(\boldsymbol{x}\right)$ satisfies the condition:

$$-\frac{\hbar^{2}}{2m}\boldsymbol{\nabla}^{2}u_{k}\left(\boldsymbol{x}\right)+V\left(\boldsymbol{x}\right)u_{k}\left(\boldsymbol{x}\right)=E_{k}u_{k}\left(\boldsymbol{x}\right)$$

Where we have restricted our system to have a time-independent potential $V\left(\boldsymbol{x}\right)$. The $u_{k}\left(\boldsymbol{x}\right)$ are eigenfunctions of a Hermitian operator, so they are orthogonal, and they form a complete set. We could have expanded our field operators in any set of orthogonal functions (sines/cosines, etc.), but we chose the $u_{k}\left(\boldsymbol{x}\right)$ functions because we want to diagonalize the Hamiltonian.

Recall that the Hamiltonian density was given by

$$\mathcal{H} = \frac{\hbar^2}{2m}\boldsymbol{\nabla}\psi^\dagger \cdot \boldsymbol{\nabla}\psi + V\left(\boldsymbol{x}\right)\psi^\dagger\psi$$

The Hamiltonian is the integration of the density:

$$\begin{aligned} H &= \int \mathrm{d}^3x\, \mathcal{H} \\ &= \int \mathrm{d}^3x\, \hat{\psi}^\dagger \left[-\frac{\hbar^2}{2m}\boldsymbol{\nabla}^2\hat{\psi} + V\left(\boldsymbol{x}\right)\hat{\psi}\right] \end{aligned}$$

Where we have integrated by parts to pull out the $\hat{\psi}^\dagger$. If we now insert our expanded field operators into this, and use the fact that $u_k\left(\boldsymbol{x}\right)$ are eigenfunctions of the Hamiltonian, we will be able to reduce the whole expression.

Suppose we take the equation of motion, and substitute in our expanded operators:

$$-\frac{\hbar^2}{2m}\boldsymbol{\nabla}^2\hat{\psi} + V\left(\boldsymbol{x}\right)\hat{\psi} = i\hbar\frac{\partial\hat{\psi}}{\partial t}$$

Before doing this, we note that

$$i\hbar\frac{d\hat{\hat{a}}_k}{dt} = E_k\hat{\hat{a}}_k$$

From which we find that

$$\hat{\hat{a}}_k\left(t\right) = \hat{a}_k e^{-iE_kt/\hbar}$$

From this, we have that our expanded operators take the form:

$$\begin{aligned} \hat{\psi}\left(\boldsymbol{x},t\right) &= \sum_k \hat{a}_k u_k\left(\boldsymbol{x}\right)e^{-iE_kt/\hbar} \\ \hat{\psi}^\dagger\left(\boldsymbol{x},t\right) &= \sum_k \hat{a}_k^\dagger u_k^*\left(\boldsymbol{x}\right)e^{iE_kt/\hbar} \end{aligned}$$

We see that we can extract the time dependence from our Fourier coefficient operators. Now we can invert this equations (just as one does to find Fourier coefficients) to find $\hat{a}_k$ and $\hat{a}_k^\dagger$, via the fact that the u_k functions provide an orthogonal basis of functions. Let us assume that we have a normalization condition:

$$\int \mathrm{d}^3x\, u_k^*\left(\boldsymbol{x}\right)u_l\left(\boldsymbol{x}\right) = \delta_{kl}$$

Inverting our expansions, we find that

$$\begin{aligned} \hat{a}_k &= \int \mathrm{d}^3\boldsymbol{x}\, \hat{\psi}\left(\boldsymbol{x},t\right)u_k^*\left(\boldsymbol{x}\right)e^{iE_kt/\hbar} \\ \hat{a}_k^\dagger &= \int \mathrm{d}^3\boldsymbol{x}\, \hat{\psi}^\dagger\left(\boldsymbol{x},t\right)u_k\left(\boldsymbol{x}\right)e^{-iE_kt/\hbar} \end{aligned}$$

Since we know the commutation relations between $\hat{\psi}$ and $\hat{\psi}^\dagger$, we can use these to find the commutation relations between $\hat{a}_k$ and $\hat{a}_k^\dagger$:

$$\left[\hat{a}_k, \hat{a}_l\right] = 0$$

$$\left[\hat{a}_k^\dagger, \hat{a}_l^\dagger\right] = 0$$

These follow from the fact that $\hat{a}$ and $\hat{a}^\dagger$ only depend on $\hat{\psi}$ and $\hat{\psi}^\dagger$ respectively, and those commute with each other. The final commutator requires a bit more work:

$$\begin{aligned}
\left[\hat{a}_k, \hat{a}_l^\dagger\right] &= \int \mathrm{d}^3x \int \mathrm{d}^3x' \, e^{i(E_k - E_l)t/\hbar} u_k^* \left(\boldsymbol{x}\right) u_l \left(\boldsymbol{x}'\right) \overbrace{\left[\hat{\psi}\left(\boldsymbol{x}, t\right), \hat{\psi}^\dagger \left(\boldsymbol{x}', t\right)\right]}^{\delta^3(\boldsymbol{x}-\boldsymbol{x}')} \\
&= e^{i(E_k - E_l)t/\hbar} \int \mathrm{d}^3x \, u_k^* \left(\boldsymbol{x}\right) u_l \left(\boldsymbol{x}\right) \\
&= \delta_{kl}
\end{aligned}$$

Now let us return to the Hamiltonian and number operator. Let us rewrite these in terms of $\hat{a}_k$ and $\hat{a}_k^\dagger$:

$$\begin{aligned}
\hat{N} &= \int \mathrm{d}^3x \, \hat{\psi}^\dagger \left(\boldsymbol{x}, t\right) \hat{\psi} \left(\boldsymbol{x}, t\right) \\
&= \int \mathrm{d}^3x \sum_l \hat{a}_l^\dagger u_l^* \left(\boldsymbol{x}\right) e^{iE_l t/\hbar} \sum_k \hat{a}_k u_k \left(\boldsymbol{x}\right) e^{-iE_k t/\hbar} \\
&= \sum_{l,k} \hat{a}_l^\dagger \hat{a}_k e^{i(E_l - E_k)t/\hbar} \int \mathrm{d}^3x \, u_k \left(\boldsymbol{x}\right) u_l^* \left(\boldsymbol{x}\right) \\
&= \sum_k \hat{a}_k^\dagger \hat{a}_k
\end{aligned}$$

Where we note that the only terms that depend on $\boldsymbol{x}$ in the integral are the functions $u_l^* \left(\boldsymbol{x}\right)$ and $u_k \left(\boldsymbol{x}\right)$, so we can use the orthogonality relation to remove the integral and get a Kronecker delta, and then use that to collapse the two sums into one. It is also useful to define an operator $\hat{N}_k$:

$$\hat{N}_k = \hat{a}_k^\dagger \hat{a}_k$$

From which we have

$$\hat{N} = \sum_k \hat{N}_k$$

Let us do the same process for the Hamiltonian:

$$\begin{aligned}
\hat{H} &= \int \mathrm{d}^3\boldsymbol{x} \, \frac{\hbar^2}{2m} \boldsymbol{\nabla}^2 \hat{\psi}^\dagger \cdot \boldsymbol{\nabla} \hat{\psi} + V\left(\boldsymbol{x}\right) \hat{\psi}^\dagger \hat{\psi} \\
&= \sum_{k,l} \hat{a}_k^\dagger \hat{a}_l e^{i(E_k - E_l)t/\hbar} \int \mathrm{d}^3\boldsymbol{x} \left(\frac{\hbar^2}{2m} \boldsymbol{\nabla} u_k^* \cdot \boldsymbol{\nabla} u_l + u_k^* V\left(\boldsymbol{x}\right) u_l\right) \\
&= \sum_{k,l} \hat{a}_k^\dagger \hat{a}_l e^{i(E_k - E_l)t/\hbar} \int \mathrm{d}^3\boldsymbol{x} \, u_k^* \left(-\frac{\hbar^2}{2m} \boldsymbol{\nabla}^2 u_l + V\left(\boldsymbol{x}\right) u_l\right) \\
&= \sum_{k,l} \hat{a}_k^\dagger \hat{a}_l e^{i(E_k - E_l)t/\hbar} \int \mathrm{d}^3\boldsymbol{x} \, u_k^* \left(E_l u_l\right) \\
&= \sum_{k,l} \hat{a}_k^\dagger \hat{a}_l e^{i(E_k - E_l)t/\hbar} E_l \delta_{kl}
\end{aligned}$$

$$
\begin{aligned}
&= \sum_k E_k \hat{a}_k^\dagger \hat{a}_k \\
&= \sum_k E_k \hat{N}_k
\end{aligned}
$$

Where we integrate by parts, and drop the boundary term, under the assumption that the eigenfunctions vanish at infinity. We then use the fact that u_l is by definition an eigenfunction of the Schrodinger equation, and then we use the orthogonality condition to get a Kronecker delta and collapse the integral and two sums into a single sum.

Now how do we find the eigenvalues of the Hamiltonian? First, let us note that $\left[\hat{N}_i, \hat{N}_j\right] = 0$. Because of this, we can diagonalize them simultaneously. Thus, finding the eigenvalues of the Hamiltonian boils down to finding the eigenvalues of $\hat{N}_k$. Now recall the harmonic oscillator from quantum mechanics. We can write the oscillator Hamiltonian as:

$$
\hat{H}_{\text{osc}} = \hbar\omega\left(a^\dagger a + \frac{1}{2}\right)
$$

The raising and lowering operators satisfied the same commutation relations that we have for our Schrodinger field, and in the case of the oscillator, eigenvalues of $a^\dagger a$ were integers. Thus, if we can reproduce this argument, we can get the eigenvalues of our Schrodinger field Hamiltonian.

Consider the commutator:

$$
\begin{aligned}
\left[\hat{N}_i, \hat{a}_i\right] &= \left[\hat{a}_i \hat{a}_i, \hat{a}_i\right] \\
&= \left[\hat{a}_i^\dagger, \hat{a}_i\right]\hat{a}_i \\
&= -\hat{a}_i
\end{aligned}
$$

Similarly, we can compute the commutator with $\hat{a}_i^\dagger$:

$$
\left[\hat{N}_i, \hat{a}_i^\dagger\right] = \hat{a}_i^\dagger
$$

Now let $|n_i\rangle$ be an eigenstate of $\hat{N}_i$, with eigenvalue n_i:

$$
\hat{N}_i |n_i\rangle = n_i |n_i\rangle
$$

Now, consider applying $\hat{N}_i$ to the eigenstate that has $\hat{a}_i$ applied to it:

$$
\begin{aligned}
\hat{N}_i\left(\hat{a}_i |n_i\rangle\right) &= \left[\hat{N}_i, \hat{a}_i\right]|n_i\rangle + \hat{a}_i \hat{N}_i |n_i\rangle \\
&= -\hat{a}_i |n_i\rangle + n_i \hat{a}_i |n_i\rangle \\
&= \left(n_i - 1\right)\hat{a}_i |n_i\rangle
\end{aligned}
$$

From this, we see that $\hat{a}_i |n_i\rangle$ is an eigenstate of $\hat{N}_i$, with eigenvalue $n_i - 1$, and therefore this is the state $|n_i - 1\rangle$. From here, we can keep applying $\hat{a}_i$, and continuing lowering the eigenvalue and the eigenstate label. To prove that n_i must be a positive integer, we note that $\langle \hat{N}_i \rangle_\psi$ must be positive, $\hat{N}_i$ is a positive definite operator[14]:

$$
\langle \psi | \hat{N}_i | \psi \rangle = \langle \psi | \hat{a}_i^\dagger \left(\hat{a}_i |\psi\rangle\right)
$$

[14] Not all Hermitian matrices are positive definite!

$$= \langle \phi | \phi \rangle \geq 0$$

Thus, by repeatedly applying $\hat{a}_i$ to $|n_i\rangle$, it appears that one can continue decreasing the eigenvalue infinitely. However, since $\hat{N}_i$ is positive definite, all of its eigenvalues are positive. Therefore, the only way that we resolve this is if we have a state labelled $|0\rangle_i$, with eigenvalue 0:

$$\hat{a}_i |0_i\rangle = 0 |0_i\rangle$$

Then, applying $\hat{N}_i$ will also give us zero:

$$\hat{N}_i |0_i\rangle = 0 |0_i\rangle$$

So $|0_i\rangle$ is an eigenvector of $\hat{N}_i$ with eigenvalue 0. Due to the unit spacing of the eigenvalues, the eigenvalues of $\hat{N}_i$ are $0, 1, 2, \ldots$, and the corresponding eigenvectors are $|0_i\rangle, |1_i\rangle, |2_i\rangle, \ldots$. Since all the $\hat{N}_i$s can be independently diagonalized, an eigenstate of all of the $\hat{N}_i$ operators as:

$$|n_1, n_2, \ldots, n_i, \ldots\rangle$$

Given this state, we can see what the action of a single one of the $\hat{N}_i$ operators will be:

$$\hat{N}_i |n_1, n_2, \ldots, n_i, \ldots\rangle = n_i |n_1, n_2, \ldots, n_i, \ldots\rangle$$

From this, we have that the eigenvalues of the full $\hat{N}$ operator are given by:

$$\hat{N} |n_1, n_2, \ldots, n_i, \ldots\rangle = \sum_i n_i |n_1, n_2, \ldots, n_i, \ldots\rangle$$

Which immediately gives us the eigenvalues of the Hamiltonian:

$$\hat{H} |n_1, n_2, \ldots, n_i, \ldots\rangle = \left(\sum_i n_i E_i\right) |n_1, n_2, \ldots, n_i, \ldots\rangle$$

How do we interpret this? We interpret n_1 as the number of particles in energy level 1, and generalizing, n_i is the number of particles in the ith energy level. From this, the eigenvalue of $\hat{N}$ gives us the total number of particles in our system.

We solved this for an arbitrary potential $V(\boldsymbol{x})$. All of the particles in the system are interacting with the potential, but they are not interacting with each other. The problem that we have solved is therefore analogous to $\sum_i n_i$ noninteracting particles distributed among the energy eigenstates generated by the potential $V(\boldsymbol{x})$. We see that even though we started with a single particle Schrodinger equation, the solution gave us the dynamics of an *arbitrary* number of particles in the system.

Now we return to our $\hat{a}_i$ operator, and we see that it can decrement the number of particles in the ith energy level:

$$\hat{a}_i |n_1, n_2, \ldots n_i, \ldots\rangle \propto |n_1, n_2, \ldots, n_i - 1, \ldots\rangle$$

Similarly, $\hat{a}_i^\dagger$ increments the number of particles in the ith energy level:

$$\hat{a}_i^\dagger |n_1, n_2, \ldots, n_i, \ldots\rangle \propto |n_1, n_2, \ldots, n_i + 1, \ldots\rangle$$

For this reason, $\hat{a}_i$ is known as the annihilation operator, and $\hat{a}_i^\dagger$ is known as the creation operator.

The state with no particles, $|0,0,0\ldots\rangle$ is known as the vacuum.

Now suppose we want to take what we have just found, and return to the usual way we think about quantum mechanics. Let us consider the action of the operators $\hat{\psi}\left(x\right)$ and $\hat{\psi}^\dagger\left(x\right)$ on the vacuum. Let us suppose this state is denoted $|\boldsymbol{x},t\rangle$:

$$\hat{\psi}^\dagger\left(\boldsymbol{x},t\right)|0\rangle = |\boldsymbol{x},t\rangle$$

Note that the action of $\psi\left(\boldsymbol{x},t\right)$ on the vacuum must be zero, since it contains just a $\hat{a}$. Let us now consider the time derivative of this state:

$$i\hbar\frac{\partial}{\partial t}|\boldsymbol{x},t\rangle = i\hbar\frac{\partial}{\partial t}\hat{\psi}^\dagger\left(\boldsymbol{x},t\right)|0\rangle$$

Now using the Heisenberg picture evolution of the operator $\hat{\psi}^\dagger$, where $|0\rangle$ is constant in time:

$$\begin{aligned} i\hbar\frac{\partial}{\partial t}\hat{\psi}^\dagger\left(\boldsymbol{x},t\right)|0\rangle &= \left[\hat{\psi}^\dagger,\hat{H}\right]|0\rangle \\ &= -\hat{H}\hat{\psi}^\dagger\left(\boldsymbol{x},t\right)|0\rangle \\ &= -\hat{H}\,|\boldsymbol{x},t\rangle \end{aligned}$$

We see that this is very reminiscent of the Schrodinger equation, but we have the wrong sign. Now let us consider the inner product of two of these states:

$$\begin{aligned} \langle\boldsymbol{x}',t|\boldsymbol{x},t\rangle &= \langle 0|\hat{\psi}\left(\boldsymbol{x}',t\right)\hat{\psi}^\dagger\left(\boldsymbol{x},t\right)|0\rangle \\ &= \langle 0|\left[\hat{\psi}\left(\boldsymbol{x}',t\right),\hat{\psi}^\dagger\left(\boldsymbol{x},t\right)\right]|0\rangle \\ &= \delta^3\left(\boldsymbol{x}-\boldsymbol{x}'\right) \end{aligned}$$

Where we note that we can replace the product of our two operators with the commutator since the second term in the commutator vanishes, $\hat{\psi}$ acting on the vacuum is zero. The object that satisfies the Schrodinger equation with the wrong sign and has this normalization condition is the position basis vector in the Heisenberg picture. This suggests that $|\boldsymbol{x},t\rangle$ are basis vectors for single particle states in the Heisenberg picture. Given a time-independent single particle state vector $|\psi\rangle$, the position space wavefunction is given by

$$\langle\boldsymbol{x},t\,|\psi\rangle = \psi\left(\boldsymbol{x},t\right)$$

Let us confirm this by constructing the wavefunction of the statevector

$$|0,0,\ldots 1_i,\ldots\rangle$$

which is a single particle in the ith energy eigenstate. We relabel this state as $|i\rangle$:

$$|i\rangle = \hat{a}_i^\dagger|0\rangle$$

The wavefunction of $|i\rangle$ is:

$$\begin{aligned} \langle\boldsymbol{x},t|i\rangle &= \langle 0|\hat{\psi}\left(\boldsymbol{x},t\right)\hat{a}_i^\dagger|0\rangle \\ &= \left\langle 0\left|\hat{\psi}\left(\boldsymbol{x},t\right)\int \mathrm{d}^3\boldsymbol{x}\,\hat{\psi}^\dagger\left(\boldsymbol{x}',t\right)u_i\left(\boldsymbol{x}\right)e^{-iE_it/\hbar}\right|0\right\rangle \end{aligned}$$

$$
\begin{aligned}
&= e^{-iE_i t/\hbar}\int \mathrm{d}^3\boldsymbol{x}\, u_i\left(\boldsymbol{x}\right) \underbrace{\langle 0|\left[\hat{\psi}\left(\boldsymbol{x},t\right),\hat{\psi}^{\dagger}\left(\boldsymbol{x}',t\right)\right]|0\rangle}_{\delta^3\left(\boldsymbol{x}-\boldsymbol{x}'\right)} \\
&= u_i\left(\boldsymbol{x}\right) e^{-iE_i t/\hbar}
\end{aligned}
$$

Which is exactly the time-dependent wavefunction for a particle in the ith energy eigenstate.

Thus we have shown that we can take a state and construct its wavefunction. Suppose instead we had a wavefunction, and we wanted to construct the state with that particular wavefunction at some time $t = t'$. Consider the operator $\hat{\psi}_s^{\dagger}\left(\boldsymbol{x},t,t'\right)$, defined as:

$$
\hat{\psi}_s^{\dagger}\left(\boldsymbol{x},t,t'\right) = e^{-i\hat{H}(t-t')/\hbar}\hat{\psi}^{\dagger}\left(\boldsymbol{x},t\right)e^{i\hat{H}(t-t')/\hbar}
$$

The s subscript denotes Schrodinger, as we can see that this object does not evolve in time. Thus, when we apply this to the vacuum, we create states that do not evolve in time, which is what we want when working in the Heisenberg picture. Let us show that this is time independent:

$$
\begin{aligned}
i\hbar\frac{\partial}{\partial t}\hat{\psi}_s^{\dagger}\left(\boldsymbol{x},t,t'\right) &= e^{-i\hat{H}(t-t')/\hbar}\left(\hat{H}\hat{\psi}^{\dagger} + i\hbar\frac{\partial\hat{\psi}^{\dagger}}{\partial t} - \hat{\psi}^{\dagger}\hat{H}\right)E^{i\hat{H}(t-t')/\hbar} \\
&= e^{-i\hat{H}(t-t')/\hbar}\left(\hat{H}\hat{\psi}^{\dagger} + \left[\hat{\psi}^{\dagger},\hat{H}\right] - \hat{\psi}^{\dagger}\hat{H}\right)E^{i\hat{H}(t-t')/\hbar} \\
&= 0
\end{aligned}
$$

Where we have applied the Heisenberg equation of motion for $\hat{\psi}^{\dagger}$. Thus we see that $\hat{\psi}_s^{\dagger}$ is actually independent of t:

$$
\hat{\psi}_s^{\dagger}\left(\boldsymbol{x},t,t'\right) = \hat{\psi}_s^{\dagger}\left(\boldsymbol{x},t'\right)
$$

Now consider the statevector given by the action of this operator on the vacuum

$$
\hat{\psi}_s^{\dagger}\left(\boldsymbol{x},t'\right)|0\rangle
$$

We claim that this represents a single particle state localized at $\boldsymbol{x}$, at time $t = t'$. To see this, we can project it against the basis vectors:

$$
\langle\boldsymbol{x}',t|\hat{\psi}_s^{\dagger}\left(\boldsymbol{x},t'\right)|0\rangle = \left\langle 0\left|\hat{\psi}\left(\boldsymbol{x}',t\right)e^{-i\hat{H}(t-t')/\hbar}\hat{\psi}^{\dagger}\left(\boldsymbol{x},t\right)e^{i\hat{H}(t-t')/\hbar}\right|0\right\rangle
$$

At time $t = t'$, this is exactly a delta function, $\delta^3\left(\boldsymbol{x}-\boldsymbol{x}'\right)$. What if we look at the time derivative of this object?

$$
\begin{aligned}
i\hbar\frac{\partial}{\partial t}\langle\boldsymbol{x}',t|\hat{\psi}_s^{\dagger}\left(\boldsymbol{x},t'\right)|0\rangle &= i\hbar\frac{\partial}{\partial t}\langle 0|\hat{\psi}\left(\boldsymbol{x}',t\right)\hat{\psi}_s\left(\boldsymbol{x},t'\right)|0\rangle \\
&= \left\langle 0\left|i\hbar\frac{\partial}{\partial t}\hat{\psi}\left(\boldsymbol{x}',t\right)\hat{\psi}_s^{\dagger}\left(\boldsymbol{x},t'\right)\right|0\right\rangle \\
&= \left\langle 0\left|\left[-\frac{\hbar^2}{2m}\boldsymbol{\nabla}'^2\hat{\psi} + V\left(\boldsymbol{x}'\right)\hat{\psi}\right]\hat{\psi}_s^{\dagger}\left(\boldsymbol{x},t'\right)\right|0\right\rangle \\
&= \left[-\frac{\hbar^2}{2m}\boldsymbol{\nabla}'^2 + V\left(\boldsymbol{x}'\right)\right]\langle 0|\hat{\psi}\left(\boldsymbol{x}',t\right)\hat{\psi}_s^{\dagger}\left(\boldsymbol{x},t'\right)|0\rangle \\
&= \left[-\frac{\hbar^2}{2m}\boldsymbol{\nabla}'^2 + V\left(\boldsymbol{x}'\right)\right]\langle\boldsymbol{x}',t|\hat{\psi}_s^{\dagger}\left(\boldsymbol{x},t'\right)|0\rangle
\end{aligned}
$$

We see that the state evolves in time according the Schrodinger equation, which is exactly what we would expect. Now that we have proven that it has the correct time evolution, as well as the correct localization, we have shown that this state is indeed a single particle state localized at $\boldsymbol{x}$, at time $t = t'$.

Suppose we now want an arbitrary one-particle state $|f\rangle$ with wavefunction $f(\boldsymbol{x})$ at time $t = t'$. We can construct this via the delta function state:

$$|f\rangle = \int \mathrm{d}^3\boldsymbol{x}\, f(\boldsymbol{x})\, \hat{\psi}_s^\dagger(\boldsymbol{x}, t')\, |0\rangle$$

It is straightforward to verify that the wavefunction of $|f\rangle$, $\langle \boldsymbol{x}, t|f\rangle$ is indeed $f(\boldsymbol{x})$ at time $t = t'$ and satisfies the Schrodinger wave equation, by following the exact same steps we used in the delta function case.

Now suppose we looked at $\langle \boldsymbol{x}, t|f\rangle$:

$$\begin{aligned}
\langle \boldsymbol{x}, t|f\rangle &= \langle 0|\hat{\psi}(\boldsymbol{x}, t)\,|f\rangle \\
&= \langle 0|\hat{\psi}(\boldsymbol{x}, t)\left(\int \mathrm{d}^3\boldsymbol{x}\ |\boldsymbol{x}', t'\rangle\langle \boldsymbol{x}', t'|\right)|f\rangle \\
&= \int \mathrm{d}^3\boldsymbol{x}\ \langle 0|\hat{\psi}(\boldsymbol{x}, t)\,\hat{\psi}^\dagger(\boldsymbol{x}', t')\,|0\rangle\langle \boldsymbol{x}', t'|f\rangle
\end{aligned}$$

We see that the second term is the wavefunction at time t', and the other inner product is the Greens function, known as the propagator:

$$G(\boldsymbol{x}, \boldsymbol{x}'; t, t') = \langle 0|\hat{\psi}(\boldsymbol{x}, t)\,\hat{\psi}^\dagger(\boldsymbol{x}', t')\,|0\rangle \tag{11}$$

The propagator encompasses the time dependence of the state. Suppose we rewrite the propagator in the energy eigenbasis, using $\hat{a}_k$ and $\hat{a}_k^\dagger$. If we did this, we would recover the familiar form of the non-relativistic propagator:

$$G(\boldsymbol{x}, \boldsymbol{x}'; t, t') = \sum_k u_k(\boldsymbol{x})\, u_k^*(\boldsymbol{x}')\, e^{-iE_k(t-t')/\hbar}$$

In a certain sense we have not done anything new here, this is a mathematically equivalent way of doing quantum mechanics, everything we have done could be done by working in the traditional quantum mechanics formalism.

Now let us turn our attention to multi-particle states, starting with 2 particle states. For any 2 particle state $|f\rangle$, the wavefunction can be found by using two $\hat{\psi}$ operators, rather than just one:

$$f(\boldsymbol{x}_1, \boldsymbol{x}_2, t) = \langle 0|\hat{\psi}(\boldsymbol{x}_1, t)\,\hat{\psi}(\boldsymbol{x}_2, t)\,|f\rangle$$

This satisfies the Schrodinger equation:

$$\begin{aligned}
i\hbar\frac{\partial}{\partial t}\langle 0|\hat{\psi}(\boldsymbol{x}_1, t)\,\hat{\psi}(\boldsymbol{x}_2, t)\,|f\rangle &= \langle 0|i\hbar\frac{\partial}{\partial t}\hat{\psi}(\boldsymbol{x}_1, t)\,\hat{\psi}(\boldsymbol{x}_2, t)\,|f\rangle + \langle 0|\hat{\psi}(\boldsymbol{x}_1, t)\, i\hbar\frac{\partial}{\partial t}\hat{\psi}(\boldsymbol{x}_2, t)\,|f\rangle \\
&= \left[-\frac{\hbar^2}{2m}\left(\boldsymbol{\nabla}_1^2 + \boldsymbol{\nabla}_2^2\right) + V(\boldsymbol{x}_1) + V(\boldsymbol{x}_2)\right]\langle 0|\hat{\psi}(\boldsymbol{x}_1, t)\,\hat{\psi}(\boldsymbol{x}_2, t)\,|f\rangle
\end{aligned}$$

Where we have inserted the equations of motion for both single-particle wavefunctions. We see that this object satisfies the two particle Schrodinger equation, as we would expect. Also note that the

$\hat{\psi}$ operators commute, and therefore the object is symmetric under interchange of $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$, and therefore this can only describe bosonic particles. This constraint was introduced by the canonical commutation relations that we imposed on the fields.

Thus we have shown that given a statevector, we can extract its wavefunction. To do the converse, where we want to construct a state with wavefunction $f(\boldsymbol{x}_1, \boldsymbol{x}_2)$ at time t', we can generalize the single particle case, by introducing another $\hat{\psi}_s^\dagger$, and integrating over another set of coordinates:

$$\int \mathrm{d}^3\boldsymbol{x}_1 \int \mathrm{d}^3\boldsymbol{x}_2\, f(\boldsymbol{x}_1, \boldsymbol{x}_2)\, \hat{\psi}_s^\dagger(\boldsymbol{x}_1, t')\, \hat{\psi}_s^\dagger(\boldsymbol{x}_2, t')\, |0\rangle$$

The generalization to higher particle count states is straightforward. Notice that the $\hat{\psi}_s^\dagger$ operators commute, so again this will be symmetric in $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$, regardless of whether $f(\boldsymbol{x}_1, \boldsymbol{x}_2)$ is symmetric under interchange. Suppose we chose an f that does not have symmetry. If we then wrote this object down and exchanged $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$, and then used the fact that the operators commute and we can switch the order of integration:

$$\int \mathrm{d}^3\boldsymbol{x}_2 \int \mathrm{d}^3\boldsymbol{x}_1\, f(\boldsymbol{x}_2, \boldsymbol{x}_1)\, \hat{\psi}_s^\dagger(\boldsymbol{x}_2, t')\, \hat{\psi}_s^\dagger(\boldsymbol{x}_1, t')\, |0\rangle = \int \mathrm{d}^3\boldsymbol{x}_1 \int \mathrm{d}^3\boldsymbol{x}_2\, f(\boldsymbol{x}_2 . \boldsymbol{x}_1)\, \hat{\psi}_s^\dagger(\boldsymbol{x}_1, t')\, \hat{\psi}_s^\dagger(\boldsymbol{x}_2, t')\, |0\rangle$$

So in fact the only part of f that survives is the symmetrized portion:

$$\frac{1}{2}\left[f(\boldsymbol{x}_1, \boldsymbol{x}_2) + f(\boldsymbol{x}_2, \boldsymbol{x}_1)\right]$$

4.1. Fermionic Second Quantization

We have seen that our construction of the state automatically takes into account the statistics of the bosons, since we imposed the canonical commutation relations. How can we recover Fermi-Dirac statistics? Recall the two particle wavefunction:

$$f(\boldsymbol{x}_1, \boldsymbol{x}_2, t) = \langle 0|\hat{\psi}(\boldsymbol{x}_1, t)\, \hat{\psi}(\boldsymbol{x}_2, t)\, |f\rangle$$

Suppose we required that the $\hat{\psi}$ operators *anti*commute, rather than commute. In that case, we would recover the fermionic antisymmetry of the wavefunction, exchanging $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ would introduce a minus sign. The issue is whether or not this anticommutation requirement is consistent with all of the rest of the theory that we have worked through, under the assumption that the operators commute. Instead of imposing the canonical commutation relations, let us impose the canonical *anticommutation* relations (equal time):

$$\begin{aligned}
\{\hat{\psi}(\boldsymbol{x}, t), \hat{\psi}(\boldsymbol{x}', t)\} &= 0 \\
\{\hat{\pi}(\boldsymbol{x}, t), \hat{\pi}(\boldsymbol{x}', t)\} &= 0 \\
\{\hat{\psi}(\boldsymbol{x}, t), \hat{\pi}(\boldsymbol{x}', t)\} &= i\hbar\delta^3(\boldsymbol{x} - \boldsymbol{x}')
\end{aligned} \tag{12}$$

Let us start from the same Lagrangian density (10):

$$\mathcal{L} = i\hbar\psi^\dagger \frac{\partial \psi}{\partial t} - \frac{\hbar^2}{2m}\boldsymbol{\nabla}\psi^\dagger \cdot \boldsymbol{\nabla}\psi - V(\boldsymbol{x})\,\psi^\dagger\psi$$

Where ψ is a complex scalar field. Recall that the equation of motion is:

$$i\hbar\frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m}\boldsymbol{\nabla}^2\psi + V\psi$$

And we have a conjugate momentum $\pi = \frac{\partial \mathcal{L}}{\partial \dot{\psi}} = i\hbar\psi^\dagger$, and this gives us the Hamiltonian density:

$$\begin{aligned}\mathcal{H} &= \pi\dot{\psi} - \mathcal{L} \\ &= \frac{\hbar^2}{2m}\boldsymbol{\nabla}\psi\,\dagger\cdot\boldsymbol{\nabla}\psi + V\left(\boldsymbol{x}\right)\psi^\dagger\psi\end{aligned}$$

So far, everything has been classical. Now let us impose our canonical anticommutation relations (12), and let us make note of the fact that the constant in front the delta function vanishes:

$$\left\{\psi\left(\boldsymbol{x},t'\right),\psi^\dagger\left(\boldsymbol{x}',t'\right)\right\} = \delta^3\left(\boldsymbol{x}-\boldsymbol{x}'\right)$$

We will skip many of the derivations that are analogous to the bosonic case[15]. If we combine our imposed anticommutation relations with the Heisenberg equation of motion:

$$i\hbar\frac{\partial\hat{\psi}}{\partial t} = \left[\hat{\psi},\hat{H}\right]$$

we obtain the same equation of motion that we obtained from the Lagrangian. This is a good start!

Let us once again define an operator $\hat{N}$:

$$\hat{N} = \int \mathrm{d}^3\boldsymbol{x}\,\hat{\psi}^\dagger\left(x\right)\hat{\psi}\left(x\right)$$

And we can once again show that $\left[\hat{N},\hat{H}\right] = 0$, under the anticommutation relations. Thus, we again see that $\hat{N}$ and $\hat{H}$ can be simultaneously diagonalized. Let us go to a basis where they are both diagonal. We can expand $\hat{\psi}\left(\boldsymbol{x},t\right)$ in terms of basis vectors that are eigenstates of the Hamiltonian:

$$\hat{\psi}\left(\boldsymbol{x},t\right) = \sum_k \hat{a}_k u_k\left(\boldsymbol{x}\right)e^{-iE_kt/\hbar}$$

Just like before, $u_k\left(\boldsymbol{x}\right)$ is an eigenstate of the Schrodinger equation, with eigenvalue E_k. Just like in the bosonic case, we can obtain an expression for the Fourier coefficients $\hat{a}_k$ and $\hat{a}_k^\dagger$, and then find their corresponding anticommutation relations:

$$\begin{aligned}\left\{\hat{a}_j,\hat{a}_k\right\} &= 0 \\ \left\{\hat{a}_j^\dagger,\hat{a}_k^\dagger\right\} &= 0 \\ \left\{\hat{a}_j,\hat{a}_k^\dagger\right\} &= \delta_{jk}\end{aligned}$$

We can express $\hat{N}$ and $\hat{H}$ in terms of these operators:

$$\begin{aligned}\hat{N} &= \sum_k \hat{N}_k \\ \hat{H} &= \sum_k E_k\hat{N}_k\end{aligned}$$

Where $\hat{N}_k = \hat{a}_k^\dagger\hat{a}_k$. So far, everything has been very similar to the bosonic case. However, now we want to find the eigenvalues of $\hat{N}_k$, and we see that there is a difference from the bosonic case, we

[15] Exercises left to the homework-doer.

found the eigenvalues via the commutation relations, but now we have anticommutation relations. We note that if we square $\hat{N}_k$, we get the same thing back, via the anticommutation relations:

$$\hat{N}_k^2 = \hat{N}_k$$

Thus, it immediately follows that the eigenvalues of $\hat{N}_k$ are 0 or 1. This is exactly what we expect, fermionic sites cannot support occupations higher than 1, unlike bosonic modes.

If we have an arbitrary eigenstate of all the $\hat{N}_i$ operators, we can label it as:

$$|n_1, n_2, \ldots, n_i, \ldots\rangle$$

Where $n_i \in \{0, 1\}$. We can see that the action of $\hat{N}_i$ on this state is:

$$\hat{N}_i \,|n_1, n_2, \ldots n_i, \ldots\rangle = n_i \,|n_1, n_2, \ldots n_i, \ldots\rangle$$

where n_i is the number of particles in the ith energy level.

The eigenvalues of $\hat{N}$ and $\hat{H}$ are then N and E, the total number of fermions and the total energy repsectively:

$$\begin{aligned} N &= \sum_i n_i \\ E &= \sum_i n_i E_i \end{aligned}$$

We see that the crucial difference from the bosonic case is the restriction on n_i to be either 0 or 1, rather than the infinite range of positive integers.

Now looking at the $\hat{a}$ and $\hat{a}^\dagger$ operators, we see that

$$\begin{aligned} \left[\hat{N}_i, \hat{a}_i\right] &= -\hat{a}_i \\ \left[\hat{N}_i, \hat{a}_i^\dagger\right] &= \hat{a}_i^\dagger \end{aligned}$$

$\hat{a}_i$ is again an annihilation operator for particles in the ith energy level, and $\hat{a}_i^\dagger$ is again a creation operator for particles in the ith energy level.

Now, just as we did in the bosonic case, we can move back to the position basis, where we again define $|\boldsymbol{x}, t\rangle$:

$$|\boldsymbol{x}, t\rangle = \hat{\psi}^\dagger\left(\boldsymbol{x}, t\right)|0\rangle$$

This is a basis vector in position space in the Heisenberg picture. Similarly, we can define our $\hat{\psi}_s\left(\boldsymbol{x}, t'\right)$ operator:

$$\hat{\psi}_s\left(\boldsymbol{x}, t'\right) = e^{-i\hat{H}(t-t')/\hbar}\hat{\psi}\left(\boldsymbol{x}, t\right)e^{i\hat{H}(t-t')/\hbar}$$

Note that again this is independent of t. As with the bosonic case, $\hat{\psi}_s^\dagger\left(\boldsymbol{x}', t'\right)$ acting on the vacuum creates a single particle state localized at $\boldsymbol{x}'$ at time t'. The subsequent time evolution of the state is governed by the Schrodinger equation. An arbitrary single particle state with wavefunction $f\left(\boldsymbol{x}\right)$ at time $t = t'$ can be written as:

$$|f\rangle = \int \mathrm{d}^3\boldsymbol{x}\, f\left(\boldsymbol{x}\right)\hat{\psi}_s^\dagger\left(\boldsymbol{x}, t'\right)|0\rangle$$

We see that everything from the bosonic case carries over to the fermionic case for single particle states (as we would expect). The differences emerge in the case of states with multiple particles.

The wavefunction of an arbitrary two particle state $|f\rangle$ can be obtained as:

$$\phi\left(\boldsymbol{x}_1, \boldsymbol{x}_2, t\right) = \langle 0|\hat{\psi}\left(\boldsymbol{x}_1, t\right)\hat{\psi}\left(\boldsymbol{x}_2, t\right)|f\rangle$$

Becuase of the anticommutation properties of $\hat{\psi}$, exchanging $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ gives the wavefunction a minus sign, producing the expected fermionic antisymmetry.

We can show that this wavefunction satisfies the two particle Schrodinger equation:

$$i\hbar\frac{\partial}{\partial t}\phi\left(\boldsymbol{x}_1, \boldsymbol{x}_2, t\right) = -\frac{\hbar^2}{2m}\left(\boldsymbol{\nabla}_1^2 + \boldsymbol{\nabla}_2^2\right)\phi\left(\boldsymbol{x}_1, \boldsymbol{x}_2, t\right) + \left[V\left(\boldsymbol{x}_1\right) + V\left(\boldsymbol{x}_2\right)\right]\phi\left(\boldsymbol{x}_1, \boldsymbol{x}_2, t\right)$$

We have shown that the Schrodinger field theory quantized with canonical anticommutation relations is completely equivalent to the quantum mechanics of particles obeying Fermi-Dirac statistics.

5. Klein-Gordon Theory

Relativistic quantum fields can have different spins, just like in regular quantum mechanics. We will systematically go higher in spin, starting with spin-0, then spin-1/2, and then spin-1 particles.

Let us begin with the Klein-Gordon field, for spin-0 particles. For a real scalar field ϕ of mass m, consider the Lagrangian density:

$$\mathcal{L} = \frac{1}{2}\left(\partial_\mu\phi\partial^\mu\phi - m^2\phi^2\right)$$

Let us take this classical Lagrangian density, which leads to the Klein-Gordon equation (5) for the field ϕ (rather than the wavefunction ψ):

$$\partial_\mu\partial^\mu\phi + m^2\phi = 0$$

The canonical momentum density conjugate to ϕ is given by:

$$\begin{aligned}\pi &= \frac{\partial\mathcal{L}}{\partial\dot{\phi}} \\ &= \dot{\phi}\end{aligned}$$

We can then write the Hamiltonian density:

$$\begin{aligned}\mathcal{H} &= \pi\dot{\phi} - \mathcal{L} \\ &= \frac{1}{2}\left[\dot{\phi}^2 + \left(\boldsymbol{\nabla}\phi\right)^2 + m^2\phi^2\right]\end{aligned}$$

Let us now quantize this theory. $\hat{\phi}$ and $\hat{\pi}$ are now operators satisfying commutation relations:

$$\begin{aligned}\left[\phi\left(x\right), \hat{\phi}\left(x'\right)\right] &= 0 \\ \left[\hat{\pi}\left(x\right), \hat{\pi}\left(x'\right)\right] &= 0 \\ \left[\hat{\phi}\left(x\right), \hat{\pi}\left(x'\right)\right] &= i\delta^3\left(\boldsymbol{x} - \boldsymbol{x}'\right)\end{aligned}$$

Note that we are using natural units, $\hbar = 1$. If we now write out our Hamiltonian density:

$$\hat{\mathcal{H}} = \frac{1}{2}\left[\hat{\pi}^2 + \left(\boldsymbol{\nabla}\hat{\phi}\right)^2 + m^2\hat{\phi}^2\right]$$

This is how we **define** our quantum theory, since we have imposed the canonical commutation relations[16], and we have a Hamiltonian density. Now let us find the equation of motion for $\hat{\phi}$ via the Heisenberg picture:

$$\begin{aligned}
i\frac{\partial\hat{\phi}}{\partial t} &= \left[\hat{\phi}, \hat{H}\right] \\
&= \left[\hat{\phi}\left(\boldsymbol{x}\right), \int \mathrm{d}^3\boldsymbol{x}' \left[\frac{1}{2}\hat{\pi}^2\left(\boldsymbol{x}'\right) + \frac{1}{2}\left(\boldsymbol{\nabla}'\hat{\phi}\left(\boldsymbol{x}'\right)\right)^2 + \frac{1}{2}m^2\hat{\phi}^2\left(\boldsymbol{x}'\right)\right]\right] \\
&= \int \mathrm{d}^3\boldsymbol{x}' \left[\hat{\phi}\left(\boldsymbol{x}\right), \frac{1}{2}\hat{\pi}^2\left(\boldsymbol{x}'\right)\right] \\
&= i\hat{\pi}\left(\boldsymbol{x}\right)
\end{aligned}$$

This is just the relation between $\hat{\phi}$ and $\hat{\pi}$, not the quantum version of the Klein-Gordon equation of motion that we wanted. However, we can still look at the equation of motion for $\hat{\pi}$:

$$\begin{aligned}
i\frac{\partial\hat{\pi}}{\partial t} &= \left[\hat{\pi}, \hat{H}\right] \\
&= \left[\hat{\pi}\left(\boldsymbol{x}\right), \int \mathrm{d}^3\boldsymbol{x}'\,\mathcal{H}\left(\boldsymbol{x}'\right)\right] \\
&= i\left[\boldsymbol{\nabla}^2\hat{\phi}\left(\boldsymbol{x}\right) - m^2\hat{\phi}\left(\boldsymbol{x}\right)\right]
\end{aligned}$$

where we skip a lot of the steps (since its messy, unlike the $\hat{\phi}$ equation of motion). If we now replace $\hat{\pi}$ with $\frac{\partial\hat{\phi}}{\partial t}$, the relation that we found via the equation of motion for $\hat{\phi}$, we end up with

$$\partial_\mu\partial^\mu\hat{\phi} + m^2\hat{\phi} = 0 \tag{13}$$

Which is the Klein-Gordon equation, except this time derived completely within our quantized quantum theory. We see that we needed both equations of motion to recover the Klein-Gordon equation, we couldn't just use one of them (or use the classical relation between ϕ and π).

Now let us move to a basis where our Hamiltonian is diagonal. We expand $\hat{\phi}\left(\boldsymbol{x}\right)$ in Fourier modes:

$$\hat{\phi}\left(\boldsymbol{x}, t\right) = \int \frac{\mathrm{d}^3\boldsymbol{p}}{\left(2\pi\right)^3}\hat{f}\left(\boldsymbol{p}, t\right)e^{i\boldsymbol{p}\cdot\boldsymbol{x}}$$

If we now insert this into the Klein-Gordon equation, we find that

$$\begin{aligned}
\partial_t^2\hat{f}\left(\boldsymbol{p}, t\right) &= -\left(\boldsymbol{p}^2 + m^2\right)\hat{f}\left(\boldsymbol{p}, t\right) \\
&= -\omega_p^2\hat{f}\left(\boldsymbol{p}, t\right)
\end{aligned}$$

Where we define

$$\omega_p = \sqrt{\boldsymbol{p}^2 + m^2}$$

[16] Note that if we instead imposed the anticommutation relations to try to get fermions, it would not lead to a consistent theory (as one might expect, we cannot have spin-0 fermions).

This is the harmonic oscillator, so we can write $\hat{f}(\boldsymbol{p},t)$ in terms of complex exponentials:

$$\hat{f}(\boldsymbol{p},t) = \hat{c}_{\boldsymbol{p}} e^{-i\omega_p t} + \hat{d}_{\boldsymbol{p}} e^{i\omega_p t}$$

Now using the fact that $\hat{\phi} = \hat{\phi}^\dagger$, since we chose $\hat{\phi}$ to be a real field, we have a restriction on $\hat{f}$:

$$\begin{aligned}\hat{f}(\boldsymbol{p},t) &= \hat{f}^\dagger(-\boldsymbol{p},t)\\ &= \hat{c}^\dagger_{-\boldsymbol{p}} e^{i\omega_p t} + \hat{d}^\dagger_{-\boldsymbol{p}} e^{-i\omega_p t}\end{aligned}$$

Equating this to the original definition, we have that

$$\begin{aligned}\hat{d}^\dagger_{-\boldsymbol{p}} &= \hat{c}_{\boldsymbol{p}}\\ \hat{d}_{\boldsymbol{p}} &= \hat{c}^\dagger_{-\boldsymbol{p}}\end{aligned}$$

We can then get rid of the $\hat{d}$ operators, and write everything in terms of the $\hat{c}$ operators[17]:

$$\begin{aligned}\hat{\phi}(\boldsymbol{x},t) &= \int \frac{\mathrm{d}^3\boldsymbol{p}}{(2\pi)^3}\left[\hat{c}_{\boldsymbol{p}} \underbrace{e^{-i\omega_p t} e^{i\boldsymbol{p}\cdot\boldsymbol{x}}}_{e^{-ip\cdot x}} + \hat{c}^\dagger_{-\boldsymbol{p}} e^{i\omega_p t} e^{i\boldsymbol{p}\cdot\boldsymbol{x}}\right]\\ &= \int \frac{\mathrm{d}^3\boldsymbol{p}}{(2\pi)^3}\left[\hat{c}_{\boldsymbol{p}} e^{-ip\cdot x} + \hat{c}^\dagger_{\boldsymbol{p}} e^{ip\cdot x}\right]\end{aligned}$$

Where we have written the two exponentials as a four vector dot product, $p = (\omega_p, \boldsymbol{p})$, and using the fact that $\boldsymbol{p}$ is a dummy variable, so we switch the sign of $\boldsymbol{p}$ in the second term of the integral. We now rescale the operators to write this in the standard convention:

$$\hat{a}_{\boldsymbol{p}} = \hat{c}_{\boldsymbol{p}}\sqrt{2\omega_p}$$

From which we find:

$$\hat{\phi}(\boldsymbol{x},t) = \int \frac{\mathrm{d}^3\boldsymbol{p}}{(2\pi)^3}\frac{1}{\sqrt{2\omega_p}}\left[\hat{a}_{\boldsymbol{p}} e^{-ip\cdot x} + \hat{a}^\dagger_{\boldsymbol{p}} e^{ip\cdot x}\right]$$

This is the standard Fourier expansion of our field. Let us compare this to the Schrodinger case. We see that in this case, we have both positive and negative frequency terms ($e^{-i\omega t}$ and $e^{i\omega t}$ respectively), whereas in the Schrodinger case, we only had the positive frequency term. This is a consequence of the fact that our theory is relativistic, it comes from the fact that the Klein-Gordon equation, unlike the Schrodinger equation, is second order in time.

Let us now find the commutation relations between $\hat{a}$ and $\hat{a}^\dagger$, by taking the Fourier transform of our field to invert the relation between our new operators and our field:

$$\int \mathrm{d}^3\boldsymbol{x}\hat{\phi}(\boldsymbol{x},t)\, e^{-i\boldsymbol{p}'\cdot\boldsymbol{x}} = \frac{1}{\sqrt{2\omega_p'}}\left[\hat{a}_{\boldsymbol{p}'} e^{-i\omega_{p'} t} + \hat{a}^\dagger_{-\boldsymbol{p}'} e^{i\omega_{p'} t}\right]$$

Now doing the same thing for our other field, $\hat{\pi}$:

$$\int \mathrm{d}^3\boldsymbol{x}\,\hat{\pi}(\boldsymbol{x},t)\, e^{-i\boldsymbol{p}'\cdot\boldsymbol{x}} = -i\sqrt{\frac{\omega_{p'}}{2}}\left[\hat{a}_{\boldsymbol{p}'} e^{-i\omega_{p'} t} - \hat{a}^\dagger_{-\boldsymbol{p}'} e^{i\omega_{p'} t}\right]$$

[17] In the complex $\hat{\phi}$ case, we could not do this step.

This now allows us to solve for our operators in terms of our fields:

$$\hat{a}_{\boldsymbol{p}} = \int \mathrm{d}^3\boldsymbol{x}\, \frac{1}{\sqrt{2\omega_p}} \left[\omega_p \hat{\phi} + i\hat{\pi}\right] e^{ip\cdot x}$$
$$\hat{a}^\dagger_{\boldsymbol{p}} = \int \mathrm{d}^3\boldsymbol{x}\, \frac{1}{\sqrt{2\omega_p}} \left[\omega_p \hat{\phi} - i\hat{\pi}\right] e^{-ip\cdot x}$$

Using these, we can find the commutation relations between our two operators:

$$\left[\hat{a}_{\boldsymbol{p}}, \hat{a}^\dagger_{\boldsymbol{p}'}\right] = \iint \mathrm{d}^3\boldsymbol{x}\, \mathrm{d}^3\boldsymbol{x}'\, \frac{1}{2\sqrt{\omega_p \omega_{p'}}} \Big[- i\omega_p \underbrace{\left[\hat{\phi}\left(x\right), \hat{\pi}\left(x'\right)\right]}_{i\delta^3(\boldsymbol{x}-\boldsymbol{x}')} e^{ip\cdot x} e^{-ip'\cdot x'} + i\omega_{p'} \underbrace{\left[\hat{\pi}\left(x\right), \hat{\phi}\left(x'\right)\right]}_{-i\delta^3(\boldsymbol{x}-\boldsymbol{x}')} e^{ip\cdot x} e^{-ip'\cdot x'} \Big]$$
$$= \left(2\pi\right)^3 \delta^3\left(\boldsymbol{p} - \boldsymbol{p}'\right)$$

We can also show that the other commutation relations are zero:

$$\left[\hat{a}_{\boldsymbol{p}}, \hat{a}_{\boldsymbol{p}'}\right] = 0$$
$$\left[\hat{a}^\dagger_{\boldsymbol{p}}, \hat{a}^\dagger_{\boldsymbol{p}'}\right] = 0$$

Now let us express the Hamiltonian in the basis of these new $\hat{a}$ and $\hat{a}^\dagger$ operators:

$$\begin{aligned}
\hat{H} &= \int \mathrm{d}^3\boldsymbol{x}\, \mathcal{H} \\
&= \int \mathrm{d}^3\boldsymbol{x}\, \left[\frac{1}{2}\hat{\pi}^2 + \frac{1}{2}\left(\boldsymbol{\nabla}\hat{\phi}\right)^2 + \frac{1}{2}m^2\hat{\phi}^2\right] \\
&= \frac{1}{2}\int \frac{\mathrm{d}^3\boldsymbol{p}}{\left(2\pi\right)^3} \left[\omega_p \left(\hat{a}_{\boldsymbol{p}}\hat{a}^\dagger_{\boldsymbol{p}} + \hat{a}^\dagger_{\boldsymbol{p}}\hat{a}_{\boldsymbol{p}}\right)\right]
\end{aligned}$$

Let us compare this to the Schrodinger case. First off, we didn't have the factor of $\frac{1}{2}$, and we also had discrete energy levels, instead of an integral over $\boldsymbol{p}$, we had a sum over k. We also did not have the $\hat{a}_{\boldsymbol{p}}\hat{a}^\dagger_{\boldsymbol{p}}$ term. We can rewrite our Hamiltonian to become closer to the Schrodinger case:

$$\hat{H} = \int \frac{\mathrm{d}^3\boldsymbol{p}}{\left(2\pi\right)^3} \omega_p \left[\hat{a}^\dagger_{\boldsymbol{p}}\hat{a}_{\boldsymbol{p}} + \frac{1}{2}\left(2\pi\right)^3 \delta^3\left(0\right)\right]$$

Now we note that the second term is infinite, for several reasons, the first being due to the delta function, but let us ignore this. The second issue is that we are integrating a constant over all momenta, which is necessarily infinite. Where was this infinity when we did the Schrodinger theory? In fact, we were cheated when we quantized the Schrodinger equation, we always wrote $\overline{\psi}$ to the left of ψ. This was a *choice* of ordering, classically there is no difference based on ordering since the two commute. Infinities such as this one are resolved via renormalization, where it can be shown that they do not affect the physical theory. We would do this by adding a constant to the Hamiltonian, whose value exactly cancels the infinity that we introduce based on the choice of ordering.

Let us for now, temporarily dodge this problematic infinity, by writing the theory in momentum space, in terms of the a and $a^\dagger$ terms, and then impose a choice of ordering, *before* we quantize our theory. If we do this, we will see that we can remove the infinity. Thus, let us proceed as if the second term is not there:

$$\hat{H} = \int \frac{\mathrm{d}^3\boldsymbol{p}}{\left(2\pi\right)^3} \omega_p \left[\hat{a}^\dagger_{\boldsymbol{p}}\hat{a}_{\boldsymbol{p}} + \cancel{\frac{1}{2}\left(2\pi\right)^3 \delta^3\left(0\right)}\right]$$

We can now look at the spatial component of momentum in terms of our $\hat{a}$ operators:

$$\begin{aligned}
p^i &= \int \mathrm{d}^3\boldsymbol{x}\, T^{0i} \\
&= \int \mathrm{d}^3\boldsymbol{x}\, \hat{\pi}\partial^i\hat{\phi} \\
&= \frac{1}{2}\int \mathrm{d}^3\boldsymbol{p}\, \boldsymbol{p}\left[\hat{a}^\dagger_{\boldsymbol{p}}\hat{a}_{\boldsymbol{p}} + \hat{a}_{\boldsymbol{p}}\hat{a}^\dagger_{\boldsymbol{p}}\right] \\
&= \int d^3\boldsymbol{p}\, \boldsymbol{p}\left[\hat{a}^\dagger_{\boldsymbol{p}}\hat{a}_{\boldsymbol{p}} + \cancel{\frac{1}{2}\left[\hat{a}_{\boldsymbol{p}}, \hat{a}^\dagger_{\boldsymbol{p}}\right]}\right]
\end{aligned}$$

Where we again remove a term that goes to infinity, again by choosing a normal ordering[18].

We can also define a new operator:

$$\hat{n}_{\boldsymbol{p}} = \frac{1}{(2\pi)^3}\hat{a}^\dagger_{\boldsymbol{p}}\hat{a}_{\boldsymbol{p}}$$

Using this, we can rewrite the Hamiltonian and our momentum:

$$\begin{aligned}
\hat{H} &= \int \mathrm{d}^3\boldsymbol{p}\, \omega_p\left(\hat{n}_{\boldsymbol{p}} + \cancel{\text{const.}}\right) \\
\hat{\boldsymbol{p}} &= \int \mathrm{d}^3\boldsymbol{p}\, \boldsymbol{p}\left(\hat{n}_{\boldsymbol{p}} + \cancel{\text{const.}}\right)
\end{aligned}$$

Now let us show that the eigenvalues of $\hat{n}_{\boldsymbol{p}}$ are positive integers. Since the $\hat{n}_{\boldsymbol{p}}$ operator commutes across momentum:

$$\left[\hat{n}_{\boldsymbol{p}}, \hat{n}_{\boldsymbol{p}'}\right] = 0$$

We can simultaneously diagonalize all the $\hat{n}_{\boldsymbol{p}}$ operators. Let us define an operator $\hat{N}$, which will end up being the total particle number:

$$\hat{N} = \int \mathrm{d}^3\boldsymbol{p}\, \hat{n}_{\boldsymbol{p}}$$

From this, we can derive commutation rules for this new operator[19]:

$$\left[\hat{N}, \hat{a}_{\boldsymbol{p}}\right] = -\hat{a}_{\boldsymbol{p}}\delta^3(0)$$

From this, if $|N\rangle$ is an eigenstate of $\hat{N}$, with eigenvalue N, then $\hat{a}_{\boldsymbol{p}}|N\rangle$ is an eigenstate of $\hat{N}$ with eigenvalue $N-1$. Now by the same argument as the harmonic oscillator case, we must have an eigenstate with eigenvalue 0, since $\hat{N}$ is a positive definite operator, and therefore must have positive eigenvalues. This state is the "vacuum", and is also an eigenstate of all the $\hat{n}_{\boldsymbol{p}}$ operators, with eigenvalue 0, and can thus be represented as:

$$|0_{\boldsymbol{p}_1}, 0_{\boldsymbol{p}_2}, 0_{\boldsymbol{p}_3}, \ldots 0_{\boldsymbol{p}_i}, \ldots\rangle$$

[18] One way to think about this infinity is that it stems from the zero point energy of the infinitely many coupled harmonic oscillators that we based our field paradigm on. We are essentially shifting the energy spectrum of the oscillators down so that the ground state energy is labelled as zero.

[19] Chacko wasn't sure about this extra delta function term.

Now for any $\boldsymbol{p}$, we have a commutation relation between $\hat{n}$ and $\hat{a}$:

$$\left[\hat{n}_{\boldsymbol{p}}, \hat{a}_{\boldsymbol{p}}^{\dagger}\right] = \delta^3(0)\, \hat{a}_{\boldsymbol{p}}^{\dagger}$$

We can construct an arbitrary eigenstate of $\hat{n}_{\boldsymbol{p}}$ by applying $\hat{a}_{\boldsymbol{p}}^{\dagger}$ an arbitrary number of times on the vacuum. The resulting state will have an integer eigenvalue equal to the number of times we applied $\hat{a}_{\boldsymbol{p}}^{\dagger}$ (up to a factor of $\delta^3(0)$):

$$\hat{n}_{\boldsymbol{p}} \left|n_{\boldsymbol{p}}\right\rangle = \underbrace{n_{\boldsymbol{p}}}_{\in \mathbb{Z}^+} \delta^3(0) \left|n_{\boldsymbol{p}}\right\rangle$$

An arbitrary state in the Hilbert space which is an eigenstate of all the $\hat{n}_{\boldsymbol{p}}$ can be specified via an integer $\hat{n}_{\boldsymbol{p}}$ for every value of $\boldsymbol{p}$:

$$\left|n_{\boldsymbol{p}_1}, n_{\boldsymbol{p}_2}, \ldots n_{\boldsymbol{p}_i}, \ldots\right\rangle$$

This is an arbitrary eigenvector of $\hat{N}$ and $\hat{H}$, so let us find the eigenvalues:

$$\begin{aligned}
\hat{N}\left|n_{\boldsymbol{p}_1}, n_{\boldsymbol{p}_2}, \ldots n_{\boldsymbol{p}_i}, \ldots\right\rangle &= \text{const.} \cdot \int \mathrm{d}^3\boldsymbol{p}\, \hat{n}_{\boldsymbol{p}} \left[\left(\hat{a}_{\boldsymbol{p}_1}^{\dagger}\right)^{n_1} \left(\hat{a}_{\boldsymbol{p}_2}^{\dagger}\right)^{n_2} \left(\hat{a}_{\boldsymbol{p}_3}^{\dagger}\right)^{n_3} \cdots \left(\hat{a}_{\boldsymbol{p}_i}^{\dagger}\right)^{n_i} \cdots\right] |0\rangle \\
&= \sum_{\boldsymbol{k}} n_{\boldsymbol{p}_k} \left|n_{\boldsymbol{p}_1}, n_{\boldsymbol{p}_2}, \ldots n_{\boldsymbol{p}_i}, \ldots\right\rangle
\end{aligned}$$

Where we have used the commutation relation between $\hat{n}$ and $\hat{a}^{\dagger}$. We can similarly look at the eigenvalues of $\hat{H}$:

$$\hat{H}\left|n_{\boldsymbol{p}_1}, n_{\boldsymbol{p}_2}, \ldots n_{\boldsymbol{p}_i}, \ldots\right\rangle = \sum_{\boldsymbol{k}} \omega_{\boldsymbol{p}_k} n_{\boldsymbol{p}_k} \left|n_{\boldsymbol{p}_1}, n_{\boldsymbol{p}_2}, \ldots n_{\boldsymbol{p}_i}, \ldots\right\rangle$$

This allows us to use the interpretation of the state as a state with $n_{\boldsymbol{p}_1}$ particles with momentum $\boldsymbol{p}_1$, and similarly for the rest of the momenta.

Let us now discuss the normalization of states. Each state is labelled by the occupation number in each momentum mode. In non-relativistic quantum mechanics, we normalize our states to 1. In QFT, there is no standard normalization, different textbooks use different normalizations, and likewise in research. This is because different normalizations allow for convenience depending on which types of computation you are performing. We will be following Peskin's convention (collider physics). The vacuum state is normalized to 1[20]:

$$\langle 0|0\rangle = 1$$

Single particle states are normalized via a factor of ω_p:

$$|\boldsymbol{p}\rangle = \sqrt{2\omega_p}\, \hat{a}_{\boldsymbol{p}}^{\dagger} |0\rangle$$

Which implies that

$$\langle \boldsymbol{p}|\boldsymbol{q}\rangle = 2\omega_p (2\pi)^3 \delta^3(\boldsymbol{p} - \boldsymbol{q})$$

[20] Everyone agrees on this one.

The reason for this convention is that had we normalized it to one, the states would not be Lorentz invariant, whereas this convention makes the states invariant. This is why people doing non-relativistic physics dislike this convention, since they never transform their states, and thus the Lorentz invariance of the single particle state is not useful.

This normalization changes some of the usual quantum mechanics formulae, such as the completeness relation, which picks up a factor of $1/\left(2\omega_p\right)$:

$$\hat{I}_{1\text{ particle}} = \int \frac{\mathrm{d}^3\boldsymbol{p}}{(2\pi)^3} \frac{|\boldsymbol{p}\rangle\langle\boldsymbol{p}|}{2\omega_p}$$

5.1. Causality

> *"You always have to make sure you can't go back in time and kill your mother before you're born. There's also the issue of whether you'd want to kill your mother..."*
>
> Chacko

Let us now discuss causality. Let us say $\hat{A}$ and $\hat{B}$ are two Hermitian operators, with eigenvalues :

$$\hat{A}\,|a_1\rangle = a_1\,|a_1\rangle$$
$$\hat{A}\,|a_2\rangle = a_2\,|a_2\rangle$$
$$\hat{B}\,|b_1\rangle = b_1\,|b_1\rangle$$
$$\hat{B}\,|b_2\rangle = b_2\,|b_2\rangle$$

Now suppose that $\left[\hat{A},\hat{B}\right] \neq 0$, the operators do not commute. If we start in state $|a_1\rangle$, and we measure $\hat{A}$ first, we will always get a_1. If we then measured $\hat{B}$, we could get either b_1 or b_2. However, had we started by measuring $\hat{B}$, we would get either b_1 or b_2, and then if we measure $\hat{A}$, we could get either a_1 or a_2. These are different results! The order of measurement matters. How does this relate to causality? Suppose we measure $\hat{A}$ and $\hat{B}$ at spacetime locations that are space-like separated. In this case, depending on the frame of reference, the time-ordering of events is not well-defined. We expect that operators at space-like separation must commute for a Lorentz invariant theory to be causal.

We have been discussing a real scalar field, $\hat{\phi}\left(x\right)$, this is a Hermitian operator, so in principle we could measure this operator at every point in space. Thus, it must be true that

$$\left[\hat{\phi}\left(x\right),\hat{\phi}\left(y\right)\right] = 0 \text{ if } x, y \text{ are space-like separated.}$$

Since our other operators are built out of these $\hat{\phi}$ operators, if this condition is satisfied, then the rest of the operators will also commute, and therefore respect causality.

Let us now show that this commutator is zero for space-like separated events in our theory. Let us define a new quantity:

$$D\left(x-y\right) = \left\langle 0 \left| \hat{\phi}\left(x\right)\hat{\phi}\left(y\right) \right| 0 \right\rangle$$

$$
\begin{aligned}
&= \left\langle 0 \middle| \int \frac{\mathrm{d}^3\boldsymbol{p}}{(2\pi)^3} \int \frac{\mathrm{d}^3\boldsymbol{p'}}{(2\pi)^3} \frac{1}{2\sqrt{\omega_p \omega_{p'}}} \hat{a}_{\boldsymbol{p}} e^{-ip\cdot x} \hat{a}^\dagger_{\boldsymbol{p'}} e^{ip'\cdot y} \middle| 0 \right\rangle \\
&= \int \frac{\mathrm{d}^3\boldsymbol{p}}{(2\pi)^3} \frac{1}{2\omega_p} e^{-ip\cdot(x-y)}
\end{aligned}
$$

We now claim that this object is a scalar under Lorentz transformations. This is because the vacuum is a scalar, and the field is also a scalar, and thus the definition of $D(x-y)$ forces it to be a scalar. Let us rewrite what we found to make this more clear:

$$
D(x-y) = \int \frac{d^4p}{(2\pi)^3} \underbrace{\delta\left(p^2 - m^2\right)}_{\delta\left((p^0)^2 - \omega_p^2\right)} \Theta\left(p^0\right) e^{-ip\cdot(x-y)}
$$

Where we have used the fact that

$$
\delta\left(x^2 - a^2\right) = \frac{1}{2a}\left(\delta(x+a) + \delta(x-a)\right)
$$

The $\Theta\left(p^0\right)$ restricts the delta function (which has support for $\pm p^0$) to only have support at the positive p^0. Now we can see that each term is Lorentz invariant, except for the $\Theta\left(p^0\right)$ term, but we see that this term will never change sign. Thus if we boost a positive energy state, we will always have it remain a positive energy state, we can never change the sign of the energy of the state.

Now consider the commutator we began with:

$$
\begin{aligned}
\left[\hat{\phi}(x), \hat{\phi}(y)\right] &= \int \frac{\mathrm{d}^3\boldsymbol{p}}{(2\pi)^3} \int \frac{\mathrm{d}^3\boldsymbol{p'}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \frac{1}{\sqrt{2\omega_{p'}}} \left[\left(\hat{a}_{\boldsymbol{p}} e^{-ip\cdot x} + \hat{a}^\dagger_{\boldsymbol{p}} e^{ip\cdot x}\right), \left(\hat{a}_{\boldsymbol{p'}} e^{-ip'\cdot y} + \hat{a}^\dagger_{\boldsymbol{p'}} e^{ip'\cdot y}\right)\right] \\
&= \int \frac{\mathrm{d}^3\boldsymbol{p}}{(2\pi)^3} \frac{1}{2\omega_p} \left[e^{-ip\cdot(x-y)} - e^{ip\cdot(x-y)}\right] \\
&= D(x-y) - D(y-x)
\end{aligned}
$$

We expect this to vanish, if x and y are space-like. We know that $D(x-y)$ is Lorentz invariant. Let us go to a frame where $x^0 = y^0$, which is always possible if x and y are space-like (by definition, there exists a Lorentz frame in which they occur at the same time). Then:

$$
\begin{aligned}
\left[\hat{\phi}(x), \hat{\phi}(y)\right] &= \int \frac{\mathrm{d}^3\boldsymbol{p}}{(2\pi)^3} \frac{1}{2\omega_p} \left[e^{-i\boldsymbol{p}\cdot(\boldsymbol{x}-\boldsymbol{y})} - e^{i\boldsymbol{p}\cdot(\boldsymbol{x}-\boldsymbol{y})}\right] \\
&= 0
\end{aligned}
$$

Where we note that the integrand is odd under $\boldsymbol{p} \to -\boldsymbol{p}$, and therefore the integral is zero. From this, we have that our theory is indeed causal.

5.2. Klein-Gordon Propagator

A very useful quantity is the Feynman propagator, defined as:

$$
D_F(x-y) = \begin{cases} D(x-y) & x^0 > y^0 \\ D(y-x) & x^0 < y^0 \end{cases}
$$

Recall that

$$D(x-y) = \int \frac{\mathrm{d}^3 p}{(2\pi)^3} \frac{1}{2\omega_p} e^{-ip\cdot(x-y)}$$

A useful representation of $D_F(x-y)$ is as a contour integral:

$$D_F(x-y) = \int_C \frac{\mathrm{d}^4 p}{(2\pi)^4} \frac{i}{p^2 - m^2} e^{-ip\cdot(x-y)} \tag{14}$$

Where p^0 is integrated over a specific contour, the real line, avoiding the two poles, $-\omega_p$ and ω_p, specifically going under $-\omega_p$ and over $+\omega_p$, as shown in Figure 1.

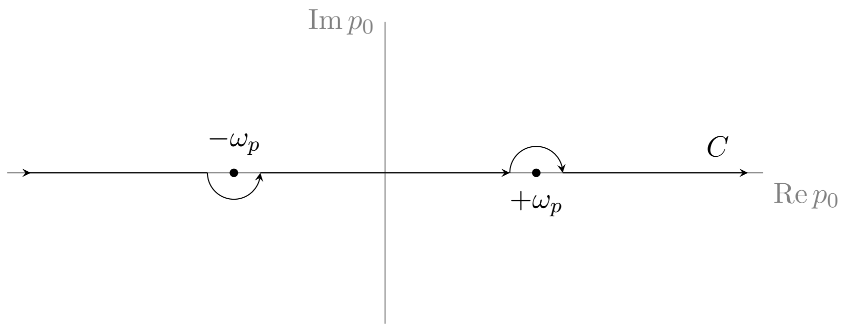
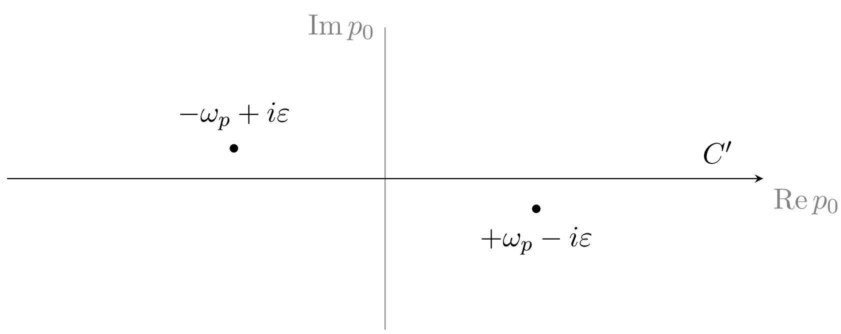


Figure 1: Choice of contour for the Feynman propagator $D_F(x-y)$ (14).

For $x^0 > y^0$, we close the contour below, picking up a factor of $D(x-y)$, while for $x^0 < y^0$, we close the contour above, picking up a factor of $D(y-x)$.

What is the advantage of writing the propagator this way? If we write it this way, it looks like a Lorentz scalar. On the other hand, there is another, slightly different way we write the propagator. Imagine we integrate along the real line but displace the poles by $i\varepsilon$, as shown in Figure 2.

Figure 2: Alternative method of writing the Feynman propagator, via the $i\varepsilon$ method of shifting the poles.



The corresponding integral is then:

$$D_F(x-y) = \int_{C'} \frac{d^4 p}{(2\pi)^4} \frac{i}{p^2 - m^2 + i\varepsilon} e^{-ip\cdot(x-y)}$$

Where we close the contour in the same way, based on which of x^0 or y^0 is larger.

Another way of writing the Feynman propagator is via the time ordering operator, $\hat{T}$:

$$D_F(x-y) = \langle 0|\hat{T}\left[\hat{\phi}(x)\hat{\phi}(y)\right]|0\rangle$$

Where we define the time ordered product as:

$$\hat{T}\left[\hat{\phi}(x)\hat{\phi}(y)\right] = \begin{cases} \hat{\phi}(x)\hat{\phi}(y) & x^0 > y^0 \\ \hat{\phi}(y)\hat{\phi}(x) & y^0 > x^0 \end{cases}$$

5.3. Complex Klein-Gordon Field

Let us now consider a theory of two real scalar fields, with the same mass. The Lagrangian is then just the sum of two Klein-Gordon Lagrangians, one for each field:

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi_1\partial^\mu\phi_1 + \frac{1}{2}\partial_\mu\phi_2\partial^\mu\phi_2 - \frac{1}{2}m^2\phi_1^2 + \frac{1}{2}m^2\phi_2^2$$

One way that this could be handled is by quantizing both fields, and then following the same steps we did in the single field case. Since the fields never talk to each other, this could be done. Instead, let us define a new field,

$$\begin{aligned} \phi &= \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2) \\ \phi^\dagger &= \frac{1}{\sqrt{2}}(\phi_1 - i\phi_2) \end{aligned}$$

Which then leaves the Lagrangian as:

$$\mathcal{L} = \partial_\mu\phi^\dagger\partial^\mu\phi - m^2\phi^\dagger\phi$$

Where ϕ is now a *complex* scalar field. The canonical momenta conjugate to ϕ and $\phi^\dagger$ are then:

$$\begin{aligned} \pi_\phi &= \dot{\phi}^\dagger \\ \pi_{\phi^\dagger} &= \dot{\phi} \end{aligned}$$

Which leads to the Hamiltonian:

$$\begin{aligned} \mathcal{H} &= \pi_\phi\dot{\phi} + \pi_{\phi^\dagger}\dot{\phi}^\dagger - \mathcal{L} \\ &= \dot{\phi}^\dagger\dot{\phi} + (\boldsymbol{\nabla}\phi)^\dagger\cdot(\boldsymbol{\nabla}\phi) + m^2\phi^\dagger\phi \end{aligned}$$

We can now quantize the theory, by imposing the canonical commutation relations, and then expand $\hat{\phi}$ in plane waves:

$$\hat{\phi}(\boldsymbol{x},t) = \int\frac{\mathrm{d}^3p}{(2\pi)^3}\frac{1}{\sqrt{2\omega_p}}\left(\hat{a}_{\boldsymbol{p}}e^{-ip\cdot x} + \hat{b}_{\boldsymbol{p}}e^{ip\cdot x}\right)$$

Note that in the real Klein-Gordon field, we had a single $\hat{a}$, but in this case, we have $\hat{a}$ and $\hat{b}$, which are independent. The commutation relations can then be found:

$$\left[\hat{a}_{\boldsymbol{p}}, \hat{a}^\dagger_{\boldsymbol{p}'}\right] = (2\pi)^3\delta^3(\boldsymbol{p}-\boldsymbol{p}')$$

$$\left[\hat{b}_{\boldsymbol{p}}, \hat{b}^{\dagger}_{\boldsymbol{p}'}\right] = (2\pi)^3 \delta^3 \left(\boldsymbol{p} - \boldsymbol{p}'\right)$$

In this basis, the Hamiltonian is given by (after normal ordering)

$$\hat{H} = \int \frac{\mathrm{d}^3 p}{(2\pi)^3} \left(\hat{a}^{\dagger}_{\boldsymbol{p}} \hat{a}_{\boldsymbol{p}} + \hat{b}^{\dagger}_{\boldsymbol{p}} \hat{b}_{\boldsymbol{p}}\right) \omega_{\boldsymbol{p}}$$

If we look at our Lagrangian, we can see that there is an obvious symmetry, under rephasing:

$$\phi \to \phi e^{i\alpha}$$

The conserved charge (after it is promoted to an operator in the quantum theory), is given by

$$\hat{Q} = \text{const} \int \frac{\mathrm{d}^3 \boldsymbol{p}}{(2\pi)^3} \left(\hat{a}^{\dagger}_{\boldsymbol{p}} \hat{a}_{\boldsymbol{p}} - \hat{b}^{\dagger}_{\boldsymbol{p}} \hat{b}_{\boldsymbol{p}}\right)$$

Now noting that this is the difference of the two number operators, this tells us that the conserved charge is based on the difference between the number of a particles and b particles, which leads to the natural labelling of b particles as antiparticles.

6. Dirac Theory

Let us now consider the spin-1/2 case. We will be second quantizing the Dirac field. Let us begin with the classical Lagrangian density:

$$\mathcal{L} = \overline{\psi} \left(i\gamma^{\mu} \partial_{\mu} - m\right) \psi$$

Which leads us to the Dirac equation:

$$\left(i\gamma^{\mu} \partial_{\mu} - m\right) \psi = 0$$

We can find the canonical momentum conjugate to ψ:

$$\pi = i\psi^{\dagger}$$

We can then write out the Hamiltonian density:

$$\begin{aligned}\mathcal{H} &= \pi\dot{\psi} - \mathcal{L} \\ &= -i\overline{\psi} \left(\boldsymbol{\gamma} \cdot \boldsymbol{\nabla}\right) \psi + m\overline{\psi}\psi\end{aligned}$$

Everything so far has been classical, we have introduced no quantum mechanics yet. Now let us impose the canonical anticommutation relations, because we want this to be a fermionic theory. The anticommutation relations are of the form:

$$\{\hat{\psi}_a \left(\boldsymbol{x}, t\right), \hat{\psi}^{\dagger}_b \left(\boldsymbol{x}', t\right)\} = \delta^3 \left(\boldsymbol{x} - \boldsymbol{x}'\right) \delta_{ab}$$

Note the indices on the ψ operators, which are needed because ψ is a four-component object. Now that we have our anticommutation relations, and our Hamiltonian operator:

$$\hat{H} = \int \mathrm{d}^3 \boldsymbol{x}\, \hat{\mathcal{H}}$$

We can use the Heisenberg equation of motion:

$$\left[\hat{H}, \hat{\psi}\right] = -i\frac{\partial\hat{\psi}}{\partial t}$$

From which we find that[21]

$$i\gamma^\mu\partial_\mu\hat{\psi} - m\hat{\psi} = 0$$

This is the same result as the classical Dirac equation, but $\hat{\psi}$ is now an operator. As usual, we want to go to a basis in which the Hamiltonian is diagonal, so we expand $\hat{\psi}$ and $\hat{\psi}^\dagger$ in plane waves:

$$\begin{aligned}\hat{\psi}\left(\boldsymbol{x}, t\right) &= \int \frac{\mathrm{d}^3 p}{\left(2\pi\right)^3}\frac{1}{\sqrt{2\omega_p}}\sum_{s=1}^{2}\left[\hat{a}^s_{\boldsymbol{p}}\hat{u}^s\left(\boldsymbol{p}\right)e^{-ip\cdot x} + \hat{b}^{s\dagger}_{\boldsymbol{p}}v^s\left(\boldsymbol{p}\right)e^{ip\cdot x}\right]\\ \hat{\psi}^\dagger\left(\boldsymbol{x}, t\right) &= \int \frac{\mathrm{d}^3 p}{\left(2\pi\right)^3}\frac{1}{\sqrt{2\omega_p}}\sum_{s}\left[\hat{b}^s_{\boldsymbol{p}}\hat{\overline{v}}^s\left(\boldsymbol{p}\right)e^{-ip\cdot x} + \hat{a}^{s\dagger}_{\boldsymbol{p}}\overline{u}^s\left(\boldsymbol{p}\right)e^{ip\cdot x}\right]\end{aligned}$$

Where we use the Dirac plane wave solutions. We can now find the commutation relations between our 4 operators, $\hat{a}$, $\hat{a}^\dagger$, $\hat{b}$, and $\hat{b}^\dagger$. To do this, we can solve for the operators in terms of the fields:

$$\begin{aligned}\hat{b}^{r\dagger}_{\boldsymbol{p}'} &= \frac{1}{\sqrt{2\omega_{p'}}}\int \mathrm{d}^3\boldsymbol{x}\, v^{r\dagger}\left(\boldsymbol{p}'\right)\hat{\psi}\left(\boldsymbol{x}\right)e^{-ip'\cdot x}\\ \hat{a}^{r}_{\boldsymbol{p}'} &= \frac{1}{\sqrt{2\omega_{p'}}}\int \mathrm{d}^3\boldsymbol{x}\, u^{r\dagger}\left(\boldsymbol{p}'\right)\hat{\psi}\left(\boldsymbol{x}\right)e^{ip'\cdot x}\end{aligned}$$

Where we use the plane wave solution orthogonality condition for the Dirac equation. From these, we can see some of the anticommutation relations:

$$\{\hat{a}, \hat{a}\} = \{\hat{b}, \hat{b}\} = \{\hat{a}, \hat{b}^\dagger\} = \{\hat{a}^\dagger, \hat{b}\} = 0$$

And after a bit of algebra, one can find that

$$\{\hat{a}, \hat{b}\} = 0$$

And finally, one can show that

$$\left\{\hat{a}^r_{\boldsymbol{p}}, \hat{a}^{s\dagger}_{\boldsymbol{p}'}\right\} = \left(2\pi\right)^3\delta^3\left(\boldsymbol{p} - \boldsymbol{p}'\right)\delta^{rs}$$

And similarly for the $\hat{b}$ and $\hat{b}^\dagger$ anticommutator. We can now write out the Hamiltonian:

$$\begin{aligned}\hat{H} &= \int \mathrm{d}^3\boldsymbol{x}\left[-\hat{\overline{\psi}}\left(i\gamma^i\partial_i - m\right)\hat{\psi}\right]\\ &= \int \mathrm{d}^3\boldsymbol{x}\, i\hat{\psi}^\dagger\partial_0\hat{\psi}\end{aligned}$$

Where we have insert the Dirac equation, since we know it is satisfied. This leaves us with the Hamiltonian:

$$\hat{H} = \int \frac{\mathrm{d}^3\boldsymbol{p}}{\left(2\pi\right)^3}\omega_p\sum_r\left(\hat{a}^{r\dagger}_{\boldsymbol{p}}\hat{a}^r_{\boldsymbol{p}} - \hat{b}^r_{\boldsymbol{p}}\hat{b}^{r\dagger}_{\boldsymbol{p}}\right)$$

[21] We skip this derivation, but a useful identity is $\left[\hat{A}\hat{B}, \hat{C}\right] = \hat{A}\{\hat{B}, \hat{C}\} - \{\hat{A}, \hat{C}\}\hat{B}$.

Now using the anticommutation relations to swap the $\hat{b}$ and $\hat{b}^\dagger$, picking up a sign and an infinite constant from the delta function (which we throw away):

$$\hat{H} = \int \frac{\mathrm{d}^3\boldsymbol{p}}{(2\pi)^3}\omega_p \sum_r \left(\hat{a}^{r\dagger}_{\boldsymbol{p}}\hat{a}^r_{\boldsymbol{p}} + \hat{b}^{r\dagger}_{\boldsymbol{p}}\hat{b}^r_{\boldsymbol{p}}\right) + \cancel{\text{const.}}$$

Analogously to the complex scalar field case, we can assign the label of particles to the $\hat{a}$s, and antiparticles to the $\hat{b}$s, and they correspond to positive and negative energy solutions to the Dirac equation respectively.

We can define the momentum $\hat{p}_i$:

$$\begin{aligned}\hat{p}_i &= \int \mathrm{d}^3\boldsymbol{x}\,\hat{T}^{0i}\\ &= -\int \mathrm{d}^3\boldsymbol{x}\,\hat{\pi}\partial_i\hat{\psi}\\ &= \int \frac{\mathrm{d}^3\boldsymbol{p}}{(2\pi)^3}\boldsymbol{p}\sum_r \left[\hat{a}^{r\dagger}_{\boldsymbol{p}}\hat{a}^r_{\boldsymbol{p}} + \hat{b}^{r\dagger}_{\boldsymbol{p}}\hat{b}^r_{\boldsymbol{p}}\right] + \cancel{\text{const.}}\end{aligned}$$

Thus we have some expectation that $\hat{a}^{r\dagger}_{\boldsymbol{p}}\hat{a}^r_{\boldsymbol{p}}$ corresponds to the number of particles of type a:

$$\begin{aligned}\hat{N}^a &= \sum_r \int \frac{\mathrm{d}^3\boldsymbol{p}}{(2\pi)^3}\hat{a}^{r\dagger}_{\boldsymbol{p}}\hat{a}^r_{\boldsymbol{p}}\\ \hat{N}^b &= \sum_r \int \frac{\mathrm{d}^3\boldsymbol{p}}{(2\pi)^3}\hat{b}^{r\dagger}_{\boldsymbol{p}}\hat{b}^r_{\boldsymbol{p}}\end{aligned}$$

Now we can note that $\hat{N}^a$ is a positive definite operator, and it has the following commutation relation with the $\hat{a}$ operator:

$$\left[\hat{N}^a, \hat{a}^r_{\boldsymbol{p}}\right] = -\hat{a}^r_{\boldsymbol{p}}$$

Now we use the same argument that we have seen multiple times, $\hat{N}^a$ must have an eigenvector $|0_a\rangle$ with eigenvalue 0 such that

$$\hat{a}^r_{\boldsymbol{p}}|0_a\rangle = 0,\ \forall r, \forall \boldsymbol{p}$$

We can now define another operator:

$$\hat{n}^{ar}_{\ \ \boldsymbol{p}} = \frac{1}{(2\pi)^3}\left(\hat{a}^{r\dagger}_{\boldsymbol{p}}\hat{a}^r_{\boldsymbol{p}}\right)$$

This is also positive definite, and also has 0 as one of its eigenvalues. If we now square this operator:

$$\begin{aligned}\hat{n}^{ar}_{\ \ \boldsymbol{p}}\hat{n}^{ar}_{\ \ \boldsymbol{p}} &= \frac{1}{(2\pi)^6}\left(\hat{a}^{r\dagger}_{\boldsymbol{p}}\hat{a}^r_{\boldsymbol{p}}\right)\left(\hat{a}^{r\dagger}_{\boldsymbol{p}}\hat{a}^r_{\boldsymbol{p}}\right)\\ &= \hat{n}^{ar}_{\ \ \boldsymbol{p}}\delta^3(0)\end{aligned}$$

The eigenvalues of $\hat{n}^{ar}_{\ \ \boldsymbol{p}}$ are either zero or $\delta^3(0)$. All of the $\hat{n}^a$ operators commute with each other, and therefore they are all simultaneously diagonalizable. An arbitrary eigenstate of $\hat{n}^{ar}_{\ \ \boldsymbol{p}}$ can be labelled by $|n^{ar}_{\ \ \boldsymbol{p}}\rangle$ and has the property that:

$$\hat{n}^{ar}_{\ \ \boldsymbol{p}}|n^{ar}_{\ \ \boldsymbol{p}}\rangle = n^{ar}_{\ \ \boldsymbol{p}}\delta^3(0)|n^{ar}_{\ \ \boldsymbol{p}}\rangle$$

Where $n^{ar}_{\boldsymbol{p}} = 0, 1$.

In an identical fashion, we can say the same for $\hat{n}^{br}_{\boldsymbol{p}}$:

$$\hat{n}^{br}_{\boldsymbol{p}} = \frac{1}{(2\pi)^3}\hat{b}^{r\dagger}_{\boldsymbol{p}}\hat{b}^{r}_{\boldsymbol{p}}$$

Which, following the same chain of logic, gives us:

$$\hat{n}^{br}_{\boldsymbol{p}} \left|n^{br}_{\boldsymbol{p}}\right\rangle = n^{br}_{\boldsymbol{p}}\delta^3(0)\left|n^{br}_{\boldsymbol{p}}\right\rangle$$

Recall that the as denote particles, the bs denote antiparticles, r denotes the spin ($r = 1, 2$), and $\boldsymbol{p}$ denotes the momentum. We can put these all together, since they are simultaneously diagonalizable:

$$\left|n^{a1}_{\boldsymbol{p}_1}, n^{a1}_{\boldsymbol{p}_2}, \ldots, n^{a2}_{\boldsymbol{p}_1}, n^{a2}_{\boldsymbol{p}_2}, \ldots, n^{b1}_{\boldsymbol{p}_1}, n^{b1}_{\boldsymbol{p}_2}, \ldots, n^{b2}_{\boldsymbol{p}_1}, n^{b2}_{\boldsymbol{p}_2}, \ldots\right\rangle$$

This covers all momenta, both particles and antiparticles, as well as both spin configurations. All elements of this ket are either 0 or 1. This is a general eigenstate of $\hat{N}^a$, $\hat{N}^b$, and $\hat{H}$. We can define a total number operator, $\hat{N} = \hat{N}^a + \hat{N}^b$, and then look at the eigenvalues of several operators:

$$\begin{aligned}
\hat{N}\left|n^{a1}_{\boldsymbol{p}_1}, n^{a1}_{\boldsymbol{p}_2}, \ldots, n^{a2}_{\boldsymbol{p}_1}, n^{a2}_{\boldsymbol{p}_2}, \ldots, n^{b1}_{\boldsymbol{p}_1}, n^{b1}_{\boldsymbol{p}_2}, \ldots, n^{b2}_{\boldsymbol{p}_1}, n^{b2}_{\boldsymbol{p}_2}, \ldots\right\rangle &= \sum_{r,k} n^{ar}_{\boldsymbol{p}_k} + n^{br}_{\boldsymbol{p}_k} \\
\hat{H}\left|n^{a1}_{\boldsymbol{p}_1}, n^{a1}_{\boldsymbol{p}_2}, \ldots, n^{a2}_{\boldsymbol{p}_1}, n^{a2}_{\boldsymbol{p}_2}, \ldots, n^{b1}_{\boldsymbol{p}_1}, n^{b1}_{\boldsymbol{p}_2}, \ldots, n^{b2}_{\boldsymbol{p}_1}, n^{b2}_{\boldsymbol{p}_2}, \ldots\right\rangle &= \sum_{r,k} \omega_{p_k}\left(n^{ar}_{\boldsymbol{p}_k} + n^{br}_{\boldsymbol{p}_k}\right) \\
\hat{\boldsymbol{p}}\left|n^{a1}_{\boldsymbol{p}_1}, n^{a1}_{\boldsymbol{p}_2}, \ldots, n^{a2}_{\boldsymbol{p}_1}, n^{a2}_{\boldsymbol{p}_2}, \ldots, n^{b1}_{\boldsymbol{p}_1}, n^{b1}_{\boldsymbol{p}_2}, \ldots, n^{b2}_{\boldsymbol{p}_1}, n^{b2}_{\boldsymbol{p}_2}, \ldots\right\rangle &= \sum_{r,k} \boldsymbol{p}_k\left(n^{ar}_{\boldsymbol{p}_k} + n^{br}_{\boldsymbol{p}_k}\right)
\end{aligned}$$

This allows the interpretation of $n^{ar}_{\boldsymbol{p}_k}$ ($n^{br}_{\boldsymbol{p}_k}$) as the number of particles (antiparticles) with spin r and momentum $\boldsymbol{p}_k$ in the state. $\hat{N}^a$ measures the number of particles, $\hat{N}^b$ the number of antiparticles, and $\hat{N}$ the sum of the two.

The state with no particles is the vacuum state, labelled $|0\rangle$. We normalize the vacuum to 1:

$$\langle 0|0\rangle = 1$$

We can construct single particle states via a creation operator:

$$|\boldsymbol{p}, s\rangle = \sqrt{2\omega_p}\,\hat{a}^{s\dagger}_{\boldsymbol{p}}|0\rangle$$

The advantage of this normalization factor is that the inner product is Lorentz invariant:

$$\langle \boldsymbol{p}, r|\boldsymbol{p}', s\rangle = 2\omega_p (2\pi)^3\,\delta^3\left(\boldsymbol{p} - \boldsymbol{p}'\right)\delta^{rs}$$

How do we know that we have a particle and antiparticle, rather than just two types of particles? To show this, we first note that the Dirac theory is invariant under rephasing:

$$\psi \rightarrow e^{i\alpha}\psi$$

The corresponding Noether charge $\hat{Q}$ will show us that the particles have a different charge when compared to the other type of particle (We will see this when we discuss coupling E&M to matter, in QED).

6.1. Weyl and Majorana Spinors

Let us return to Dirac's logic for introducing a matrix equation to deal with the issues with the Klein-Gordon equation. Dirac took the square root of the following equation:

$$E^2 = c^2p^2 + m^2c^4$$

Which left him with

$$E = c\boldsymbol{\alpha} \cdot \boldsymbol{p} + mc^2\beta$$

As we have seen, the α^i and β anticommute, and there is no such set of 2×2 matrices, so we went to 4×4 matrices. Suppose we instead consider a massless particle. In this case, we just need 3 such matrices, and the Pauli matrices satisfy our requirements:

$$\alpha^i = \sigma^i$$

However, note that the negatives of the Pauli matrices also work:

$$\alpha^i = -\sigma^i$$

Let us consider the Dirac equation in this massless case:

$$E\psi = \pm c\boldsymbol{\sigma} \cdot \boldsymbol{p}\psi$$

Making this quantum mechanical:

$$i\frac{\partial \psi}{\partial t} = \mp i\boldsymbol{\sigma} \cdot \boldsymbol{\nabla}\psi$$

Which leads us to

$$i\left(\partial_0\psi \pm \boldsymbol{\sigma} \cdot \boldsymbol{\nabla}\psi\right) = 0$$

Note that ψ in this equation is a 2-component object, not a 4-component object. Let us take the Dirac equation that we already have, and take the massless limit, and see if we can find a connection to the above equation. The Dirac equation (7) is:

$$i\left(\gamma^\mu\partial_\mu - m\right)\psi = 0$$

Which in the massless limit is:

$$i\gamma^\mu\partial_\mu\psi = 0$$

Now suppose we break ψ (4-component) into a left-handed and right-handed objects (each 2-component objects):

$$\psi = \begin{pmatrix}\psi_L \\ \psi_R\end{pmatrix}$$

We can then write the massless Dirac equation as two independent equations, one for ψ_L and one for ψ_R:

$$i\left(\partial_0 - \boldsymbol{\sigma} \cdot \boldsymbol{\nabla}\right)\psi_L = 0$$

$$i\left(\partial_0+\boldsymbol{\sigma}\cdot\boldsymbol{\nabla}\right)\psi_R=0$$

These are known as the Weyl equations, and the spinors $\psi_{L/R}$ are known as Weyl spinors. Now consider the two component spinor χ:

$$\chi=i\sigma^2\psi_L^*$$

Let us now insert this into the second Weyl equation:

$$\begin{aligned}i\left(\partial_0+\boldsymbol{\sigma}\cdot\boldsymbol{\nabla}\right)\chi&=i\sigma^2\underbrace{\left(i\left(\partial_0-\boldsymbol{\sigma}^*\cdot\boldsymbol{\nabla}\right)\right)\psi_L^*}_{0}\\&=0\end{aligned}$$

Where we note that we have recovered the complex conjugate of the first Weyl equation, which must be zero, and therefore the whole expression is zero. Thus, we have shown that χ satisfies the same equation of motion as ψ_R. Thus we have shown that using ψ_L, we can construct an object that transforms like ψ_R under Lorentz transformations.

We can then define a 4-component spinor

$$\begin{pmatrix}\psi_L\\\chi\end{pmatrix}=\begin{pmatrix}\psi_L\\i\sigma^2\psi_L^*\end{pmatrix}$$

This is known as a Majorana spinor. When we usually define a 4-component object, the objects are all independent, but in this case, we only have 2 independent components, rather than 4. We constructed this object for the massless case, but we can in fact define this object in the massive case, and write a mass term in our Lagrangian. We define a general Majorana spinor as:

$$\psi_m=\begin{pmatrix}\psi_L\\i\sigma^2\psi_L^*\end{pmatrix}$$

Now we can define a mass term:

$$\overline{\psi_m}\psi_m=-\psi_L^T\left(i\sigma_2\right)\psi_L+\psi_L^\dagger\left(i\sigma^2\right)\psi_L^*$$

In such a theory, we can see that we no longer have a conserved charge, and the particles in the theory are their own antiparticles, since particles in the theory have no charge.

6.2. Dirac Propagator

We previously discussed the Klein-Gordon propagator (5.2), let us see what this looks like in the Dirac theory. In this case, we have two different quantities we could look at:

$$\langle 0|\hat{\psi}_a\left(x\right)\hat{\overline{\psi}}_b\left(y\right)|0\rangle \qquad \langle 0|\hat{\overline{\psi}}_b\left(y\right)\hat{\psi}_a\left(x\right)|0\rangle$$

Let us first consider the first of these two. We have a $\hat{a}$ in the first term, and a $\hat{a}^\dagger$ in the second term, which is the only surviving term:

$$\begin{aligned}\langle 0|\hat{\psi}_a\left(x\right)\hat{\overline{\psi}}_b\left(y\right)|0\rangle&=\int\frac{\mathrm{d}^3p}{\left(2\pi\right)^3}\frac{1}{2\omega_p}\sum_r u_a^r\left(\boldsymbol{p}\right)\overline{u}_b^r\left(\boldsymbol{p}\right)e^{-ip\cdot(x-y)}\\&=\int\frac{\mathrm{d}^3\boldsymbol{p}}{\left(2\pi\right)^3}\frac{1}{2\omega_p}\left(\not{p}+m\right)_{ab}e^{-ip\cdot(x-y)}\end{aligned}$$

If we instead consider the second of the two expressions, we have something that looks almost the same thing:

$$\langle 0|\hat{\bar{\psi}}_b(y)\,\hat{\psi}_a(x)\,|0\rangle = \int \frac{\mathrm{d}^3\boldsymbol{p}}{(2\pi)^3}\frac{1}{2\omega_p}\left(\not{p}-m\right)_{ab}e^{-ip\cdot(y-x)}$$

Thus we see that the ordering of the fields produces different results. Thus we define the Feynman propagator:

$$S_F(x-y) = \begin{cases}\langle 0|\hat{\psi}(x)\,\hat{\bar{\psi}}(y)\,|0\rangle & x^0 > y^0\\ -\langle 0|\hat{\bar{\psi}}(y)\,\hat{\psi}(x)\,|0\rangle & y^0 > x^0\end{cases}$$

Note that we pick up a minus sign, since these are fermionic fields, when we swap them we must pick up a minus sign. We can condense this notation via the time ordering operator, $\hat{T}$:

$$S_F(x-y) = \langle 0|\hat{T}\left[\hat{\psi}(x)\,\hat{\bar{\psi}}(y)\right]|0\rangle$$

In order to make the Lorentz invariance of this object more clear, we usually write the Feynman propagator for the Dirac theory as:

$$S_F(x-y) = \int \frac{\mathrm{d}^4p}{(2\pi)^4}\frac{i\left(\not{p}+m\right)}{p^2-m^2+i\varepsilon}e^{-ip\cdot(x-y)}$$

Where the integral is along the real line, and the $i\varepsilon$ shifts the poles slightly off the real line. Comparing this to the scalar Feynman propagator, we have this extra factor of $\not{p}+m$ in the integrand.

7. Free Electromagnetism

So far we have quantized the spin-0 and spin-1/2 theories, let us now move to the spin-1 case. Let us begin with the classical E &M Lagrangian:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$$

Where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. The classical equation of motion from this equation is:

$$\partial_\mu F^{\mu\nu} = 0$$

We can find the canonical momentum conjugate to A^μ:

$$\begin{aligned}\pi^\mu &= \frac{\partial\mathcal{L}}{\partial\dot{A}_\mu}\\ &= -F^{\mu 0}\end{aligned}$$

This is the first place that we run into a problem. For the case where $\mu = i$, we have well-defined conjugate momenta. However, if $\mu = 0$, then $\pi^0 = -F^{00}$, which is zero, there is no canonical momentum conjugate to A^0, even in the classical theory! The reason for this is the gauge redundancy in classical electromagnetism, if we know the vector potential and the scalar potential at a point in space, we can specify $\boldsymbol{E}$ and $\boldsymbol{B}$, which means that the 6 variables in $\boldsymbol{E}$ and $\boldsymbol{B}$ have some degree of redundancy, which is removed when we consider A^μ. However, we still have gauge invariance for A^μ, we can add $\partial_\mu \Lambda(x)$ for a scalar function $\Lambda(x)$ that provides the same physical fields. This is what we are finding, treating the 4 components of A^μ as independent variables exposes this redundancy. If we want to sensibly quantize the theory, we will have to account for this redundancy.

From here, there are several things that we can do. One choice is to get rid of A^0, and work with just the vector potential. This is known as the Coulomb gauge. However, in doing so, we will violate Lorentz invariance during intermediate steps of our derivations. Instead, we will choose to use a covariant gauge[22]. The covariant (or Lorenz) gauge is defined by the choice:

$$\partial_\mu A^\mu = 0$$

We then subtract from the Lagrangian a term which, in our choice of gauge, is zero:

$$\mathcal{L} \to \mathcal{L} - \frac{1}{2}\left(\partial_\mu A^\mu\right)^2$$

Our new Lagrangian is then given by:

$$\mathcal{L}' = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2}\left(\partial_\mu A^\mu\right)^2$$

If we now expand our $F_{\mu\nu}$s:

$$-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} = -\frac{1}{2}\left(\partial_\mu A^\nu\right)\left(\partial_\mu A^\nu\right) + \frac{1}{2}\left(\partial_\nu A^\mu\right)\left(\partial_\mu A^\nu\right)$$

Now integrating the second term by parts in the action integral, we find $\frac{1}{2}\left(\partial_\mu A^\mu\right)^2$, which exactly cancels the additional term that we subtracted from the Lagrangian. Thus, when the dust settles, we are left with:

$$S' = \int \mathrm{d}^4x\, \mathcal{L}_{\mathrm{EM}}$$

Where

$$\mathcal{L}_{\mathrm{EM}} = -\frac{1}{2}\left(\partial^\mu A^\nu\right)\left(\partial_\mu A_\nu\right)$$

This looks exactly like the kinetic term for the scalar field case, but we have 4 indices, so we have 4 kinetic terms. However, we have a minus sign out front, which was not there in the scalar field case. However, we note that the spatial indices will have the correct sign.

With this Lagrangian, we essentially have 4 copies of the scalar field, but one of them has the wrong sign in the kinetic energy. Once we choose this Lagrangian, we can define a sensible canonical momentum for each component:

$$\pi^\mu = -\dot{A}^\mu$$

And the field equation is:

$$\partial_\nu \partial^\nu A^\mu = 0$$

[22] This choice is taken from the book by Mandl and Shaw.

This is equivalent to:

$$\begin{aligned}\partial_\mu F^{\mu\nu} &= \partial_\mu \left(\partial^\mu A^\nu - \partial^\nu A^\mu\right) \\ \partial_\mu \partial^\mu A^\nu &= 0\end{aligned}$$

Since the second term $\partial_\mu \partial^\nu A^\mu$ is zero.

Even with the Lorenz condition $\partial_\mu A^\mu = 0$, there is residual gauge invariance:

$$A^\mu \to A^\mu + \partial^\mu f$$

which leaves the equations of motion and the Lorenz condition unchanged, if $\Box f = 0$. This gauge invariance does not end up causing a problem.

We can proceed in the same way we did with the scalar case, we can expand in plane waves:

$$A_\mu\left(x\right) = \int \frac{\mathrm{d}^3\boldsymbol{p}}{\left(2\pi\right)^3} \frac{1}{\sqrt{2\omega_p}} \sum_{r=0}^{3} \left[\hat{a}^r_{\boldsymbol{p}} \varepsilon^r_\mu\left(\boldsymbol{p}\right) e^{-ip\cdot x} + \hat{a}^{r\dagger}_{\boldsymbol{p}} \varepsilon^r_\mu\left(\boldsymbol{p}\right) e^{ip\cdot x}\right]$$

Where ε^r_μ are polarization vectors, they satisfy orthonormality and completeness relations:

$$\begin{aligned}\varepsilon^r_\mu\left(\boldsymbol{p}\right) \varepsilon^{s\mu}\left(\boldsymbol{p}\right) &= -\xi_r \delta^{rs} \\ \sum_r \xi_r \varepsilon^r_\mu\left(\boldsymbol{p}\right) \varepsilon^{r\nu}\left(\boldsymbol{p}\right) &= -\eta^{\mu\nu}\end{aligned}$$

Where $\xi_0 = -1$, $\xi_1 = \xi_2 = \xi_3 = 1$.

We make a specific choice of basis:

$$\begin{aligned}\varepsilon^0_\mu\left(\boldsymbol{p}\right) &= \left(1, 0, 0, 0\right) \\ \varepsilon^r_\mu\left(\boldsymbol{p}\right) &= \left(0, \boldsymbol{\varepsilon}^r\left(\boldsymbol{p}\right)\right)\end{aligned}$$

With this definition, we have that $\boldsymbol{\varepsilon}^1$ and $\boldsymbol{\varepsilon}^2$ are orthogonal to $\boldsymbol{p}$, and $\boldsymbol{\varepsilon}^0$ and $\boldsymbol{\varepsilon}^3$ are unphysical.

Now let us quantize the theory, by imposing the canonical commutation relations:

$$\begin{aligned}\left[\hat{A}^\mu\left(\boldsymbol{x}, t\right), \hat{A}^\nu\left(\boldsymbol{x}', t\right)\right] &= 0 \\ \left[\dot{\hat{A}}^\mu\left(\boldsymbol{x}, t\right), \dot{\hat{A}}^\nu\left(\boldsymbol{x}', t\right)\right] &= 0 \\ \left[\hat{A}^\mu\left(\boldsymbol{x}, t\right), \dot{\hat{A}}^\nu\left(\boldsymbol{x}', t\right)\right] &= -i\eta^{\mu\nu}\delta^3\left(\boldsymbol{x} - \boldsymbol{x}'\right)\end{aligned}$$

We can obtain commutation relations between $\hat{a}$ and $\hat{a}^\dagger$:

$$\begin{aligned}\left[\hat{a}^r_{\boldsymbol{p}}, \hat{a}^{s\dagger}_{\boldsymbol{p}'}\right] &= \xi_r \delta^{rs} \left(2\pi\right)^3 \left(\boldsymbol{p} - \boldsymbol{p}'\right) \\ \left[\hat{a}^r_{\boldsymbol{p}}, \hat{a}^s_{\boldsymbol{p}'}\right] &= \left[\hat{a}^{r\dagger}_{\boldsymbol{p}'}, \hat{a}^{s\dagger}_{\boldsymbol{p}'}\right] = 0\end{aligned}$$

Note the ξ_r, which picks up a sign for the spatial terms.

The vacuum $|0\rangle$ is defined as the state without any photons. As in the Klein-Gordon theory, we can generate a state with arbitrarily many photons by applying $\hat{a}^\dagger$ to the vacuum. We can write the Hamiltonian:

$$\hat{H} = \int \mathrm{d}^3x \left[-\dot{\hat{A}}_\mu \dot{\hat{A}}^\mu - \mathcal{L}_{\mathrm{EM}}\right]$$

Expressing this in terms of $\hat{a}$ and $\hat{a}^\dagger$, we can rewrite the Hamiltonian:

$$\hat{H} = \int \frac{\mathrm{d}^3\boldsymbol{p}}{(2\pi)^3} \sum_r \omega_p \xi_r \hat{a}^{r\dagger}_{\boldsymbol{p}} \hat{a}^r_{\boldsymbol{p}}$$

Now let us compare this to the case with 4 scalar fields. We would have obtained the same result, but now we have this factor of ξ_r, which provides a sign for the time component, the scalar polarization.

This theory has a serious problem; consider the norm of the single particle state:

$$\sqrt{2\omega_p}\hat{a}^{r\dagger}_{\boldsymbol{p}} |0\rangle = |\boldsymbol{p}, r\rangle$$

The norm of this state is given by:

$$\begin{aligned}\langle \boldsymbol{p}, r|\boldsymbol{p}, r\rangle &= 2\omega_p \langle 0|\hat{a}^r_{\boldsymbol{p}}\hat{a}^{r\dagger}_{\boldsymbol{p}}|0\rangle \\ &= 2\omega_p \xi_r (2\pi)^3 \delta^3(0)\end{aligned}$$

We see that this norm is not positive definite, the ξ_r can make this norm negative! The source of the problem is that we have not yet impose the Lorenz gauge condition on the states. When we impose a gauge fixing condition, we have to introduce some constraint on the which states are physical. Can we just impose $\partial_\mu \hat{A}^\mu = 0$ in the quantum theory? It turns out that we cannot do this! Let us see why not.

We can write out the commutator of the field operators at different points in terms of the propagator:

$$\left[\hat{A}^\mu(x), \hat{A}^\nu(y)\right] = D^{\mu\nu}(x-y) - D^{\mu\nu}(y-x)$$

Where $D^{\mu\nu}$ is defined as:

$$D^{\mu\nu}(x-y) = -\eta^{\mu\nu} D(x-y)\Big|_{m\to 0}$$

We can also look at the commutator with a derivative:

$$\begin{aligned}\left[\partial_\mu \hat{A}^\mu(x), \hat{A}^\nu(y)\right] &= \partial_\mu D^{\mu\nu}(x-y) - \partial_\mu D^{\mu\nu}(y-x) \\ &\neq 0 \text{ in general}\end{aligned}$$

Thus we cannot impose the condition on the operators, we must impose a condition on the states themselves. The condition was figure out in 1951, by Gupta and Bleuler. For any physical state, we require that

$$\partial_\mu \hat{A}^{\mu+}(x) |\psi\rangle = 0$$

Where $\hat{A}^{\mu+}$ is the positive frequency part. The Lorenz condition holds between physical states:

$$\begin{aligned}\langle \psi'|\partial_\mu \hat{A}^\mu|\psi\rangle &= \langle \psi'|\partial_\mu \hat{A}^{\mu-} + \partial_\mu \hat{A}^{\mu+}|\psi\rangle \\ &= 0\end{aligned}$$

In momentum space, the condition $\left(\partial_\mu \hat{A}^{\mu+}\right)|\psi\rangle = 0$ becomes

$$\left[\hat{a}^3_{\boldsymbol{p}} - \hat{a}^0_{\boldsymbol{p}}\right]|\psi\rangle = 0$$

This is a condition on the longitudinal and scalar photons, imposing that they do not contribute to any physical process.

Longitudinal and Scalar Photons

We have claimed that the condition we impose on the states leads to the longitudinal and scalar photons not contributing to anything physical. As an example, let us consider the expectation value of the Hamiltonian in any physical state:

$$\langle\psi|H|\psi\rangle$$

We can insert our diagonalized Hamiltonian:

$$\langle\psi|H|\psi\rangle = \left\langle \psi \left| \int \frac{\mathrm{d}^3\boldsymbol{p}}{(2\pi)^3} \sum_r \omega_p \xi_r \hat{a}_{\boldsymbol{p}}^{r\dagger} \hat{a}_{\boldsymbol{p}}^r \right| \psi \right\rangle$$

Now let us consider the expectation value of a subset of this expression, only the terms in the summation with $r = 0, 3$:

$$\begin{aligned}\langle\psi|\hat{a}_{\boldsymbol{p}}^{3\dagger}\hat{a}_{\boldsymbol{p}}^3 - \hat{a}_{\boldsymbol{p}}^{0\dagger}\hat{a}_{\boldsymbol{p}}^0|\psi\rangle &= \frac{1}{2}\langle\psi|\left(\hat{a}_{\boldsymbol{p}}^{3\dagger} + \hat{a}_{\boldsymbol{p}}^{0\dagger}\right)\left(\hat{a}_{\boldsymbol{p}}^3 - \hat{a}_{\boldsymbol{p}}^0\right)|\psi\rangle + \frac{1}{2}\langle\psi|\left(\hat{a}_{\boldsymbol{p}}^{3\dagger} - \hat{a}_{\boldsymbol{p}}^{0\dagger}\right)\left(\hat{a}_{\boldsymbol{p}}^3 + \hat{a}_{\boldsymbol{p}}^0\right)|\psi\rangle \\ &= 0\end{aligned}$$

since $\langle\psi|\left(\hat{a}_{\boldsymbol{p}}^{3\dagger} - \hat{a}_{\boldsymbol{p}}^{0\dagger}\right) = 0$, and $\left(\hat{a}_{\boldsymbol{p}}^3 - \hat{a}_{\boldsymbol{p}}^0\right)|\psi\rangle = 0$. Thus we see that the contribution from the two unphysical polarizations cancel when we impose our condition:

$$\langle\psi|H|\psi\rangle = \left\langle \psi \left| \int \frac{\mathrm{d}^3\boldsymbol{p}}{(2\pi)^3} \sum_{r=1}^{2} \omega_p \xi_r \hat{a}_{\boldsymbol{p}}^{r\dagger} \hat{a}_{\boldsymbol{p}}^r \right| \psi \right\rangle$$

The price that we pay for choosing the covariant gauge is that we have extra, unphysical polarizations.

To finish our discussion of free electromagnetism, we can define the Feynman propagator:

$$\begin{aligned}D_F^{\mu\nu}\left(x - x'\right) &= \langle 0|\hat{T}\left[\hat{A}^\mu(x), \hat{A}^\nu\left(x'\right)\right]|0\rangle \\ &= -\eta^{\mu\nu} D_F\left(x - x'\right) \\ &= -\eta^{\mu\nu}\int \frac{\mathrm{d}^4p}{(2\pi)^4}\frac{ie^{-ip\cdot(x-x')}}{p^2 + i\varepsilon}\end{aligned}$$

This is essentially four copies of the scalar Feynman propagator, but with a metric out front. Note that we have only discussed massless vector particles, for a discussion of massive vector particles (such as the ρ meson), see Coleman's notes.

8. Interacting Theories

So far, we have discussed Lagrangian that contain only quadratic terms, which led to linear equations of motion. To introduce interactions between particles in our theories, we must introduce non-linear terms in $\mathcal{L}$. The restriction that we place on these new terms is that they are *local*, we do not allow terms that cause interactions between different spacetime points, for example, a term like $\phi^3(x)\,\phi(y)$ is not allowed. In general, theories like this will introduce violations of causality. The *conjecture* is that the only Lorentz invariant field theories are those with local interactions, but this is not known as a fact. Instead, we will focus on three types of local interactions (since they show up in nature):

1. ϕ^4 theory: Higgs in the Standard Model
2. Yukawa theory: How fermions obtain their masses
3. Quantum Electrodynamics

Let us quickly write down the Lagrangian for each of these, starting with the ϕ^4 theory:

$$\mathcal{L} = \frac{1}{2}\left(\partial_\mu \phi\right)^2 - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4 \tag{15}$$

Which will lead to the equation of motion:

$$\partial_\mu \partial^\mu \phi + m^2\phi + \frac{\lambda}{3!}\phi^3 = 0$$

Which is very clearly nonlinear.

The Lagrangian for the Yukawa theory is given by the Dirac theory plus the Klein-Gordon theory, and an interaction term that couples the two:

$$\mathcal{L}_{\text{Yukawa}} = \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{KG}} - g\overline{\psi}\psi\phi \tag{16}$$

Finally, QED has the Lagrangian:

$$\mathcal{L}_{\text{QED}} = \overline{\psi}\left(i\partial\!\!\!/ - m\right)\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - e\left(\overline{\psi}\gamma^\mu\psi\right)A_\mu \tag{17}$$

Note that the sign of the coupling constant e depends on convention, Peskin has $e < 0$. We can also rewrite the Lagrangian:

$$\mathcal{L}_{\text{QED}} = \overline{\psi}\left(iD\!\!\!\!/ - m\right)\psi - \frac{1}{2}F_{\mu\nu}F^{\mu\nu} \tag{18}$$

Where we define the gauge covariant derivative D_μ:

$$D_\mu = \partial_\mu + ieA_\mu \tag{19}$$

This Lagrangian is famously invariant under the gauge transformation:

$$\begin{aligned} \psi\left(x\right) &\to \psi\left(x\right)e^{i\alpha(x)} \\ A_\mu\left(x\right) &\to A_\mu\left(x\right) - \frac{1}{e}\partial_\mu\alpha\left(x\right) \end{aligned}$$

We will see that this invariance allows us to use the Gupta-Bleuler quantization method to use a covariant gauge when dealing with QED. Before we discuss any of these three theories, let us discuss what types of interaction terms we are allowed to introduce.

8.1. Renormalizability

Even after we impose locality on interaction terms, we can still introduce an infinite number of possible interaction terms, say ϕ^5, ϕ^6, etc. Why do we choose ϕ^4? We will impose the axiom of renormalizability, which will restrict the possibilities of interaction terms.

Axiom 8.1. ***Axiom of Renormalizability.*** *Any theory containing a coupling constant of mass dimension less than zero is not renormalizable.*

Recall that we are working in natural units, where $c = 1$ and $\hbar = 1$. The dimensions of c are $\left[LT^{-1}\right]$, and the dimensions of $\hbar$ are $\left[ML^2T^{-1}\right]$. Together, these allow us to define everything in terms of just mass. The dimensions of length and time in natural units are both inverse mass:

$$[L] = \left[M^{-1}\right]$$
$$[T] = \left[M^{-1}\right]$$

The action S has dimension $\hbar$, which means that it is dimensionless in natural units. From this, we can find the dimension of the Lagrangian density:

$$S = \int \mathrm{d}^4x\mathcal{L}$$

The left side is dimensionless, and d^4x has units of $\left[M^{-4}\right]$, and therefore the Lagrangian density has units of $\left[M^4\right]$.

Armed with this knowledge, let us look at some Lagrangian terms for a scalar field ϕ:

$$\frac{1}{2}\partial_\mu\phi\partial^\mu\phi \sim \frac{\phi^2}{x^2} \sim \left[M^4\right]$$

From this, we have that ϕ has dimensions of mass. We can double check this by looking at the mass term:

$$\frac{1}{2}m^2\phi^2 \sim \left[M^4\right]$$

Which clearly gives us that $[\phi] = M$. Yet another way of showing this is to look at the commutation relations:

$$\left[\phi, \dot{\phi}\right] = i\delta^3\left(\cdots\right)$$

Since $\delta^3 \sim \left[L^{-3}\right] = \left[M^3\right]$, we can again see that $[\phi] = M$.

Suppose we have an interaction term that looks like

$$\mu\phi^3$$

Since this must end up as dimension M^4, we know that $[\mu] = M$. In the case of the ϕ^4 term, we see that $[\lambda] = M^0$, the coupling constant is dimensionless.

Now let us consider a fermion field, ψ. Looking at terms in the Lagrangian:

$$\overline{\psi}\left(\gamma^\mu\partial_\mu\psi\right)$$

From which we have that $[\psi] = \frac{3}{2}$.

Similarly, looking at a gauge boson, whose Lagrangian has terms like:

$$-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$$

One can show that $[A^\nu] = M$.

Returning to renormalizability, we see that ϕ^3 and ϕ^4 are renormalizable, since the coupling constants have mass dimension 1 and 0, respectively. However, we see that if we go higher, ϕ^5, ϕ^6, etc, we see that the mass dimension is lower than 0, and therefore they are no longer renormalizable. We have therefore cut down the number of allowed interactions terms from infinity to 2.

For interacting fermions (with just fermions), we need an even number of fields (in order to maintain Lorentz invariance), and it turns out that ψ^4 is non-renormalizable, we cannot have just pure fermion field theories with interactions. However, if we introduce a scalar field, we can have one interaction:

$$\overline{\psi}\psi\phi$$

Which we note is exactly the Yukawa interaction. If we instead introduce a vector field to couple to our fermions, we have only one allowed interaction:

$$\overline{\psi}\gamma^\mu\psi A_\mu$$

Which is exactly the QED interaction term.

For gauge bosons, we can have terms of the form

$$A_\nu A^\nu \left[\partial_\mu A^\mu\right]$$

or any term of the same type, with indices mixed. Alternatively, we could add a term of the form

$$A^4$$

However, these violate gauge invariance unless the coefficients of the interaction are carefully chosen[23]. Like we saw earlier, we can only couple the gauge bosons to fermions in one way, via the QED interaction term. We can couple to scalars in multiple ways, but again the coefficients are restricted via gauge invariance:

$$A^\mu\partial_\mu\phi,\ A^\mu\phi\partial_\mu\phi,\ A_\mu A^\mu\phi^2$$

Why do we focus on renormalizable theories? We could imagine some interaction terms of the form:

$$\hat{\lambda}\frac{\phi^5}{M}, \hat{\lambda}\frac{\phi^6}{M^2}$$

Where $\hat{\lambda}$ is a dimensionless coupling constant. Suppose we want to compute something like a cross-section σ, which has dimensions of area, which in natural units corresponds to M^{-2}. In Yukawa, or ϕ^4, or QED, the cross-section scales with p^{-2}, where p is the momentum in the scattering process. If we computed this in a ϕ^5 theory, the cross-section would have to scale with M^{-2} or

[23]Which leads to non-Abelian gauge theories.

$M^{-1}p^{-1}$. In this case, if $M \gg p$, the non-renormalizable contribution is subleading. What about $p \gg M$? In this case, the theory will actually violate unitarity! The presence of a non-renormalizable term implies that there is some scale at which the theory will break down, in this case, the $p \gg M$ regime. For this reason, finding non-renormalizable terms corresponds to finding new physics.

What we have discussed so far is the modern understanding of non-renormalizable theories. Until the late 70's, non-renormalizable theories were considered sick. If we take a renormalizable theory, and we start computing processes, we see that some of them are infinite! However think back to introductory electromagnetism, we always neglect the infinite self-energy of charges, we only care about differences in potential. The infinities that arise in QFT are similar to this, what we care about are differences between these infinite quantities, which turn out to be finite. However to obtain the finite contribution, we need to know the coupling. Suppose we added a ϕ^5 term into our Lagrangian. It turns out that we could actually cook up a Feynman diagram that generates a ϕ^6 term, which generates a ϕ^8 term, which itself generates higher powers of ϕ. Back in the day, this was a really big issue, but in the modern understanding, this is fine since the higher order terms are suppressed at low energies.

8.2. Interaction Picture

We usually do non-relativistic quantum mechanics in the Schrodinger picture, and so far in this course we have mostly been working in the Heisenberg picture. However, most QFT calculations are done in the *interaction picture*. In the Schrodinger picture, the states evolve in time under the Hamiltonian:

$$i\frac{\partial}{\partial t}\left|\psi\right\rangle_S = \hat{H}\left|\psi\right\rangle_S$$

And operators are time independent:

$$i\frac{\partial}{\partial t}\hat{O}_S = 0$$

In the Heisenberg picture, it is exactly the opposite, states are independent with respect to time, and operators evolve:

$$\begin{aligned} i\frac{\partial}{\partial t}\left|\psi\right\rangle_H &= 0 \\ i\frac{\partial}{\partial t}\hat{O}_H &= \left[\hat{O}_H, \hat{H}\right] \end{aligned}$$

And regardless of what picture we choose, expectation values produce the same results.

In the interaction picture, we split the Hamiltonian into a free Hamiltonian $\hat{H}_0$ and an interacting portion $\hat{H}_{\text{int}}$:

$$\hat{H} = \hat{H}_0 + \hat{H}_{\text{int}}$$

With this split, we can now describe how states and operators evolve in time in the interaction picture; states evolve under the interacting part of the Hamiltonian, and operators evolve under the free Hamiltonian:

$$i\frac{\partial}{\partial t}\left|\psi\right\rangle_I = \hat{H}_{\text{int}}\left|\psi\right\rangle_I$$

$$i\frac{\partial}{\partial t}\hat{O}_I = \left[\hat{O}_I, \hat{H}_0\right]$$

Essentially, we consider the free theory in the Heisenberg picture, and the interactions are considered in the Schrodinger picture.

The states and operators in the interaction picture can be related back to the Schrodinger picture:

$$\begin{aligned} |\psi\rangle_I &= e^{i\hat{H}_0 t}|\psi\rangle_S \\ \hat{O}_I &= e^{i\hat{H}_0 t}\hat{O}_S e^{-i\hat{H}_0 t} \end{aligned}$$

We again have the fact that expectation values are invariant under the choice of picture:

$${}_I\langle\psi|\hat{O}_I|\psi\rangle_I = {}_S\langle\psi|\hat{O}_S|\psi\rangle_S$$

8.3. S-Matrix Expansion

We begin with a Lagrangian that has a free portion, $\mathcal{L}_0$, and an interaction term, $\mathcal{L}_{\text{int}}$, which leads to a Hamiltonian:

$$\hat{H} = \hat{H}_0 + \hat{H}_{\text{int}}$$

Using the interaction picture, our states evolve in time under the interaction term:

$$i\frac{\partial}{\partial t}|\phi(t)\rangle = \hat{H}_{\text{int}}|\phi(t)\rangle$$

Most of the time when we are doing QFT, we are interested in scattering problems. We generally start with some well-separated particles, they evolve in time, interact with each other, and then eventually become well-separated once more. This is the most common type of problem that QFT is used to describe, and thus the formalism of quantum field theory is built around problems of this type. Let us try to formalize this.

The initial state $|i\rangle$ consists of well-separated particles: they are assumed to be far enough apart that they are not interacting, so we say that this is the state of the system at some negative time extent:

$$\lim_{t\to-\infty}|\phi(t)\rangle = |i\rangle$$

After the scattering process occurs, the final state particles will again be well-separated, and non-interacting, we label this state $|\phi(\infty)\rangle$. $|\phi(\infty)\rangle$ is a linear combination of well-separated final products of the scattering processes. If we think about an e^+e^- scattering, we could end up with a e^+e^- state, but also many other possible scattering products, such as $\gamma\gamma$, $\mu^+\mu^-$, etc. Each of these outcomes will be present in $|\phi(\infty)\rangle$. We introduce the S-matrix, which relates $|\phi(\infty)\rangle$ to $|\phi(-\infty)\rangle$:

$$\begin{aligned} |\phi(\infty)\rangle &= \hat{S}|\phi(-\infty)\rangle \\ &= \hat{S}|i\rangle \end{aligned}$$

The collision can lead to many different final states, and the transition probability to end up in a specific final state $|f\rangle$ is given by

$$|\langle f|\phi(\infty)\rangle|^2$$

Note that here we are assuming that $|f\rangle$ and $|i\rangle$ are normalized to 1, as we usually do in quantum mechanics[24]. We can rewrite this transition probability in terms of the S matrix:

$$\begin{aligned}\langle f|\phi(\infty)\rangle &= \langle f|\hat{S}|i\rangle \\ &\equiv S_{fi}\end{aligned}$$

This is why we call S the S-matrix, since we care about the matrix elements S_{fi}. Now we can write the final state in terms of the matrix elements:

$$|\phi(\infty)\rangle = \sum_f S_{fi}\,|f\rangle$$

Then, if we assume that these states are normalized to 1, we have the conservation of probability:

$$\sum_f |S_{fi}|^2 = 1$$

We have mapped the solution of the scattering problem onto knowledge of the S matrix, we want to be able to find the matrix elements S_{fi}, given some states $|i\rangle$ and $|f\rangle$. To solve this problem, we will apply perturbation theory. We have an expression for how the states evolve in time:

$$i\frac{\partial}{\partial t}|\phi(t)\rangle = \hat{H}_{\text{int}}(t)\,|\phi(t)\rangle$$

Suppose we had a differential equation of the form:

$$i\frac{dy}{dt} = H_{\text{int}}y$$

We would have a solution of the form:

$$y = y_0 e^{-i\int H_{\text{int}}\,dt}$$

However, since $\hat{H}_{\text{int}}$ is an operator, we cannot commute $\hat{H}_{\text{int}}$ at different times. Instead, let us solve our operator equation iteratively. Let us integrate both sides of our equation:

$$|\phi(t)\rangle = |i\rangle + (-i)\int_{-\infty}^{t} dt_1\,\hat{H}_{\text{int}}(t_1)\,|\phi(t_1)\rangle$$

This is not a useful solution, since we have $|\phi(t)\rangle$ on the right side. However, let us now substitute in our expression into itself, which gives us:

$$|\phi(t)\rangle = |i\rangle + (-i)\int_{-\infty}^{t} dt_1\,\hat{H}_{\text{int}}(t_1)\,|i\rangle + (-i)^2\int_{-\infty}^{t} dt_1\int_{-\infty}^{t_1} dt_2\,\hat{H}_{\text{int}}(t_1)\,\hat{H}_{\text{int}}(t_2)\,|\phi(t_2)\rangle$$

Now we see that the second term is something that we can work with, there is no dependence on $|\phi(t)\rangle$, but the third term is unknown. However, we can make an argument by saying that we are working in powers of $\hat{H}_{\text{int}}$, if this is "small", then we can neglect the term that has two powers of $\hat{H}_{\text{int}}$[25]. This expansion in powers of $\hat{H}_{\text{int}}$ is known as the S-matrix expansion, and we can take this

[24]If we were using Peskin's normalization, there would be some other factors that we would have to account for.
[25]For a more rigorous statement of what this expansion criterion is, take QFT II!

to arbitrary powers of $\hat{H}_{\text{int}}$. If we take this to the limit of $t \to \infty$, we have the definition of the S matrix:

$$\hat{S} = \sum_{n=0}^{\infty} (-i)^n \int_{-\infty}^{\infty} dt_1 \int_{-\infty}^{t_1} dt_2 \cdots \int_{-\infty}^{t_{n-1}} dt_n \, \hat{H}_{\text{int}}(t_1)\, \hat{H}_{\text{int}}(t_2) \cdots \hat{H}_{\text{int}}(t_n)$$

This can be rewritten in a more useful form. Consider the second term:

$$\hat{I}_2 = -\int_{-\infty}^{\infty} dt_1 \int_{-\infty}^{t_1} dt_2 \, \hat{H}_{\text{int}}(t_1)\, \hat{H}_{\text{int}}(t_2)$$

Now let us relabel the dummy indices t_1 and t_2, by swapping them:

$$\hat{I}_2 = \int_{-\infty}^{\infty} dt_2 \int_{-\infty}^{t_2} dt_1 \hat{H}_{\text{int}}(t_2)\, \hat{H}_{\text{int}}(t_1)$$

Note that $t_2 > t_1$ in this integral.

Now let us consider what this integration region looks like. We are integrating over strips of constant t_1, and then changing t_1. However, we can also integrate over the region in another way, we can hold t_2 fixed, integrate over t_1, and then integrate over t_2. These two methods are shown in Fig. 3.

Figure 3: Two methods of integrating over the same region.

The second integration method is written as:

$$\hat{I}_2 = \int_{-\infty}^{\infty} dt_2 \int_{t_2}^{\infty} dt_1 \hat{H}_{\text{int}}(t_1)\, \hat{H}_{\text{int}}(t_2)$$

Note that $t_1 > t_2$ in this integral. If we compare this integral against the previous integral we had, we could add the two together and we would have an integral from $-\infty$ to ∞, except for the fact that the order of the Hamiltonian terms are different. However, we can apply a time ordering operator to combine the two integrals:

$$2\hat{I}_2 = \int_{-\infty}^{\infty} dt_2 \int_{-\infty}^{\infty} dt_1 \, \hat{T}\left[\hat{H}_{\text{int}}(t_1)\, \hat{H}_{\text{int}}(t_2)\right]$$

If we now switch back the dummy indices t_1 and t_2:

$$\hat{I}_2 = \frac{1}{2} \int_{-\infty}^{\infty} dt_1 \int_{-\infty}^{\infty} dt_2 \, \hat{T}\left[\hat{H}_{\text{int}}(t_1)\, \hat{H}_{\text{int}}(t_2)\right]$$

Now that both integrals are over the same region, this looks very symmetric between t_1 and t_2. Similar expressions hold for the higher order terms, which allows us to rewrite the S matrix:

$$\hat{S} = \sum_{n=0}^{\infty} \frac{(-i)^n}{n!} \int_{-\infty}^{\infty} dt_1 \int_{-\infty}^{\infty} dt_1 \cdots \int_{-\infty}^{\infty} dt_n \, \hat{T}\left[\hat{H}_{\text{int}}(t_1) \cdots \hat{H}_{\text{int}}(t_n)\right]$$

Had the time ordering operator not been there, this integrand would just be an exponential, but we now need to account for the time ordering:

$$\hat{S} = \hat{T}\left[\exp\left(-i \int_{-\infty}^{\infty} dt \, \hat{H}_{\text{int}}(t)\right)\right]$$

This is similar to what we would have expected, but now we have this time ordering operator out front. Thus, we have in principle defined the $\hat{S}$ matrix in terms of powers of the interaction term in the Hamiltonian.

Typically, we scatter two incoming particles A and B with momenta $\boldsymbol{k}_A$ and $\boldsymbol{k}_B$ into some number of final state particles (there can be more than 2). We label the outgoing particle momenta as $\boldsymbol{p}_1, \boldsymbol{p}_2, \ldots$. We can label the initial and final states:

$$|i\rangle = \text{const.}\, |\boldsymbol{k}_A, \boldsymbol{k}_B\rangle$$
$$|f\rangle = \text{const.}\, |\boldsymbol{p}_1, \boldsymbol{p}_2, \cdots\rangle$$

Let us stop to highlight a subtlety: suppose we are doing scattering in non-relativistic quantum mechanics. When we do this problem, we always assume that the potentials that cause our interactions are finite-range. The issue that arises in the field theory scattering problem is that the fields fill all of space, and we have vacuum fluctuations. Thus, if we have a electron moving through space, it is still different from an electron in a universe where the electromagnetic coupling was zero, the presence of the interaction terms in the Hamiltonian mean that the vacuum itself gives rise to interactions. These effects kick in at higher orders of our S-matrix expansion, and since we will work to leading order in this course, they will not affect our results, but they do exist!

In the case of a non-interacting theory, the S-matrix is just the identity matrix. If we want to isolate the interactions, we can define the S matrix in terms of the transfer matrix $\hat{T}$, also known as the T-matrix.

$$\hat{S} = \hat{I} + i\hat{T}$$

If we look at the matrix elements of $\hat{S}$, we find that they always contain a delta function of energy-momentum conservation, there will always be a factor of:

$$\delta^4\left(k_A + k_B - \sum_f p_f\right)$$

This provides both energy conservation and momentum conservation. Extracting this factor, we define the matrix element M:

$$\langle \boldsymbol{p}_1, \boldsymbol{p}_2, \ldots | i\hat{T} | k_A, k_B\rangle = (2\pi)^4\, \delta^4\left(k_A + k_B - \sum_f p_f\right) M\left(k_A, k_B \to p_i\right)$$

Note that by convention, we are pulling out the factor of 2π, as well as the delta function.

8.4. Wick's Theorem

To calculate a scattering process, we will have to compute matrix elements of the S matrix. In general, $\hat{S}$ will contain many products of the interaction Hamiltonian, $\hat{H}_{\text{int}}$. Note that if we start from some initial state $|i\rangle$, we can end up with many different final states. If we are interested in a particular output state, we want to keep track of which combination of operators lead to the correct final state. Wick's Theorem allows us to organize this calculation.

Suppose we want to start at $|i\rangle$, and end at some $|f\rangle$. We will need to have the destruction operator for all of the initial state particles, and we need all the creation operators for the final state particles. It would be very nice if we could have all of our annihilation operators on the right, and all of our creation operators on the left. However, we have time ordered products of the $\hat{H}_{\text{int}}$, which contains a mix of creation and annihilation operators in differing orders. Wick's theorem allows us to move all of the operators to their respective sides. Recall that this is exactly the definition of normal ordering, so Wick's theorem allows us to relate time ordered products to normal ordered products.

Let us first formally define normal ordering. Let $\hat{Q}, \hat{R}, \ldots, \hat{W}$ be operators which are linear in either creation or annihilation operators. The normal ordered product of the operators is given by:

$$\hat{N}\left(\hat{Q}\hat{R}\cdots\hat{W}\right) = (-1)^P\left(\hat{Q}'\hat{R}'\cdots\hat{W}'\right)$$

Where the prime denotes the rearranged form of the original operators, where all annihilation operators are to the right of creation operators. P is the number of interchanges of neighboring fermion operators that you need to obtain this rearrangement.

How do we write a time ordered product in terms of normal ordered products?

Consider two bosonic operators $\hat{A}$ and $\hat{B}$. Let us look at the difference between the product and the normal ordered product:

$$\begin{aligned}\hat{A}\hat{B} - \hat{N}\left(\hat{A}\hat{B}\right) &= \left(\hat{A}^+\hat{B}^+ + \hat{A}^-\hat{B}^+ + \hat{A}^+\hat{B}^- + \hat{A}^-\hat{B}^-\right) - \left(\hat{A}^+\hat{B}^+ + \hat{A}^-\hat{B}^+ + \hat{B}^-\hat{A}^+ + \hat{A}^-\hat{B}^-\right) \\ &= \left[\hat{A}^+, \hat{B}^-\right]\end{aligned}$$

If $\hat{A}$ and $\hat{B}$ had been fermionic, we would have gotten the anticommutator:

$$\hat{A}\hat{B} - \hat{N}\left(\hat{A}\hat{B}\right) = \{\hat{A}^+, \hat{B}^-\}$$

But both this commutator and the anticommutator are just numbers, so we can write that this difference that we are computing is equal to the inner product:

$$\hat{A}\hat{B} - \hat{N}\left(\hat{A}\hat{B}\right) = \langle 0|\hat{A}\hat{B}|0\rangle$$

From this, we can state that:

$$\hat{T}\left(\hat{A}\hat{B}\right) = \hat{N}\left(\hat{A}\hat{B}\right) + \langle 0|\hat{T}\left[\hat{A}\hat{B}\right]|0\rangle$$

Now we note that this second term is the Feynman propagator! Let us introduce some shorthand:

$$\langle 0|\hat{T}\left[\hat{A}\left(x_1\right)\hat{B}\left(x_2\right)\right]|0\rangle = \hat{A}(\overbracket{x_1)\hat{B}}\left(x_2\right)$$

In general, for bosonic operators:

$$\hat{\phi}\,\overbracket{(x_1)\hat{\phi}}\left(x_2\right) = D_F\left(x_1 - x_2\right)$$

We can now generalize this relation between the time ordered product and the normal ordered product to the full statement of Wick's theorem.

Theorem 8.2. *Wick's Theorem. Wick's theorem states that:*

$$\hat{T}\left(\hat{A}\left(x_1\right)\hat{B}\left(x_2\right)\ldots\hat{M}\left(x_m\right)\right) = \hat{N}\left(\hat{A}\left(x_1\right)\hat{B}\left(x_2\right)\ldots\hat{M}\left(x_m\right)\right) + \textit{ all possible contractions}$$

For example, consider a field time ordered product:

$$
\begin{aligned}
\hat{T}\Big(\hat{\phi}(x_1)\hat{\phi}(x_2)\hat{\phi}(x_3)\hat{\phi}(x_4)\Big) = \hat{N}\Big(&\hat{\phi}(x_1)\hat{\phi}(x_2)\hat{\phi}(x_3)\hat{\phi}(x_4)\Big) \\
&+ \wick{\c{\hat{\phi}}(x_1)\c{\hat{\phi}}(x_2)}\hat{N}\Big(\hat{\phi}(x_3)\hat{\phi}(x_4)\Big) + \wick{\c{\hat{\phi}}(x_1)\c{\hat{\phi}}(x_3)}\hat{N}\Big(\hat{\phi}(x_2)\hat{\phi}(x_4)\Big) \\
&+ \wick{\c{\hat{\phi}}(x_1)\c{\hat{\phi}}(x_4)}\hat{N}\Big(\hat{\phi}(x_2)\hat{\phi}(x_3)\Big) + \wick{\c{\hat{\phi}}(x_2)\c{\hat{\phi}}(x_3)}\hat{N}\Big(\hat{\phi}(x_1)\hat{\phi}(x_4)\Big) \\
&+ \wick{\c{\hat{\phi}}(x_2)\c{\hat{\phi}}(x_4)}\hat{N}\Big(\hat{\phi}(x_1)\hat{\phi}(x_3)\Big) + \wick{\c{\hat{\phi}}(x_3)\c{\hat{\phi}}(x_4)}\hat{N}\Big(\hat{\phi}(x_1)\hat{\phi}(x_2)\Big) \\
&+ \wick{\c1{\hat{\phi}}(x_1)\c1{\hat{\phi}}(x_2)\c2{\hat{\phi}}(x_3)\c2{\hat{\phi}}(x_4)} + \wick{\c1{\hat{\phi}}(x_1)\c1{\hat{\phi}}(x_3)\c2{\hat{\phi}}(x_2)\c2{\hat{\phi}}(x_4)} \\
&+ \wick{\c1{\hat{\phi}}(x_1)\c1{\hat{\phi}}(x_4)\c2{\hat{\phi}}(x_2)\c2{\hat{\phi}}(x_3)}\,.
\end{aligned}
$$

The general proof of this theorem is via induction in the number of fields in the time ordered product.

Now let us look at how we recover S-matrix elements from Feynman diagrams. Consider the scattering of two particles with momenta $\boldsymbol{p}_A$ and $\boldsymbol{p}_B$ into $\boldsymbol{p}_1$ and $\boldsymbol{p}_2$, in the ϕ^4 theory.

The relevant matrix element is

$$
\langle \boldsymbol{p}_1, \boldsymbol{p}_2 | \hat{T}\left[\exp\left(-i\int_{-\infty}^{\infty} dt\, \hat{H}_{\text{int}}(t)\right)\right] |\boldsymbol{p}_A, \boldsymbol{p}_B\rangle
$$

Let us first look at the expansion of the exponential in terms of powers of $\hat{H}_{\text{int}}$:

$$
\exp\left(-i\int_{-\infty}^{\infty} dt\, \hat{H}_{\text{int}}(t)\right) \sim \mathbb{I} + \mathcal{O}\left(\hat{H}_{\text{int}}\right) + \mathcal{O}\left(\hat{H}_{\text{int}}\right)
$$

The lowest order term is the trivial case, with the $\mathbb{I}$:

$$
\begin{aligned}
\langle \boldsymbol{p}_1, \boldsymbol{p}_2 | \boldsymbol{p}_A, \boldsymbol{p}_B\rangle &= \sqrt{2E_1 \cdot 2E_2 \cdot 2E_A \cdot 2E_B}\, \langle 0|\hat{a}_1\hat{a}_2\hat{a}_A^\dagger\hat{a}_B^\dagger|0\rangle \\
&= 2E_A \cdot 2E_B\,(2\pi)^6\left[\delta\left(\boldsymbol{p}_A - \boldsymbol{p}_1\right)\delta\left(\boldsymbol{p}_B - \boldsymbol{p}_2\right) + \delta\left(\boldsymbol{p}_A - \boldsymbol{p}_2\right)\delta\left(\boldsymbol{p}_B - \boldsymbol{p}_1\right)\right]
\end{aligned}
$$

Where we drop the momenta labels for brevity. This is just the identity up to the factor of $(2E_A)(2E_B)$, which comes from the normalization. Thus, when we look at the separation of the S matrix into the $\hat{T}$ matrix, it does not contribute to the $\hat{T}$ term.

We can denote this matrix element in terms of Feynman diagrams, as shown in Figure 4.

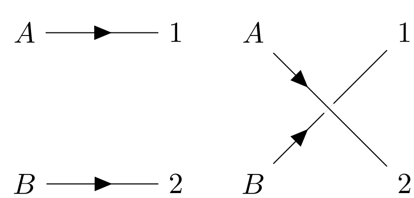


Figure 4: Feynman diagrams for the identity term of the S-matrix. The left one represents the $\delta\left(\boldsymbol{p}_A - \boldsymbol{p}_1\right)\delta\left(\boldsymbol{p}_B - \boldsymbol{p}_2\right)$ term, and the diagram on the right represents the $\delta\left(\boldsymbol{p}_A - \boldsymbol{p}_2\right)\left(\boldsymbol{p}_B - \boldsymbol{p}_1\right)$ term.

In these diagrams, each line denotes a particle. In the first diagram, particle A “becomes” particle 1, and particle B “becomes” particle 2. This represents the first set of delta functions terms. Similarly, for the second diagram, particle A “becomes” particle 2, and particle B “becomes” particle 1, all without interactions, representing the second set of delta functions. Thus the lowest order term gives us no interactions, which makes sense, since that identity term would be there even with no interaction term.

The first order expansion term is given by:

$$\begin{aligned}\langle \boldsymbol{p}_1, \boldsymbol{p}_2| - \frac{i\lambda}{4!}\hat{T}\left(\int \mathrm{d}^4x\, \hat{\phi}^4(x)\right)|\boldsymbol{p}_A, \boldsymbol{p}_B\rangle &= \langle \boldsymbol{p}_1, \boldsymbol{p}_2| - \frac{i\lambda}{4!}\hat{N}\left(\int \mathrm{d}^4x\, \hat{\phi}^4(x)\right)|\boldsymbol{p}_A\boldsymbol{p}_B\rangle \qquad (20)\\
&+ 6\,\langle \boldsymbol{p}_1, \boldsymbol{p}_2| - \frac{i\lambda}{4!}\hat{N}\left(\int \mathrm{d}^4x\, \hat{\phi}^2(x)\,\hat{\phi}\overline{(x)\hat{\phi}}(x)\right)|\boldsymbol{p}_A, \boldsymbol{p}_B\rangle \qquad (21)\\
&+ 3\,\langle \boldsymbol{p}_1, \boldsymbol{p}_2| - \frac{i\lambda}{4!}\int \mathrm{d}^4x\, \hat{\phi}\overline{(x)\hat{\phi}}(x)\hat{\phi}\overline{(x)\hat{\phi}}(x)|\boldsymbol{p}_A, \boldsymbol{p}_B\rangle \qquad (22)\end{aligned}$$

Any uncontracted $\hat{\phi}(x)$ is used to annihilation an initial state particle or create a final state particle. For example:

$$\begin{aligned}\hat{\phi}^+(x)|\boldsymbol{p}\rangle &= \underbrace{\left[\int \frac{\mathrm{d}^3k}{(2\pi)^3}\frac{1}{\sqrt{2\omega_k}}\hat{a}_k e^{-ik\cdot x}\right]}_{\hat{\phi}^+(x)}\sqrt{2\omega_p}\hat{a}^\dagger_{\boldsymbol{p}}|0\rangle\\
&= e^{-ip\cdot x}|0\rangle\end{aligned}$$

We define the contractions of field operators as

$$\begin{aligned}\hat{\phi}\overline{(x)|p}\rangle &= e^{-ip\cdot x}\\
\langle \overline{p|\hat{\phi}}(x) &= e^{ip\cdot x}\end{aligned}$$

Let us now return to our first order expansion. We can first consider the term with two contractions (22):

$$\begin{aligned}&3\,\langle \boldsymbol{p}_1, \boldsymbol{p}_2| - \frac{i\lambda}{4!}\int \mathrm{d}^4x\, \hat{\phi}\overline{(x)\hat{\phi}}(x)\hat{\phi}\overline{(x)\hat{\phi}}(x)|\boldsymbol{p}_A, \boldsymbol{p}_B\rangle\\
&= 3\left(\frac{-i\lambda}{4!}\right)\int \mathrm{d}^4x\, D_F(x-x)^2\Big[(2\pi)^6\, 2\omega_A\cdot 2\omega_B\delta^3(\boldsymbol{p}_1 - \boldsymbol{p}_A)\,\delta^3(\boldsymbol{p}_2 - \boldsymbol{p}_B)\\
&+ (2\pi)^6\, 2\omega_A\cdot 2\omega_B\delta^3(\boldsymbol{p}_1 - \boldsymbol{p}_B)\,\delta^3(\boldsymbol{p}_2 - \boldsymbol{p}_A)\Big]\end{aligned}$$

This is another trivial contribution to the scattering amplitude. In terms of Feynman diagrams, we have the same base diagrams, two non-interacting propagators, but now we have some activity in the vacuum, due to the $D_F(x-x)^2$. The two diagrams are shown in Figure 5.

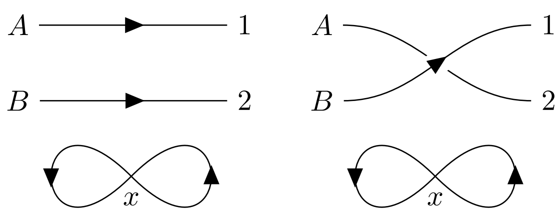


Figure 5: Two non-interacting diagrams that have vacuum fluctuations. The vacuum fluctuations arise from the $D_F\left(x-x\right)^2$ in the matrix element, which is two particles propagating from x to x. Note that the two loops at x are integrated over all x.

Comparing Figures 5 and 4, we see that they only differ in the $D_F\left(x-x\right)^2$ term, this extra set of loops in the diagram.

Let us now consider (21), which we can rewrite as:

$$6\left(-\frac{i\lambda}{4!}\right)D_F\left(x-x\right)\langle\boldsymbol{p}_1,\boldsymbol{p}_2|\int \mathrm{d}^4x\,\hat{N}\left[\hat{\phi}\left(x\right)\hat{\phi}\left(x\right)\right]|\boldsymbol{p}_A,\boldsymbol{p}_B\rangle$$

Schematically, note that only the $\hat{a}^\dagger\hat{a}$ term can contribute from this normal ordered product, so we have to contract one $\hat{\phi}$ with an incoming state and one ϕ with an outgoing state. However, there are different ways of doing this, we could contract each $\hat{\phi}$ with either one of the outgoing particles, or either one of the incoming particles. Let us just write one such possible contraction:

$$\begin{aligned}&6\left(-\frac{i\lambda}{4!}\right)D_F\left(x-x\right)\langle\boldsymbol{p}_1,\boldsymbol{p}_2|\int \mathrm{d}^4x\,\hat{N}\left[\hat{\phi}\left(x\right)\hat{\phi}\left(x\right)\right]|\boldsymbol{p}_A,\boldsymbol{p}_B\rangle\\&=6\left(-\frac{i\lambda}{4!}\right)D_F\left(x-x\right)\left(2\pi\right)^3\delta^3\left(\boldsymbol{p}_1-\boldsymbol{p}_B\right)2\omega_B\underbrace{\int \mathrm{d}^4x\,e^{-ip_A\cdot x}e^{ip_2\cdot x}}_{\left(2\pi\right)^4\delta^4\left(p_2-p_A\right)}\end{aligned}$$

However, suppose we picked a different way of contracting the terms? It turns out that we would get the same amplitude! Before we discuss this, what does this term look like in terms of Feynman diagrams? This interaction is still trivial, there is no momentum exchange between A and B, but now we have an interaction with the vacuum. There are several different ways that the interaction can occur, as shown in Figure 6. These are contributions to the trivial part of the amplitude, they contribute to the identity but not the $\hat{T}$ matrix.

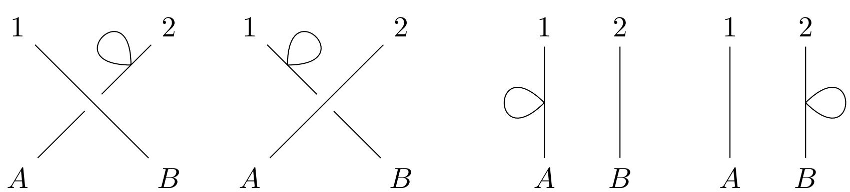


Figure 6: Feynman diagrams contributing to (21).

Let us now describe a general rule: **only fully connected diagrams contribute to the $\hat{T}$ matrix**[26].

Let us finally consider the remaining term, (20). We have a ton of different allowed contractions, any of the $\hat{\phi}(x)$ terms could contract with any of the $\boldsymbol{p}$s. In fact, we can see that there are exactly 4! possible contractions (4 choices for the first contraction, 3 for the second, etc.). Thus, we can pick a particular contraction, and then just multiply by 4!, since they all provide the same amplitude[27]. Thus we have:

$$\begin{aligned}\int \mathrm{d}^4x\, 4! \left(-\frac{i\lambda}{4!}\right) \langle \boldsymbol{p}_1, \boldsymbol{p}_2 | \hat{\phi}(x)\hat{\phi}(x)\hat{\phi}(x)\hat{\phi}(x) | \boldsymbol{p}_A, \boldsymbol{p}_B \rangle &= -i\lambda \int \mathrm{d}^4x\, e^{i(p_1+p_2-p_A-p_B)\cdot x} \\ &= -i\lambda\, (2\pi)^4\, \delta^4\left(p_1+p_2-p_A-p_B\right)\end{aligned}$$

This is of the form

$$iM\,(2\pi)^4\,\delta^4\left(p_1+p_2-p_A-p_B\right)$$

Where in this case $M = -\lambda$. This is the first nontrivial amplitude for scattering. The corresponding Feynman diagram is shown in Figure 7.

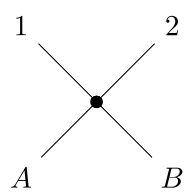


Figure 7: One-vertex diagram, the only non-trivial diagram for first order ϕ^4 $2 \to 2$ scattering.

Now, while we will not actually work out the higher order contributions, they are of the form:

$$\langle \boldsymbol{p}_1, \boldsymbol{p}_2 | \frac{(-i)^2}{2!}\left(\frac{1}{4!}\right)^2 \lambda^2 \hat{T}\left[\int \mathrm{d}^4x_1\, \hat{\phi}^4(x_1) \int \mathrm{d}^4x_2\, \hat{\phi}^4(x_2)\right] | \boldsymbol{p}_A \boldsymbol{p}_B \rangle$$

There are many different diagrams from this process, a subset of these is shown in Figure 8.

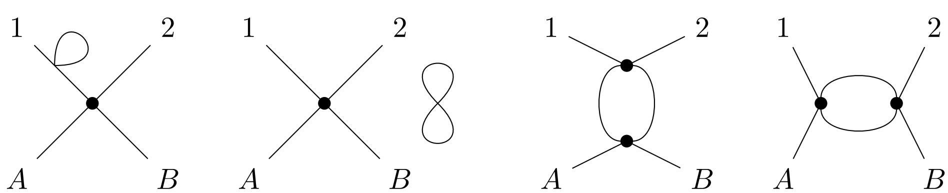


Figure 8: Second order in $\hat{H}_{\text{int}}$ diagrams for $2 \to 2$ scattering in the ϕ^4 theory.

Suppose we look at the third diagram in Figure 8. The contribution of the diagram is of the form (neglecting constant terms for brevity):

$$\langle \boldsymbol{p}_1, \boldsymbol{p}_2 | \int \mathrm{d}^4x_1\, \hat{\phi}(x_1)\hat{\phi}(x_1)\hat{\phi}(x_1)\hat{\phi}(x_1) \int \mathrm{d}^4x_2\, \hat{\phi}(x_2)\hat{\phi}(x_2)\hat{\phi}(x_2)\hat{\phi}(x_2) | \boldsymbol{p}_A, \boldsymbol{p}_B \rangle$$

[26] We will not prove this, but this is true to all orders.

[27] Note that it cancels the 4! term!

In fact, there are $2^2\,(12)^2$ such contractions, and thus the net amplitude is:

$$\frac{2^2\,(12)^2}{2!\,(4!)^2}\,(-i\lambda)^2\int \mathrm{d}^4x_1\int \mathrm{d}^4x_2\,\left[D_F\left(x_1-x_2\right)\right]^2 e^{i(p_1+p_2)\cdot x_1}e^{-i(p_A+p_B)\cdot x_2}$$

The prefactor out front results in 1/2. This factor of 2, which comes from the number of contractions that leads to the same diagram, is known as the symmetry factor of the diagram. Note that the symmetry factor is not defined by what multiplies the diagram, but instead the *reciprocal* of what is multiplying the contribution. We can do out the integrals, and explicitly write out the Feynman propagator:

$$\begin{aligned}
&\frac{1}{2}\,(-\lambda)^2\int \mathrm{d}^4x_1\int \mathrm{d}^4x_2\int\frac{\mathrm{d}^4k_1}{(2\pi)^4}\frac{ie^{ik_1\cdot(x_1-x_2)}}{k_1^2-m^2}\int\frac{\mathrm{d}^4k_2}{(2\pi)^4}\frac{ie^{ik_2\cdot(x_2-x_1)}}{k_2^2-m^2}e^{i(p_1+p_2)\cdot x_1}e^{-i(p_A+p_B)\cdot x_2}\\
&=\frac{1}{2}\,(-\lambda)^2\int\frac{\mathrm{d}^4k_1}{(2\pi)^4}\int\frac{\mathrm{d}^4k_2}{(2\pi)^4}\frac{1}{\left(k_1^2-m^2\right)\left(k_2^2-m^2\right)}\,(2\pi)^4\,\delta^4\left(p_1+p_2+k_1+k_2\right)(2\pi)^4\,\delta^4\left(k_1+k_2+p_A+p_B\right)\\
&=\frac{\lambda^2}{2}\,(2\pi)^4\,\delta^4\left(p_1+p_2-p_A-p_B\right)\int\frac{\mathrm{d}^4k}{(2\pi)^4}\frac{1}{\left(k^2-m^2\right)\left[\left(p_A+p_B+k\right)^2-m^2\right]}
\end{aligned}$$

This is again of the form

$$iM\,(2\pi)^4\,\delta^4\left(p_1+p_2-p_A-p_B\right)$$

With

$$M=-\frac{i\lambda^2}{2}\int\frac{\mathrm{d}^4k}{(2\pi)^4}\frac{1}{\left(k^2-m^2\right)\left[\left(p_A+p_B+k\right)^2-m^2\right]}$$

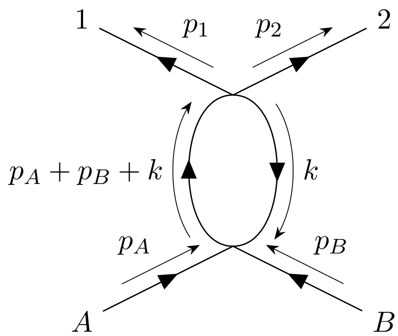
This contribution is much more complicated that the contribution from the first order terms. We obtained the contributions from evaluating the contractions, but we can find them more simply from the *Feynman rules*, which allow us to just impose momentum conservation directly. In momentum space:

$$iM=(\text{sum of diagrams})$$

where diagrams are evaluated using the following rules:

1. Propagators with momentum p contribute $\frac{i}{p^2-m^2+i\varepsilon}$.
2. Each vertex contributes $-i\lambda$.
3. External lines contribute a factor of 1.
4. Impose momentum conservation at each vertex.
5. Integrate over undetermined loop momentum.
6. Divide by the symmetry factor.

Let us recall the diagram that we are trying to evaluate:



We can write down the amplitude directly from the Feynman rules. We have 2 vertices, giving a factor of $(-i\lambda)^2$. We have one undetermined momentum, $\boldsymbol{k}$, and we have two propagators, one for each side of the loop:

$$iM = \frac{1}{2}(-i\lambda)^2 \int \frac{\mathrm{d}^4k}{(2\pi)^4} \frac{i}{k^2 - m^2} \frac{i}{\left[(p_A + p_B + k)^2 - m^2\right]}$$

Now how did we compute the symmetry factor? The fact that we are working in second order perturbation theory gives us the factor of $\frac{1}{2!}$, and the $\left(\frac{1}{4!}\right)^2$ comes from the interaction Hamiltonian coefficients. How can we find the number of contractions? We can look at the diagram we want to make, and see what "ingredients" we have to make it. We have 4 external legs, and two vertices. How many ways can we combine them to get this diagram? We first pick which vertex connects to $\boldsymbol{p}_A$ and $\boldsymbol{p}_B$. There are 2 such choices, since we have two vertices. The choice of which of the 4 legs of the chosen vertex contracts with $\boldsymbol{p}_A$ is another choice, and there are 4 options. Now that we fixed one leg, we have 3 choices for which leg contracts with $\boldsymbol{p}_B$, which gives us 3 choices. Now we need to contract the other vertex with $\boldsymbol{p}_1$, which gives us 4 choices, and then 3 choices for the contraction with $\boldsymbol{p}_2$. To finish, we need to contract the two legs from each vertex with each other. There are 2 choices for the first contraction, and then only 1 choice for the remaining contraction:

$$2 \times (4 \times 3)(4 \times 3) \times 2 = (4!)^2$$

Putting these all together, we can find the symmetry factor:

$$\frac{1}{2!}\left(\frac{1}{4!}\right)^2 \times (4!)^2 = \frac{1}{2}$$

Which matches what we originally stated.

8.5. Yukawa Theory

In the Yukawa theory, the interaction Hamiltonian is of the form:

$$H_{\text{int}} = \int \mathrm{d}^3x \, g\overline{\psi}\psi\phi$$

We have a fermion field coupled to a scalar field in the interaction term.

Let us consider 2 fermions (p,k) scattering into two fermions (p',k'). Now the question is, what is the lowest power of H_{int} that can give rise to this process? Can we do it with just one?

$$\langle p',k'| \int \mathrm{d}^4x\, g\overline{\psi}\psi\phi |p,k\rangle$$

We see that we can't do anything with the scalar field, we can't contract it since there is no scalar in the output state or input state. However, let us consider two powers of H_{int}:

$$\left\langle p',k' \middle| \hat{T}\left[\int \mathrm{d}^4x\, g\overline{\psi}\psi\phi \int \mathrm{d}^4y\, g\overline{\psi}\psi\phi\right] \middle| p,k\right\rangle$$

Now we can have the scalar fields contract with each other, and then the fermion fields contract with the fermion states, and everything works out. Let us now consider the contraction of a fermion field with a fermion with momentum $\boldsymbol{p}$ and spin s:

$$\begin{aligned}\hat{\psi}(x)\left|\boldsymbol{p},s\right\rangle &= \int \frac{\mathrm{d}^3p'}{(2\pi)^3}\frac{1}{\sqrt{2\omega_{p'}}}\sum_{s'}\hat{a}^{s'}_{\boldsymbol{p}'}u^{s'}\left(p'\right)e^{-ip'\cdot x} \underbrace{\left|\boldsymbol{p},s\right\rangle}_{\sqrt{2\omega_p}\hat{a}^{s\dagger}_{\boldsymbol{p}}|0\rangle} \\ &= e^{-ip\cdot x}u^s(p)\left|0\right\rangle\end{aligned}$$

This is what happens when we contract a fermion with an external state. In the case of two or more contractions with fermions, we have to keep track of signs. For example:

$$\hat{\psi}(y)\hat{\psi}(x)\left|p_1,s_1;p_2,s_2\right\rangle = \text{const.}\times e^{-ip_1\cdot x}e^{-ip_2\cdot y}u^{s_1}(p_1)\,u^{s_2}(p_2)$$

Where the constant is either $+1$ or -1, dependent on the order in which we construct our state:

$$\left|p_1,s_1;p_2,s_2\right\rangle \sim \hat{a}^\dagger_{\boldsymbol{p}_1}\hat{a}^\dagger_{\boldsymbol{p}_2}\left|0\right\rangle$$
or
$$\left|p_1,s_1;p_2,s_2\right\rangle \sim \hat{a}^\dagger_{\boldsymbol{p}_2}\hat{a}^\dagger_{\boldsymbol{p}_1}\left|0\right\rangle$$

It turns out that when we are looking at scattering amplitudes the choice here will not end up matter, which is why we leave it indeterminate. Suppose we instead switch the contraction order, we just pick up a relative minus sign:

$$\hat{\psi}(y)\hat{\psi}(x)\left|p_1,s_1;p_2,s_2\right\rangle = (-1)\,\hat{\psi}(y)\hat{\psi}(x)\left|p_1,s_1;p_2,s_2\right\rangle$$

Now let us return to our second order amplitude. We know that the ϕ terms must contract with each other, but the other contractions can take on different permutations, each of the ψs can contract with any of the external states. Let us single out a particular contraction:

$$\langle p',k'|\left(-\frac{g^2}{2!}\right)\int \mathrm{d}^4x\,\overline{\psi}\psi\phi\int \mathrm{d}^4y\,\overline{\psi}\psi\phi|p,k\rangle$$

This equals, up to a possible minus sign:

$$= \frac{(-ig)^2}{2}\int \mathrm{d}^4x\int \mathrm{d}^4y\,\overline{u}\left(k'\right)u(k)\,\overline{u}\left(p'\right)u(p)\,e^{ip'\cdot y}e^{ik'\cdot x}e^{-ip\cdot y}e^{-ik\cdot x}D(x-y)$$

$$= \frac{(-ig)^2}{2} \overline{u}(p') u(p) \overline{u}(k') u(k) \underbrace{\int d^4x \int d^4y \, e^{iy\cdot(p'-p)} e^{ix\cdot(k'-k)} \int \frac{d^4q}{(2\pi)^4} \frac{e^{-iq\cdot(x-y)}}{q^2 - m_\phi^2 + i\varepsilon}}_{\int \frac{d^4q}{(2\pi)^4} (2\pi)^4 \delta^4(p'-p+q)(2\pi)^4 \delta^4(k'-k-q) \frac{i}{q^2-m_\phi^2+i\varepsilon}}$$

$$= (2\pi)^4 \delta^4 (p' + k' - p - k) \left[\frac{(ig)^2}{2} \overline{u}(p') u(p) \overline{u}(k') u(k) \frac{i}{(p'-p)^2 - m_\phi^2} \right]$$

Note that in the last step, we have set $\varepsilon = 0$, because we no longer expect this to lead to divergences[28]. The nontrivial result of this expression is the contribution to the matrix element:

$$iM = \frac{-ig^2}{2} \frac{\overline{u}(p') u(p) \overline{u}(k') u(k)}{(p'-p)^2 - m_\phi^2}$$

Recall that this is just one of the possible contraction choices. However, since x and y are dummy variables, imagine flipping the contractions between the integrals. In that case, since the ϕ s are bosonic, we can swap them without picking up any extra factors. Thus, after relabelling $x \leftrightarrow y$, we have the same exact contraction, and the same exact amplitude. Therefore, this contraction appears twice. This entire process was very laborious, so let us instead see this from the Feynman rules.

We can draw the Feynman diagram for our choice of contraction, shown in Figure 9.

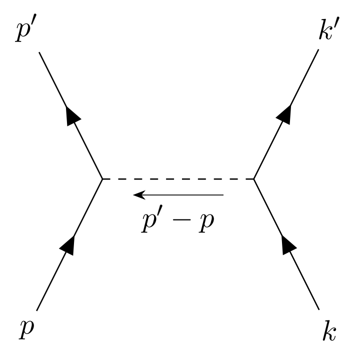


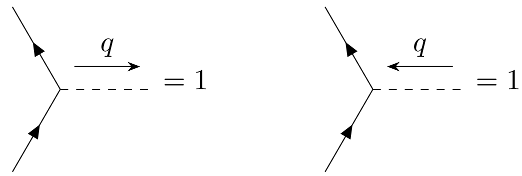
Figure 9: Diagram for 2 to 2 fermion scattering in the Yukawa theory. The dashed line denotes the scalar field contraction.

We can now apply the Feynman rules to this diagram. In this theory, the Feynman rules are as follows:

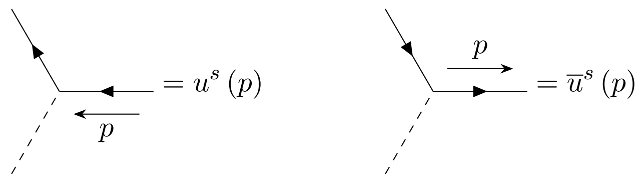
1. The scalar propagator, with momentum q, gives $\frac{i}{q^2 - m_\phi^2 + i\varepsilon}$
2. The fermion propagator, with momentum p, gives $\frac{i(\not{p}) - m_f}{p^2 - m_f^2 + i\varepsilon}$
3. Vertices give a factor of $-ig$
4. External legs are dependent on the propagator type:

[28]This is really handwavy...

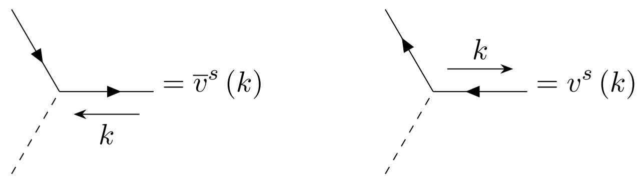
a) Scalar external legs:



b) Fermion external legs:



c) Antifermion external legs:



Note that the arrow on the fermion propagator is not the direction of the momentum, it is the direction of particle flow. For antifermions, we need a separate arrow labelling the momentum.

If we apply these rules to the diagram in Fig. 9, first, let us handle the fermion lines one by one. When handling a fermion line, we go opposite the direction of the arrow. Consider the p' propagator. We have a $\overline{u}(p')$, then a vertex $(-ig)$, and then the p propagator, $u(p)$. Now the k' propagator gives us a $\overline{u}(k')$, the vertex gives us $-ig$, and the k propagator gives us a $u(k)$. Finally, the scalar propagator gives us a $\frac{i}{(p'-p^2)-m_\phi^2}$. Putting these together:

$$\overline{u}\left(p'\right)(-ig)\,u(p)\,\overline{u}\left(k'\right)(-ig)\,u(k)\,\frac{i}{(p'-p)^2-m_\phi^2}$$

Now comparing this against what we found earlier, we see that we are off by a factor of two. Suppose we now consider the symmetry factor of this diagram. We have a factor of $\frac{1}{2!}$ since we are second order in perturbation theory, and we have 2 outgoing propagators, 2 vertices, and 2 incoming propagators. However, note that there is only 1 way to connect the vertices, since we know the scalar propagators from the vertices must connect. Thus, there are exactly 2 ways to make this diagram, which cancels the 2!. Thus, the symmetry factor is not important here, since it turns out that both equivalent contractions are actually the same diagram, so the Feynman rules provided the correct solution without even thinking about the symmetry factor.

This is not the only diagram that we can draw, there is a second diagram for this process, as shown in Figure 10.

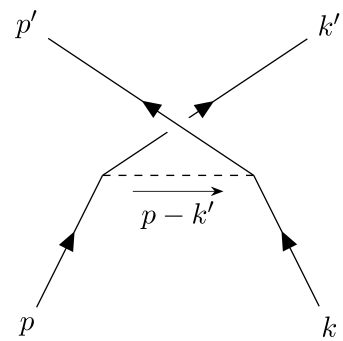


Figure 10: Second diagram for 2 to 2 fermion scattering in the Yukawa theory.

We can apply the Feynman rules to this diagram. Consider the k' propagator, which we follow down to p, and we get $\overline{u}\left(k'\right)\left(-ig\right)u\left(p\right)$. The second fermion line gives us $\overline{u}\left(p'\right)\left(-ig\right)u\left(k\right)$. We are then left with the scalar propagator, $\frac{i}{\left(p-k'\right)^2-m_\phi^2}$. All together:

$$(-1)\overline{u}\left(k'\right)\left(-ig\right)u\left(p\right)\overline{u}\left(p'\right)\left(-ig\right)u\left(k\right)\frac{i}{\left(p-k'\right)^2-m_\phi^2}$$

Note that there is a minus sign relative to the other diagram! The quick and dirty explanation for this is that we switched which fermion propagator leads to the each output state, but we can look at it at the contraction level:

$$\langle p', k'|\left(-\frac{g^2}{2!}\right)\int \mathrm{d}^4x\,\overline{\psi}\psi\phi\int \mathrm{d}^4y\,\overline{\psi}\psi\phi|p, k\rangle$$

We flipped the contractions between the ψs and the input states. This will produce a relative minus sign.

The total scattering amplitude is the sum of these two contributions:

$$iM = \left(-1g^2\right)\left[\overline{u}\left(p'\right)u\left(p\right)\frac{1}{\left(p'-p\right)^2-m_\phi^2}\overline{u}\left(k'\right)u\left(k\right) - \overline{u}\left(p'\right)u\left(k\right)\frac{1}{\left(p'-k\right)^2-m_\phi^2}\overline{u}\left(k'\right)u\left(p\right)\right]$$

Now let us consider a different process, fermion-antifermion annihilation into 2 bosons. One diagram for this process is given in Figure 11.

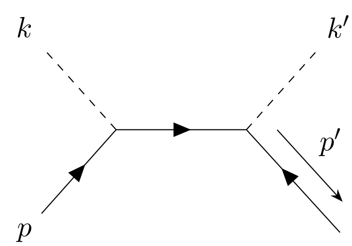


Figure 11: Diagram for fermion antifermion annihilation process.

Note that the direction of the fermion propagators must be consistent, the arrows cannot start pointing in different directions, we have to be able to trace backwards along the outgoing propagators. We can, using the Feynman rules, just write down the amplitude:

$$(-ig)^2\,\overline{v}\left(p'\right)\frac{i\left[\left(\not{p}-\not{k}\right)+m_f\right]}{(p-k)^2-m_\phi^2+i\varepsilon}u\left(p\right)$$

There is another diagram, in Figure 12.

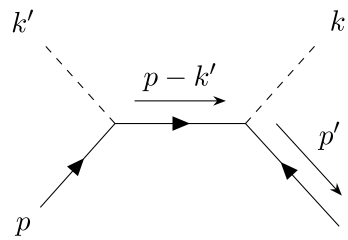


Figure 12: Second diagram for the fermion antifermion annihilation process.

This figure differs from Fig. 11 in that the order of the scalar momenta are now swapped. It has amplitude:

$$\overline{v}\left(p'\right)(-ig)\,\frac{i\left[\not{p}-\not{k}'\right]+m_f}{(p-k')^2-m_\phi^2}\,(-ig)\,u\left(p\right)$$

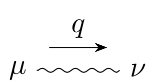
8.6. QED Feynman Rules

The interaction term in QED is given by:

$$H_{\text{int}}=\int \mathrm{d}^3x\, e\frac{1}{4}\gamma^\mu\hat{\psi}\hat{A}_\mu$$

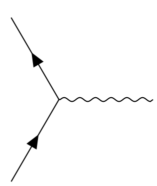
External fermion propagators are the same as in the Yukawa theory, so let us discuss the new rules for QED:

1. Photon propagators:



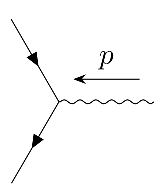
$$=-\frac{ig_{\mu\nu}}{q^2+i\varepsilon}$$

2. Vertices:

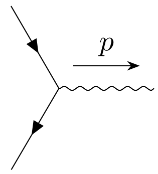


$$=-ie\gamma^\mu$$

3. External photon legs:



$$\rightarrow A_\mu\left|p\right\rangle=\varepsilon_\mu\left(p\right)$$



$$\rightarrow \langle p| A_{\mu} = \varepsilon_{\mu}^{*}$$

Recall that we expanded the electromagnetic field as:

$$A_{\mu}(x) = \int \frac{\mathrm{d}^3\boldsymbol{p}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \sum_{r=0}^{3} \left[\hat{a}_{\boldsymbol{p}}^{r} \varepsilon_{\mu}^{r}(\boldsymbol{p}) e^{-ip\cdot x} + \hat{a}_{\boldsymbol{p}}^{r\dagger} \varepsilon_{\mu}^{r}(\boldsymbol{p}) e^{ip\cdot x}\right]$$

8.7. Cross Sections and Decays

We are now (in principle) able to write down the matrix element for a process in these interacting theories. How do we go from these matrix elements to physical observables such as cross sections and decay rates? We will simply provide the expressions, derivations can be found in textbooks.

The differential cross section $d\sigma$ for the process $(p_A, p_B \to p_f)$ is given by:

$$\mathrm{d}\sigma = \frac{1}{2E_A 2E_B |v_A - v_B|} \prod_{f=1}^{N} \frac{\mathrm{d}^3 p_f}{(2\pi)^3} \frac{1}{2E_f} |M(p_A, p_B \to p_F)|^2 (2\pi)^4 \delta^4(p_A + p_B - \sum_f p_f)$$

Note that this assumes that p_A and p_B are colinear, a more general formula (where the particles are offset by an angle) can be found in Weinberg. Here, $|v_A - v_B|$ is the relative velocity of the two particles. Note that this is the nonrelativistic difference in velocity, not the relativistic relative velocity. Even so, this equation is invariant to boosts along the colinear direction.

This is a general formula, it holds for $2 \to n$ scattering, but in practice we often care about $2 \to 2$ or $2 \to 3$ scattering. In these lower particle cases, we can apply simplifications. For example, for $2 \to 2$:

$$\left(\frac{d\sigma}{d\Omega}\right)_{\mathrm{CM}} = \frac{1}{2E_A 2E_B |v_A - v_B|} \frac{|\boldsymbol{p}_1|^2}{(2\pi)^2 4E_{\mathrm{CM}}} |M(p_A, p_B \to p_1, p_2)|^2$$

Where E_{CM} is the center of mass energy of the initial state particles.

We also have a very special case where all 4 particles have the same mass (which doesn't happen very often):

$$\left(\frac{d\sigma}{d\Omega}\right)_{\mathrm{CM}} = \frac{|M|^2}{64\pi^2 E_{\mathrm{CM}}^2}$$

Now let us consider decay rates. In the rest frame of a decaying particle, the differential decay rate of the particle is given by:

$$\mathrm{d}\Gamma = \frac{1}{2m_A} \prod_{i=1}^{f} \frac{\mathrm{d}^3 p_f}{(2\pi)^3} \frac{1}{2E_f} |M(p_A \to p_f)|^2 (2\pi)^4 \delta^4(p_A - \sum_f p_f)$$

Let us discuss one final consideration, which is identical final state particles. When we integrate over the differential cross section to get the total cross section, we integrate over all final state momenta. To avoid double counting in the case of n identical particles in the final state, when integrating to find the total cross section or decay rate, we divide by $n!$.

8.8. Tree level Processes in QED

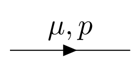
We will now consider several processes in QED. The first process will be the $e^+e^- \to \mu^+\mu^-$ process, which is the electron positron to muon antimuon process[29]. Let us write down the QED Lagrangian in the case of electrons and muons:

$$\mathcal{L} = \overline{\psi}_e \left(i\gamma^\mu \partial_\mu - m_e\right) \psi_e + \overline{\psi}_\mu \left(i\gamma^\mu \partial_\mu - m_\mu\right) \psi_\mu - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - e\overline{\psi}_e \gamma^\mu \psi_e A_\mu - e\overline{\psi}_\mu \gamma^\mu \psi_\mu A_\mu$$

Note that ψ_μ and m_μ are not Lorentz indices, but refer to the muon field and muon mass respectively.

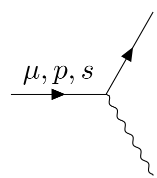
The Feynman rules for the muon are similar to those of the electron, just with a different mass, which affects both the propagator and the external leg contractions.

1. Muon propagator:

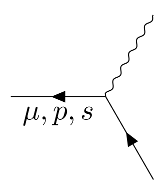


$$= \frac{i\left(\not{p} + m_\mu\right)}{p^2 - m_\mu^2 + i\varepsilon} = \frac{i}{\not{p} - m_\mu + i\varepsilon}$$

2. muon legs

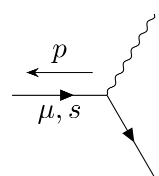


$$= u_\mu^s(p)$$

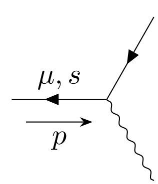


$$= \overline{u}_\mu^s(p)$$

3. antimuon legs



$$= v_\mu^s(p)$$

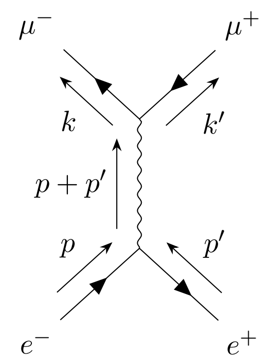


$$= \overline{v}_\mu^s(p)$$

8.8.1. $e^+e^- \to \mu^+\mu^-$

With these rules in mind, let us now compute the $e^+e^- \to \mu^+\mu^-$ process, which we chose because it only has 1 Feynman diagram:

[29] The muon is like a heavier electron, $m_\mu \sim 200 m_e$



We can look at the matrix element for this process:

$$\begin{aligned} iM &= \underbrace{\overline{v}\left(p'\right)\left(-ie\gamma^\mu\right)u\left(p\right)}_{\text{fermion}} \frac{-ig_{\mu\nu}}{\left(p+p'\right)^2+i\varepsilon} \underbrace{\overline{u}\left(k\right)\left(-ie\gamma^\nu\right)v\left(k'\right)}_{\text{muon}} \\ &= \frac{ie^2}{\left(p+p'\right)^2+i\varepsilon}\left[\overline{v}\left(p'\right)\gamma^\mu u\left(p\right)\right]_{\text{electron}}\left[\overline{u}\left(k\right)\gamma_\mu v\left(k'\right)\right]_{\text{muon}} \end{aligned}$$

Note that we are not keeping track of the superscripts on the us and vs, since we are calculating the unpolarized case. We want to compute the cross section, so we need $|M|^2$. We must first find M^*. Looking at the term

$$\begin{aligned} \left(\overline{v}\gamma^\mu u\right)^* &= \left(\overline{v}\gamma^\mu u\right)^\dagger \\ &= u^\dagger\left(\gamma^\mu\right)^\dagger\left(\gamma^0\right)^\dagger v \\ &= u^\dagger\gamma^0\gamma^\mu v \\ &= \overline{u}\gamma^\mu v \end{aligned}$$

Where we note that commuting γ^0 past γ^μ applies a $\dagger$ to the γ^μ, $\gamma^\mu\gamma^0 = \gamma^0\left(\gamma^\mu\right)^\dagger$. If we apply these tricks to the rest of M, we find that:

$$|M|^2 = \frac{e^4}{\left(p+p'\right)^4}\left[\overline{v}\left(p'\right)\gamma^\mu u\left(p\right)\overline{u}\left(p\right)\gamma^\nu v\left(p'\right)\right]_{\text{electron}}\left[\overline{u}\left(k\right)\gamma_\mu v\left(k'\right)\overline{v}\left(k'\right)\gamma_\nu u\left(k\right)\right]_{\text{muon}}$$

Recall that we are working with unpolarized electrons, so we average over initial spins s and s':

$$\left(\frac{1}{2}\sum_s\right)\left(\frac{1}{2}\sum_{s'}\right)|M|^2$$

If we are not interested in the spins of the final state particles, we just sum over spins of the final state particles:

$$\sum_r\sum_{r'}|M|^2$$

Thus, we need to compute:

$$\frac{1}{4}\sum_{s,s'}\sum_{r,r'}|M(s,s'\to r,r')|^2$$

Naively, computing this seems very difficult, but there is a trick! To compute spin sums, we use the fact that

$$\sum_s u^s(p)\,\overline{u}^s(p) = \not{p} + m$$
$$\sum_s v^s(p)\,\overline{v}^s(p) = \not{p} - m$$

How do these help us? We can write out a part of our spin sum explicitly with all of the indices:

$$\sum_{s,s'} \overline{v}_a^{s'}\left(p'\right)\gamma^\mu_{ab} u^s_b(p)\,\overline{u}^s_c(p)\,\gamma^\nu_{cd} v^{s'}_d\left(p'\right)$$

Now we note that things are just numbers, since the are all matrix elements. We then note that $u^s_b(p)\,\overline{u}^s_c(p) = \left(\not{p} + m_e\right)_{bc}$, and similarly $\overline{v}^{s'}_a(p')\,v^s_d(p') = \left(\not{p}' - m_e\right)_{da}$:

$$\left(\not{p}' - m_e\right)_{da}\gamma^\mu_{ab}\left(\not{p} + m_e\right)_{bc}\gamma^\nu_{cd}$$

Now we note that this is just a trace!

$$\left(\not{p}' - m_e\right)_{da}\gamma^\mu_{ab}\left(\not{p} + m_e\right)_{bc}\gamma^\nu_{cd} = \mathrm{Tr}\left[\left(\not{p}' - m_e\right)\gamma^\mu\left(\not{p} + m_e\right)\gamma^\nu\right]$$

Doing the same thing for the muon spin sum of M:

$$\begin{aligned}\sum_{r,r'} \overline{u}^r_a(k)\left(\gamma_\mu\right)_{ab} v^{r'}_b\left(k'\right)\overline{v}^r_c\left(k'\right)\left(\gamma_\nu\right)_{cd} u^r_d(k) &= \left(\not{k} + m_\mu\right)_{da}\left(\gamma_\mu\right)_{ab}\left(\not{k}' - m_\mu\right)_{bc}\left(\gamma_\nu\right)_{cd} \\ &= \mathrm{Tr}\left[\left(\not{k} + m_\mu\right)\gamma_\mu\left(\not{k}' - m_\mu\right)\gamma_\nu\right]\end{aligned}$$

Thus, we are left with

$$\frac{1}{4}\sum_{\text{spins}} |M|^2 = \frac{e^4}{4\left(p + p'\right)^4}\mathrm{Tr}\left[\left(\not{p}' - m_e\right)\gamma^\mu\left(\not{p} + m_e\right)\gamma^\nu\right]\mathrm{Tr}\left[\left(\not{k} + m_\mu\right)\gamma_\mu\left(\not{k}' - m_\mu\right)\gamma_\nu\right]$$

To make progress, we need to now compute these traces[30]. We utilize a set of important identities:

$$\begin{aligned}\mathrm{Tr}\left[\mathbb{I}\right] &= 4 \\ \mathrm{Tr}\left[\text{odd number of } \gamma \text{ matrices }\right] &= 0 \\ \mathrm{Tr}\left[\gamma^\mu\gamma^\nu\right] &= 4g^{\mu\nu} \\ \mathrm{Tr}\left[\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\right] &= 4\left(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho}\right) \\ \mathrm{Tr}\left[\gamma^5\right] &= 0 \\ \mathrm{Tr}\left[\gamma^\mu\gamma^\nu\gamma^5\right] &= 0 \\ \mathrm{Tr}\left[\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma^5\right] &= -4i\varepsilon^{\mu\nu\rho\sigma}\end{aligned}$$

We also have a set of identities know as the contraction identities:

$$\begin{aligned}\gamma^\mu\gamma_\mu &= 4\mathbb{I} \\ \gamma^\mu\gamma^\nu\gamma_\mu &= -2\gamma^\nu \\ \gamma^\mu\gamma^\nu\gamma^\sigma\gamma_\mu &= 4g^{\nu\sigma}\mathbb{I}\end{aligned}$$

[30]"Peskin attributes the trace trick to Feynman, probably because it was invented by Casimir."

$$\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma_\mu = -2\gamma^\sigma\gamma^\rho\gamma^\nu$$

With these identities, we can now begin to compute these traces:

$$\begin{aligned}\operatorname{Tr}\left[\left(\not{p}'-m_e\right)\gamma^\mu\left(\not{p}+m_e\right)\gamma^\nu\right] &= \underbrace{\operatorname{Tr}\left[\not{p}'\gamma^\mu\not{p}\gamma^\nu\right]}_{p'_\alpha p_\beta \operatorname{Tr}\left[\gamma^\alpha\gamma^\mu\gamma^\beta\gamma^\nu\right]} - m_e^2\underbrace{\operatorname{Tr}\left[\gamma^\mu\gamma^\nu\right]}_{4g^{\mu\nu}} \\ &= 4g^{\mu\nu}g^{\beta\nu} - 4g^{\alpha\beta}g^{\mu\nu} + 4g^{\alpha\nu}g^{\mu\beta} - 4m_e^2g^{\mu\nu} \\ &= 4\left[p'^\mu p^\nu + p'^\nu p^\mu - g^{\mu\nu}\left[\left(p\cdot p'\right)+m_e^2\right]\right]\end{aligned}$$

Looking at the other trace:

$$\operatorname{Tr}\left[\left(\not{k}+m_\mu\right)\gamma_\mu\left(\not{k}'-m_\mu\right)\gamma_\nu\right] = 4\left[k_\mu k'_\nu + k_\nu k'_\mu - g_{\mu\nu}\left[\left(k\cdot k'\right)+m_\mu^2\right]\right]$$

We can now write out the sum over spins, which is the product of these two traces:

$$\begin{aligned}\frac{1}{4}\sum_{\text{spins}}|M|^2 = \frac{e^4}{4\left(p+p'\right)^4}16[&2\left(k\cdot p'\right)\left(k'\cdot p\right) + 2\left(k\cdot p\right)\left(k'\cdot p'\right) - 2\left(p\cdot p'\right)\left(k\cdot k'\right) - 2\left(p\cdot p'\right)\left(k\cdot k'\right) \\ &+ g^{\mu\nu}g_{\mu\nu}\left(p\cdot p'\right)\left(k\cdot k'\right) - 2\left(p\cdot p'\right)m_\mu^2 + g^{\mu\nu}g_{\mu\nu}\left(p\cdot p'\right)m_\mu^2]\end{aligned}$$

Note that we have set the electron mass to zero, we treat it as negligible in this process. These terms all come from the product of the many terms in each of the traces. If we now note that $g^{\mu\nu}g_{\mu\nu}=4$, we can cancel 3 of the terms:

$$\frac{1}{4}\sum_{\text{spins}}|M|^2 = \frac{8e^4}{\left(p+p'\right)^4}\left[\left(p\cdot k\right)\left(p'\cdot k'\right) + \left(p\cdot k'\right)\left(p'\cdot k\right) + m_\mu^2\left(p\cdot p'\right)\right]$$

Since this object is Lorentz invariant (it is the sum of dot products of four-vectors), so we can choose a convenient frame, which we choose to be the center of momentum frame, to simplify the calculation. The frame is shown in Figure 13.

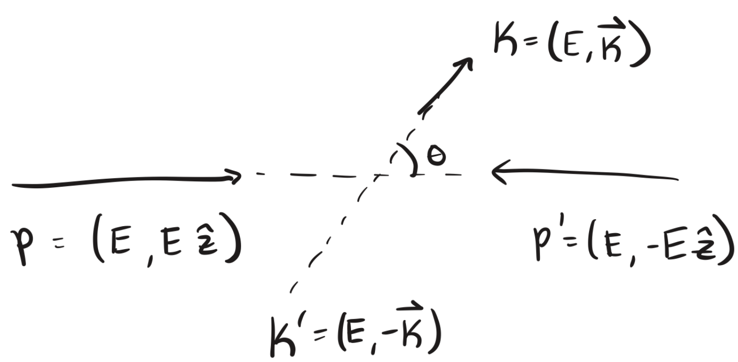


Figure 13: The scattering process in the center of momentum frame.

In this frame, we have that $(p+p')^2 = 4E^2$, $(p\cdot p')^2 = 2E^2$. We can also look at the cross terms:

$$\begin{aligned}
(p\cdot k) &= (p'\cdot k') \\
&= E^2 - \boldsymbol{k}\cdot(E\hat{z}) \\
&= E^2 - |\boldsymbol{k}|\,E\cos\theta \\
(p\cdot k') &= (p'\cdot k) \\
&= E^2 - (-\boldsymbol{k})\cdot(E\hat{z}) \\
&= E^2 + |\boldsymbol{k}|E\cos\theta
\end{aligned}$$

Inserting these relations and doing a bit of algebra:

$$\frac{1}{4}\sum_{\text{spins}} |M|^2 = e^4\left[\left(1+\cos^2\theta\right) + \frac{m_\mu^2}{E^2}\left(1-\cos^2\theta\right)\right]$$

Where we have used the fact that $|\boldsymbol{k}|^2 = E^2 - m_\mu^2$.

With this expression, we can now find the differential cross section in the center of momentum frame:

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{CM}} = \frac{1}{4E^2\,|v_A - v_B|}\frac{|\boldsymbol{k}|}{(2\pi)^2\,4\,(2E)}\left[\frac{1}{4}\sum_{\text{spins}}|M|^2\right]$$

Now noting that $|v_A - v_B| = 2$:

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{CM}} = \frac{\alpha^2}{16E^2}\sqrt{1-\frac{m_\mu^2}{E^2}}\left[\left(1+\frac{m_\mu^2}{E^2}\right)+\cos^2\theta\left(1-\frac{m_\mu^2}{E^2}\right)\right]$$

Where $\alpha = \frac{e^2}{4\pi}$.

We can now integrate this to get the total cross-section:

$$\sigma_{\text{total}} = \frac{\pi\alpha^2}{3E^2}\sqrt{1-\frac{m_\mu^2}{E^2}}\left[1+\frac{1}{2}\frac{m_\mu^2}{E^2}\right]$$

8.8.2. Compton Scattering

Compton scattering is the process $e^- + \gamma \to e^- + \gamma$. We want to find the unpolarized cross section for this process. Let us say that the incoming electron momentum is p, the incoming photon momentum is k, and the outgoing momenta are p' and k'. There are two diagrams for this process, as show in Figure 14.

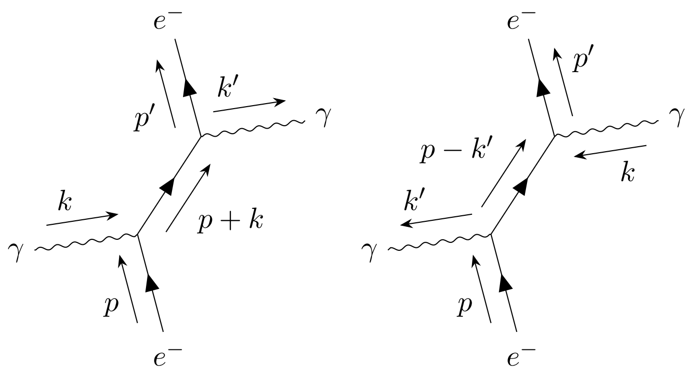


Figure 14: The two leading order Feynman diagrams for Compton scattering, $e^- + \gamma \to e^- + \gamma$.

We can look at the matrix elements for the first diagram:

$$iM_1 = \overline{u}\left(p'\right)\left(-ie\gamma^\mu\right)\varepsilon^*_\mu\left(k'\right)\frac{i\left(\not{p}+\not{k}+m\right)}{\left(p+k\right)^2-m^2}\left(-ie\gamma^\nu\right)\varepsilon_\nu\left(k\right)u\left(p\right)$$

And the matrix element for the second diagram:

$$iM_2 = \overline{u}\left(p'\right)\left(-ie\gamma^\nu\right)\varepsilon_\nu\left(k\right)\frac{i\left(\not{p}-\not{k}'+m\right)}{\left(p-k'\right)^2-m^2}\left(-ie\gamma^\mu\right)\varepsilon^*_\mu\left(k'\right)u\left(p\right)$$

The total matrix element is then:

$$\begin{aligned}
iM &= \overline{u}\left(p'\right)\left(-ie\gamma^\mu\right)\varepsilon^*_\mu\left(k'\right)\frac{i\left(\not{p}+\not{k}+m\right)}{\left(p+k\right)^2-m^2}\left(-ie\gamma^\nu\right)\varepsilon_\nu\left(k\right)u\left(p\right) \\
&\quad+\overline{u}\left(p'\right)\left(-ie\gamma^\nu\right)\varepsilon_\nu\left(k\right)\frac{i\left(\not{p}-\not{k}'+m\right)}{\left(p-k'\right)^2-m^2}\left(-ie\gamma^\mu\right)\varepsilon^*_\mu\left(k'\right)u\left(p\right) \\
&= \left[\left(-ie^2\right)\varepsilon^*_\mu\left(k'\right)\varepsilon_\nu\left(k\right)\right]\overline{u}\left(p'\right)\left[\frac{\gamma^\mu\left(\not{p}+\not{k}+m\right)\gamma^\nu}{\left(p+k\right)^2-m^2}+\frac{\gamma^\nu\left(\not{p}-\not{k}'+m\right)\gamma^\mu}{\left(p-k'\right)^2-m^2}\right]u\left(p\right)
\end{aligned}$$

Now we note that $\left(p+k\right)^2-m^2 = 2p\cdot k$, and $\left(p-k'\right)^2-m^2 = -2p\cdot k'$. We can also note that

$$\begin{aligned}
\left(\not{p}+m\right)\gamma^\nu u\left(p\right) &= \left(p_\alpha\gamma^\alpha+m\right)\gamma^\nu u\left(p\right) \\
&= \left[2p^\nu-\gamma^\nu\left(\not{p}-m\right)\right]u\left(p\right) \\
&= 2p^\nu u\left(p\right)
\end{aligned}$$

If we make all of these simplifications, we are left with

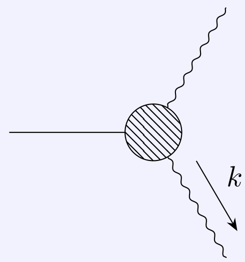
$$iM = \left[-ie^2\varepsilon^*_\mu\left(k'\right)\varepsilon_\nu\left(k\right)\right]\overline{u}\left(p'\right)\left[\frac{\gamma^\mu\not{k}\gamma^\nu+2\gamma^\mu p^\nu}{2p\cdot k}-\frac{2\gamma^\nu p^\mu-\gamma^\nu\not{k}'\gamma^\mu}{2p\cdot k'}\right]u\left(p\right)$$

We now have to square this and do the sums over the polarizations. Previously, we just had external fermions, but now we also have external photons, so we need to sum over the external photon polarizations:

$$\sum_{\text{polarizations}}\varepsilon^*_\mu\left(k\right)\varepsilon_\nu\left(k\right)\to -g_{\mu\nu}$$

We are allowed to make this substitution when computing a physical amplitude, this is not true in general, hence the lack of an equals sign. This is not an actual identity, and is only valid in a QED amplitude, $|M|^2$.

Let us discuss some of the steps that allow us to make this replacement. Consider an arbitrary QED process, involving an external photon momentum k:



We know that the external photon will give a factor of ε^*:

$$iM = iM^\mu(k)\,\varepsilon^*_\mu(k)$$

The cross section will be proportional to

$$\sum_{\text{polarizations}} \left|M^\mu(k)\,\varepsilon^*_\mu(k)\right|^2 = \sum_{\text{polarizations}} \varepsilon^*_\mu(k)\,\varepsilon_\nu(k)\,M^\mu(k)\,M^{\nu *}(k)$$

Now let us choose coordinates such that k^μ is along the z axis:

$$k^\mu = (k, 0, 0, k)$$

Then, the two polarization vectors can be chosen as*

$$\varepsilon_1^\mu = (0, 1, 0, 0)$$
$$\varepsilon_2^\mu = (0, 0, 1, 0)$$

Wtih these conventions:

$$\sum_{\text{polarizations}} \left|\varepsilon^*_\mu(k)\,M^\mu(k)\right|^2 = \left|M^1(k)\right|^2 + \left|M^2(k)\right|^2$$

Let us leave these for a moment. Let us introduce the *Ward Identity*:

$$k_\mu M^\mu(k) = 0$$

That is, the amplitude M vanishes when $\varepsilon_\mu(k)$ is replaced by k_μ. We will not prove this† For $k^\mu = (k, 0, 0, k)$, this implies that

$$kM^0(k) - kM^3(k) = 0$$

Which implies that

$$M^0(k) = M^3(k)$$

This is the key result we need from the Ward identity. Let us return to our arbitrary vertex diagram. We can now write out the amplitude as:

$$\sum_{\text{polarizations}} \varepsilon_\mu^*(k)\,\varepsilon_\nu(k)\,M^\mu(k)\,M^{\nu*}(k) = \left|M^1(k)\right|^2 + \left|M^2(k)\right|^2 + \underbrace{\left|M^3(k)\right|^2 - \left|M^0(k)\right|^2}_{0}$$
$$= -g_{\mu\nu}M^\mu(k)\,M^{\nu*}(k)$$

From this, we see that we can make the replacement

$$\sum_{\text{polarizations}} \varepsilon_\mu^*(k)\,\varepsilon_\mu(k) \to -g_{\mu\nu}$$

*Recall that only two of the polarizations are physical, the two transverse polarizations.

†To see a proof of this, please see the extra lecture, Wednesday Dec. 17th, from 11-12.

Now let us return to the case of Compton scattering. We now need to compute $|M|^2$, average over the initial spins, and sum over the final spins (since we are looking at the unpolarized cross section). If we do this, skipping a lot of work, we can use the above result to remove the polarization sums, leaving the sum over the incoming and outgoing fermion spins:

$$\frac{1}{4}\sum_{\text{spins, pol.}}|M|^2 = \frac{e^4}{4}g_{\mu\rho}g_{\nu\sigma}\sum_{r,r'}\left[\overline{u}_a^{r'}(p')\left[\frac{\gamma^\mu \not{k}\gamma^\nu + 2\gamma^\mu p^\nu}{2p\cdot k}\right]_{ab}u_b^r(p) - \overline{u}_a^{r'}(p')\left[\frac{2\gamma^\nu p^\mu - \gamma^\nu \not{k}'\gamma^\mu}{2p\cdot k'}\right]_{ab}u_b^r(p)\right]$$
$$\left[\overline{u}_{b'}^r(p)\left[\frac{\gamma^\sigma \not{k}\gamma^\rho + 2\gamma^\rho p^\sigma}{2p\cdot k}\right]_{b'a'}u_{a'}^r(p') - \overline{u}_{b'}^r(p')\left[\frac{2\gamma^\sigma p^\rho - \gamma^\rho \not{k}'\gamma^\sigma}{2p\cdot k'}\right]_{b'a'}u_{a'}^{r'}(p')\right]$$
$$= \frac{e^4}{4}g_{\mu\rho}g_{\nu\sigma}\mathrm{Tr}\Bigg[\left(\not{p}'+m\right)\left[\frac{\gamma^\mu \not{k}\gamma^\nu + 2\gamma^\mu p^\nu}{2p\cdot k} + \frac{\gamma^\nu \not{k}'\gamma^\mu - 2\gamma^\nu p^\nu}{2p\cdot k'}\right]\left(\not{p}+m\right)$$
$$\left[\frac{\gamma^\sigma \not{k}\gamma^\rho + 2\gamma^\rho p^\sigma}{2p\cdot k} + \frac{\gamma^\rho \not{k}'\gamma^\sigma - 2\gamma^\sigma p^\rho}{2p\cdot k'}\right]\Bigg]$$

In the end, we have:

$$\frac{1}{4}\sum_{\text{spins, pol.}}|M|^2 = \frac{e^4}{4}\left[\frac{\mathrm{I}}{(2p\cdot k)^2} + \frac{\mathrm{II}}{(2p\cdot k)(2p\cdot k')} + \frac{\mathrm{III}}{(2p\cdot k')(2p\cdot k)} + \frac{\mathrm{IV}}{(2p\cdot k')^2}\right]$$

Where I, II, III, and IV are all complicated traces. We now note that I $\to$ IV if we replace k with k', and therefore we only need to do one of these traces. We also note a useful identity:

$$\mathrm{Tr}\left[\gamma^\alpha\gamma^\beta\gamma^\sigma\cdots\gamma^\rho\gamma^\lambda\right] = \mathrm{Tr}\left[\gamma^\lambda\gamma^\rho,\cdots\gamma^\sigma\gamma^\beta\gamma^\alpha\right]$$

That is, we can reverse the order of the gamma matrices inside a trace[31]. With this property, we can in fact show that II and III are equivalent. Thus, instead of computing 4 traces, we just have to compute 2 of them. Let us look at a term in I:

$$\mathrm{Tr}\left[\not{p}'\gamma^\mu\not{k}\gamma^\nu\not{p}\gamma^\sigma\not{k}\gamma^\rho\right]g_{\mu\rho}g_{\nu\sigma} = \mathrm{Tr}\left[\gamma^\rho\not{p}'\gamma^\mu\not{k}\gamma^\nu\not{p}\gamma^\sigma\not{k}\right]g_{\mu\rho}g_{\nu\sigma}$$
$$= \mathrm{Tr}\left[\gamma_\mu\not{p}'\gamma^\mu\not{k}\gamma^\nu\not{p}\gamma_\nu\not{k}\right]$$

[31]This is different than the cyclic property of the trace, this is true for the gamma matrices, not in general!

$$\begin{aligned}
&= \mathrm{Tr}\left[\left(-2\not{p}'\right)\not{k}\left(-2\not{p}\right)\not{k}\right] \\
&= 4\mathrm{Tr}\left[\not{p}'\not{k}\not{p}\not{k}\right] \\
&= 4\mathrm{Tr}\left[\not{p}'\not{k}\left(2p\cdot k - \not{k}\not{p}\right)\right]
\end{aligned}$$

Now recall that $k^2 = 0$, since it is an external photon. From this, we have that the trace is equal to:

$$\begin{aligned}
4\mathrm{Tr}\left[\not{p}'\not{k}\left(2p\cdot k - \not{k}\not{p}\right)\right] &= 8\left(p\cdot k\right)\mathrm{Tr}\left[\not{p}'\not{k}\right] \\
&= 32\left(p\cdot k\right)\left(p'\cdot k\right)
\end{aligned}$$

The other terms can be evaluated in a similar fashion[32]. It is useful to remember certain relations:

$$p^2 = p'^2 = m^2$$

And some dot product relations:

$$\begin{aligned}
(p+k)^2 = \left(p'+k'\right)^2 &\to p\cdot k = p'\cdot k' \\
\left(p-k'\right)^2 = \left(p'-k\right)^2 &\to p\cdot k' = p'\cdot k
\end{aligned}$$

Finally, one obtains:

$$\frac{1}{4}\sum_{\text{spins, pol.}}|M|^2 = 2e^4\left[\frac{p\cdot k'}{p\cdot k} + \frac{p\cdot k}{p\cdot k'} + 2m^2\left(\frac{1}{p\cdot k} - \frac{1}{p\cdot k'}\right) + m^4\left(\frac{1}{p\cdot k} - \frac{1}{p\cdot k'}\right)^2\right]$$

Let us now pick a special frame. In the case of Compton scattering, we usually pick the lab frame, shown in Figure 15.

Figure 15

We choose the electron to be at rest initially, $p = (m, \mathbf{0})$, and the photon is incoming in the z direction, $k = (\omega, \omega\hat{z})$. After the collision, the electron has momentum $p' = (E', \boldsymbol{p}')$, and the photon has momentum $k' = (\omega', \omega'\sin\theta\cos\phi, \omega'\sin\theta\sin\phi, \omega'\cos\theta)$.

Energy-momentum conservation gives us:

$$p' = p + k - k' \to \left(p'\right)^2 = \left(p + k' - k'\right)^2$$

From which we can derive that

$$m^2 = m^2 + 2m\left(\omega - \omega'\right) - 2\omega\omega'\left(1 - \cos\theta\right)$$

Which leads to Compton's famous formula[33]:

$$\frac{1}{\omega'} = \frac{1}{\omega} + \frac{1}{m}\left(1 - \cos\theta\right)$$

We can rewrite this, for use later:

$$\omega' = \frac{\omega}{1 + \frac{\omega}{m}\left(1 - \cos\theta\right)}$$

[32]...

[33]Perhaps more usually written as $\Delta\lambda = \frac{h}{mc}\left(1 - \cos\theta\right)$.

Now let us consider the phase space integral in the lab frame.

$$\int \mathrm{d}\pi_2 = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \frac{1}{2\omega'} \frac{\mathrm{d}^3 p'}{(2\pi)^3} \frac{1}{2E'} (2\pi)^4 \delta^4 \left(p' + k' - p - k\right)$$

We see that we have 6 integrals, and 4 delta functions. One of the remaining integrals will be the integral over ϕ, which will end up being trivial. Let us first eliminate some of the integrals using the delta function. We will use 3 of the delta functions to eliminate the p' integral, and rewrite the integral over k as an angular integral:

$$\int \mathrm{d}\pi_2 = \int \frac{\mathrm{d}\omega'}{(2\pi)^3} \omega'^2 \mathrm{d}\Omega \frac{1}{4\omega' E'} (2\pi) \delta \left(E' + \omega' - m - \omega\right)$$

Now we can rewrite E' in terms of ω':

$$E' = \sqrt{m^2 + (\omega - \omega' \cos\theta)^2 + (\omega' \sin\theta)^2}$$

Now doing the ϕ integral, and using a delta function identity:

$$\int \mathrm{d}\pi_2 = \int \frac{\mathrm{d}(\cos\theta)}{(2\pi)} \frac{\omega'}{4E'} \frac{1}{\left|1 + \frac{\partial E'}{\partial \omega'}\right|}$$

We see that we need to calculate $\frac{\partial E'}{\partial \omega'}$:

$$\frac{\partial E'}{\partial \omega'} = \frac{\omega' - \omega\cos\theta}{E'}$$

Now recall that the delta function enforces $E' = m + \omega - \omega'$, so we have that

$$\frac{\partial E'}{\partial \omega'} = \frac{\omega' - \omega\cos\theta}{m + \omega - \omega'}$$

Inserting this into our phase space integral, we have:

$$\begin{aligned}\int \mathrm{d}\pi_2 &= \int \frac{\mathrm{d}(\cos\theta)}{2\pi} \frac{1}{4} \frac{\omega'}{m + \omega(1 - \cos\theta)} \\ &= \frac{1}{8\pi} \int \mathrm{d}(\cos\theta) \frac{\omega'^2}{m\omega}\end{aligned}$$

Now that we have the phase space integral, let us now compute the cross section. First, we note that the relative velocity is 1, $|v_A - v_B| = 1$. We can write out the differential cross section:

$$\begin{aligned}\frac{\mathrm{d}\sigma}{\mathrm{d}(\cos\theta)} &= \frac{1}{2m} \frac{1}{2\omega} \frac{1}{8\pi} \frac{\omega'^2}{m\omega} \left(\frac{1}{4} \sum_{\text{spins, pol.}} |M|^2\right) \\ &= \frac{1}{2m} \frac{1}{2\omega} \frac{1}{8\pi} \frac{\omega'^2}{m\omega} \left[2e^4 \left(\frac{\omega'}{\omega} + \frac{\omega}{\omega'} + 2m^2 \left(\frac{1}{m\omega} - \frac{1}{m\omega'}\right) + m^2 \left(\frac{1}{\omega} - \frac{1}{\omega'}\right)\right)\right]\end{aligned}$$

Now applying Compton's formula:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}(\cos\theta)} = \frac{1}{2m} \frac{1}{2\omega} \frac{1}{8\pi} \frac{\omega'^2}{m\omega} 2e^4 \left[\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - \sin^2\theta\right]$$

Which gives us the differential cross section for Compton scattering:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\left(\cos\theta\right)} = \frac{\pi\alpha^2}{m^2}\left(\frac{\omega'}{\omega}\right)^2\left[\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - \sin^2\theta\right]$$

where $\alpha = \frac{e^2}{4\pi}$ is the fine structure constant for QED. This result is known as the Klein-Nishina formula. Suppose the incoming photon energy goes to zero, $\omega \to 0$, in which case (from Compton's formula) $\omega' \to \omega$. In this limit, we have elastic scattering, and the cross section is:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\left(\cos\theta\right)} = \frac{\pi\alpha^2}{m^2}\left(1 + \cos^2\theta\right)$$

From which we can integrate to find the total cross section:

$$\sigma_{\text{total}} = \frac{8\pi\alpha^2}{3m^2}$$

This is the cross section for Thomson scattering.